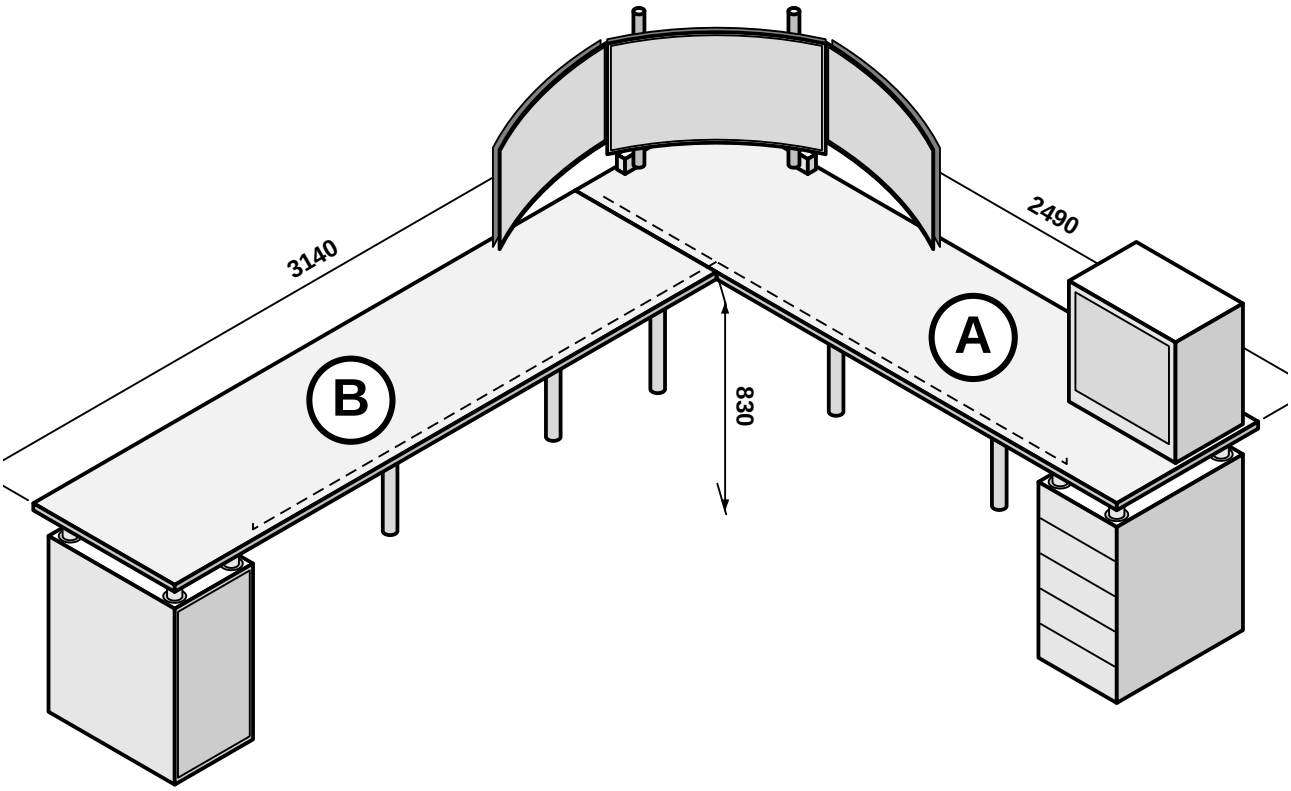
L-Shaped Corner Desk

Assembly manual



Footprint	2490 × 3140 mm
Surface height	830 mm
Tops	A along the back wall, B along the side wall (two IKEA EKBACKEN 2490 × 650 × 28)
Floor needed	2520 × 3170 mm, including the 30 mm wall gaps

All dimensions in millimetres. The rails are hidden under the front edges and are drawn dashed.

Contents

A · Finished assembly 3

B · Before you begin 5

C · Complete parts inventory 9

D · Component preparation 12

E · Steel rail fabrication and drilling 22

F · Steel corner assembly 31

G · Legs and cabinet support 35

H · Joining and positioning the countertops 44

H · 8b Wall-gap trim (optional) 48

I · Monitor mount and accessories 51

J · Final inspection and load testing 56

K · Maintenance and troubleshooting 62

Appendix · Outstanding measurements 64

A · Finished assembly

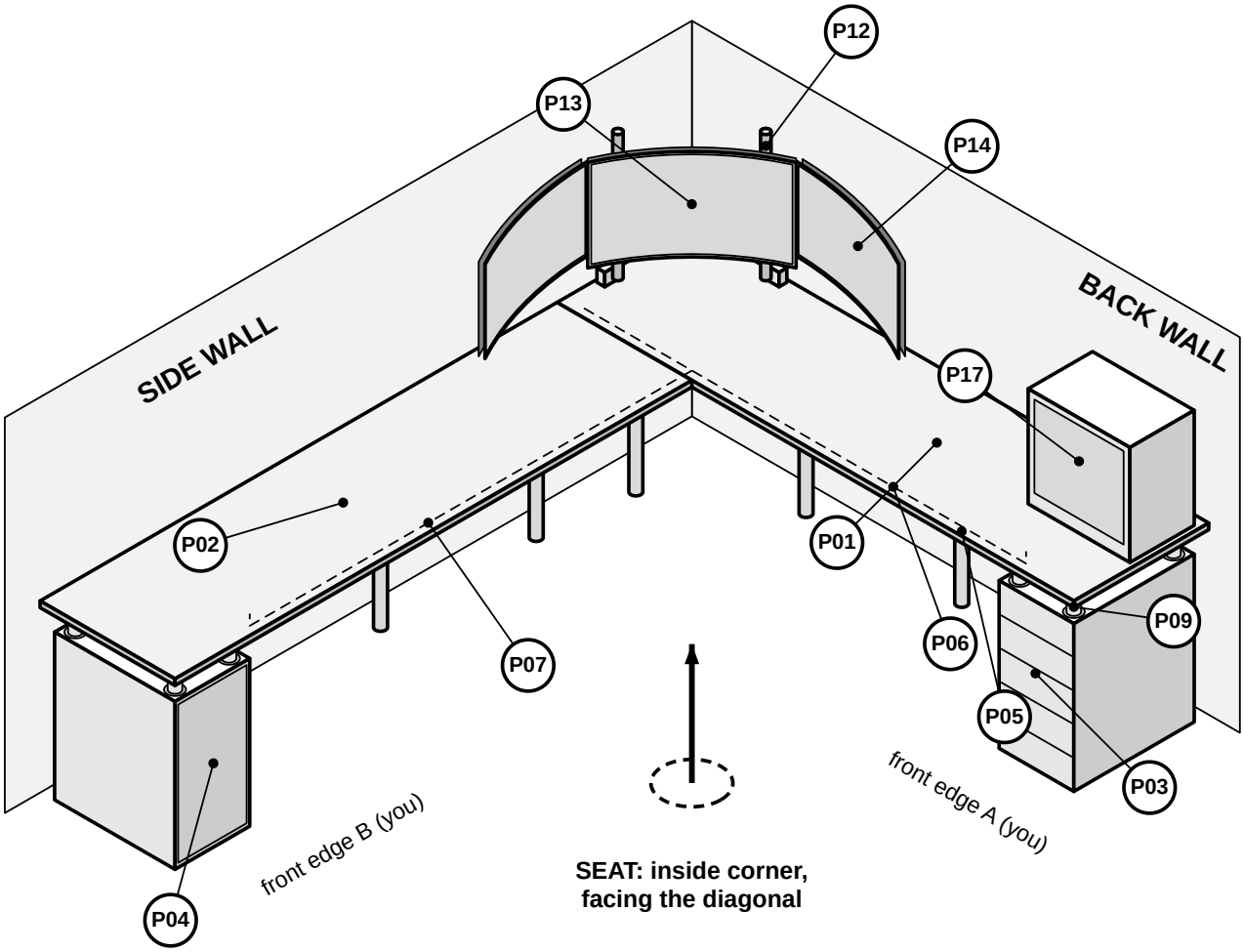


Fig. A-1 The finished desk seen from the room. Dashed lines under the front edges: the two steel rails, hidden by the tops. Left and right are as you sit at the inside corner facing the diagonal: top A is on your right, top B on your left.

No.	Part	No.	Part
P01	Top A, EKBACKEN 2490 x 650 x 28	**P07**	Rail B, steel angle 2130
P02	Top B, EKBACKEN 2490 x 650 x 28	**P09**	Round adjustable foot 4 in x 8
P03	ALEX 1, drawer unit 360 x 580 x 700	**P12**	VIVO STAND-V300CU
P04	ALEX 2, storage unit with door 360 x 580 x 700	**P13**	Samsung Odyssey G5 32 in
P05	OLOV leg x 6, 105 mm plate	**P14**	Acer XZ320QU x 2
P06	Rail A, steel angle 2130	**P17**	HYTE Y60 on printed pedestal

What this desk is

Two IKEA kitchen tops make an L. Top A lies along the back wall and keeps the corner square; top B lies along the side wall and its end butts against top A's front edge. Each top stands 30 mm off its wall, with its square, unfinished long edge toward the wall. The far end of each top rests on an ALEX cabinet through four adjustable feet: the feet are screwed to the top, and their pads drop into printed rings on the cabinet, so the desk cannot slide on the cabinets and the two heavy boxes hold the whole frame square. Six OLOV legs stand along the two wall edges, none at the front, so the space under the inside corner is open for your knees.

That leaves the two front edges with nothing under them for more than two metres, so each one is carried by a steel angle rail screwed to the underside 120 mm back from the front edge. Under the inside corner the two rails are bolted together with two gusseted steel fittings, so the corner of top B hangs from rail A. The rails never touch a cabinet: their ends load the top, and the top loads the nearest feet. The monitor stand clamps to top A's two wall edges, 400 mm from the corner; the surface is 830 mm above the floor.

A · Plan of the finished desk

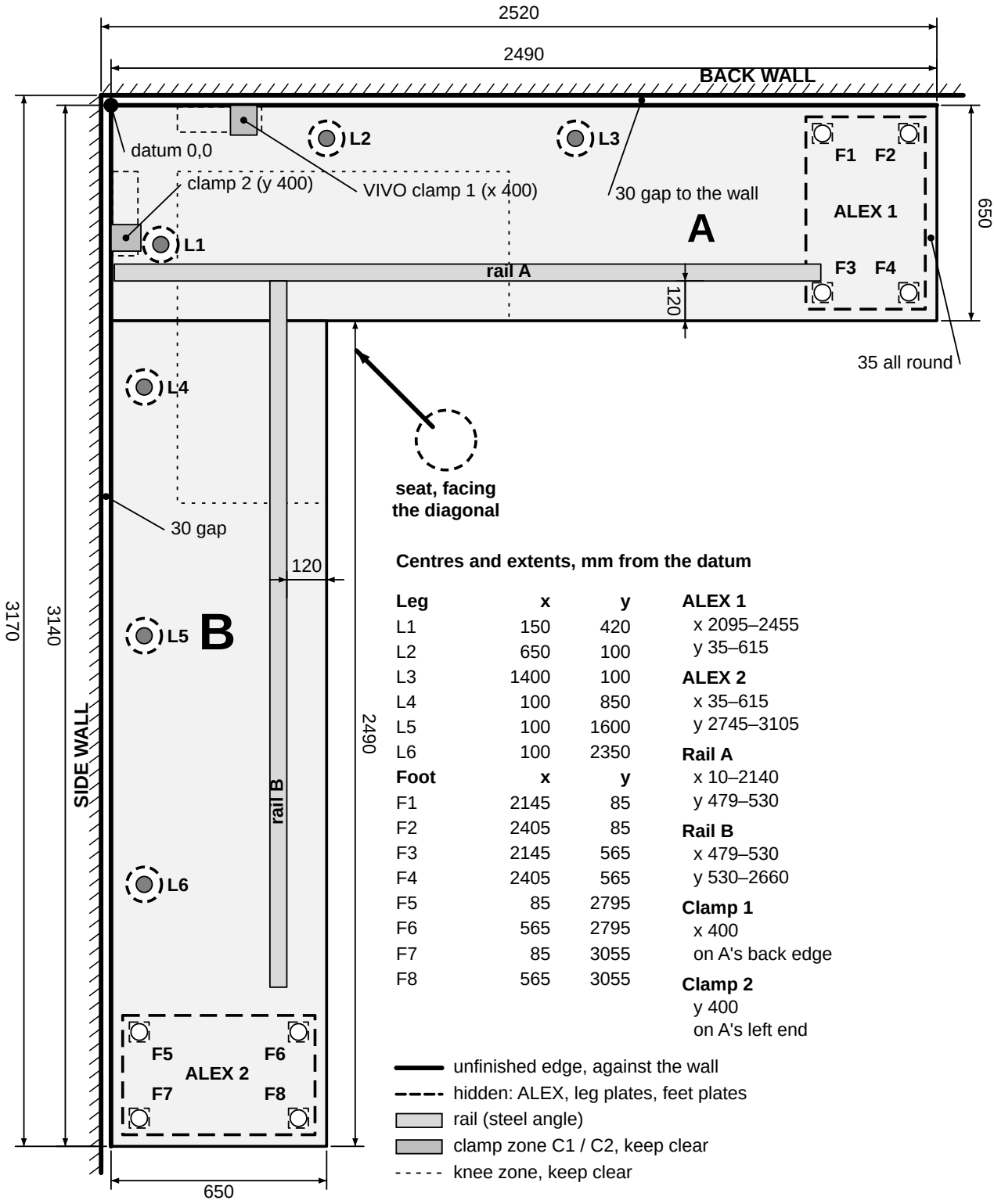


Fig. A-2 Top-down plan, seen from above. Datum: top A's back-left corner (0, 0); x runs along the back wall, y along the side wall, all in mm. Coordinates SOURCE-DEFINED (docs/build-manual.md). ALEX 1 drawers and ALEX 2 door face you.

Height above the floor	mm	Confidence
Top surface	830	SOURCE-DEFINED
Underside of the tops (OLOV setting)	802	SOURCE-DEFINED
Rail lower edge	751	SOURCE-DEFINED
ALEX top (the feet stand on it, 102 below the underside)	700	ASSUMPTION / FIELD CHECK measure your units

B · Before you begin

Read this chapter once before you pick up a drill. Budget a weekend: the steel on day one, the assembly on day two.

Safety

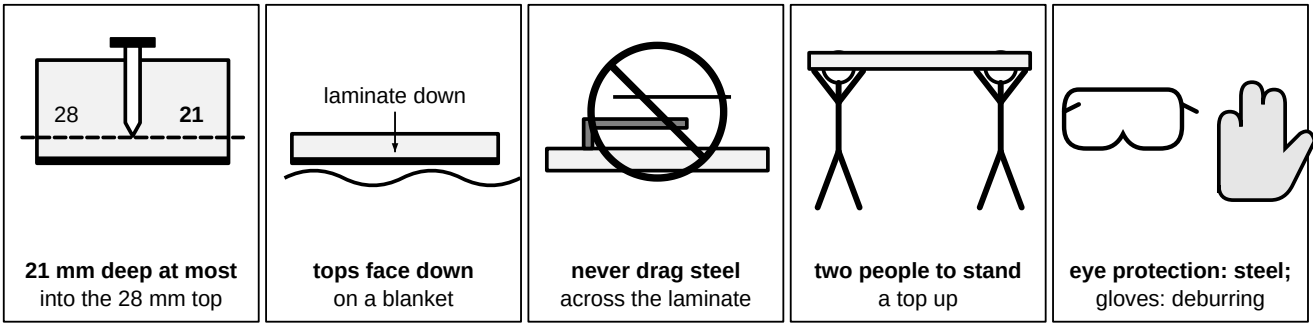
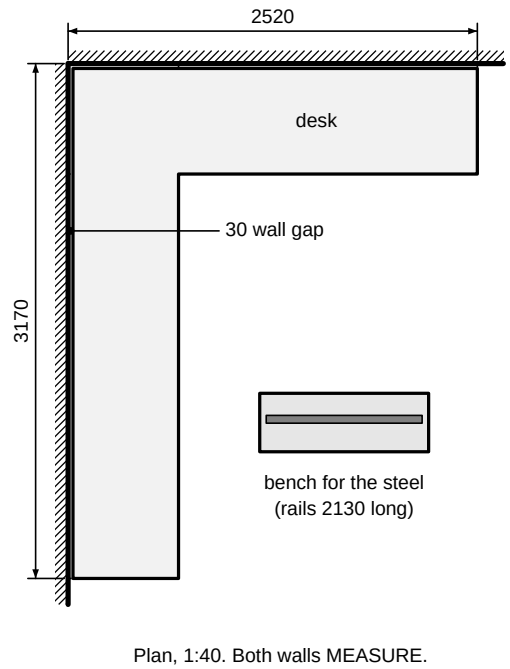


Fig. B-1 · The five rules that protect the tops and you.

> **NEVER A FASTENER DEEPER THAN 21 MM INTO A TOP**
> The EKBACKEN tops are 28 mm particleboard with a thin laminate skin. A screw or pilot that reaches the skin lifts it into a bump you will see and feel for ever. Every screw in this manual stops at least 7 mm short of the top face (28 − 21). If your tops measure 27 mm instead of 28, shorten every screw into the top by 1 mm. If you ever substitute a screw, keep this rule. Chapter D shows every depth.

- **Laminate protection.** Spread a blanket on the floor first. The tops lie face down on it, laminate on the blanket, for every marking and drilling step. Never slide or drag steel (rails, fittings, the clamp plates) across the laminate: lift it on and off.
- **Eye protection** for every hole drilled in steel; drill at low speed with cutting oil. **Gloves** while you deburr and file the steel: drilled edges are sharp.
- **Two people** to turn a top over and stand it on its legs (2490 long). A helper is also needed to slide the finished desk off and back to the walls.

Work area



Plan, 1:40. Both walls MEASURE.

Fig. B-2 · Floor needed: 2520 × 3170, including the 30 mm gaps to both walls.

The finished desk needs **2520 × 3170 mm** of clear floor in the corner: each top is 2490 long and stands 30 mm off its wall. MEASURE ON HARDWARE Measure both walls (wall to wall) and the baseboard thickness before you start.

You also need floor space for one top at a time lying face down on the blanket (2490 × 650 plus room to kneel all round), and a **bench** with two clamps for drilling the two 2130 mm steel rails. All steel is drilled on the bench, never under the desk.

Clear the corner before chapter G: the ALEX units are placed by the walls before the tops go on.

Workshop tools

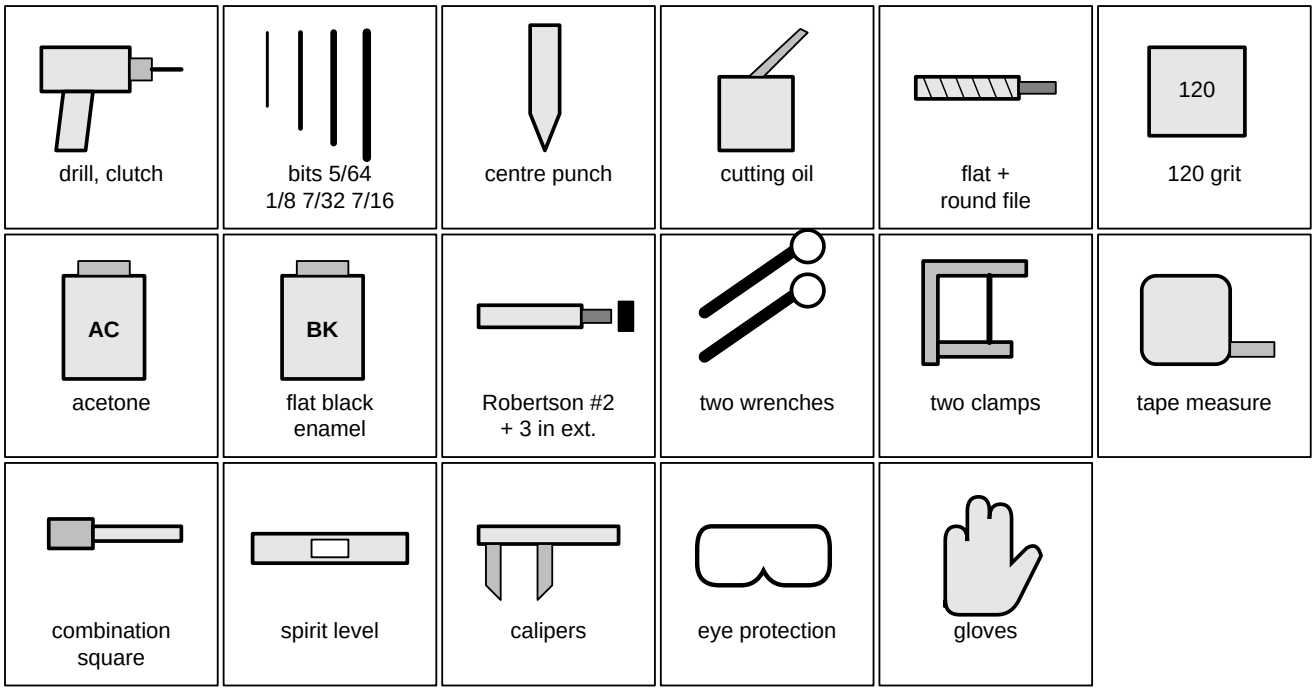


Fig. B-3 · Tools and protective equipment. Not to scale.

Tool	For
Corded or 18 V drill, clutch set low	all holes and screws
Bits: 5/64, 1/8, 7/32 in HSS; 7/16 in cobalt	ring and trim pilots; top and steel pilots; rail screw holes; fitting bolt holes
Centre punch (or a nail)	only for a mark the jig or plug cannot reach
Cutting oil or 3-in-1	every hole in steel
Flat file, round file, 120 grit, acetone, flat black enamel	deburr, ease a tight hole, scuff, degrease, paint the rails
Robertson #2 bit + 3 in magnetic extension	rail screws beside the vertical leg (the chuck would hit the leg otherwise)
Two wrenches or sockets for the 3/8-16 bolts (one for the head, one for the nut)	the corner fittings
Two clamps, 6 in or larger	hold each rail on its line; hold a fitting as a drilling template
Tape measure, combination square, masking tape, pencil	layout and the tape labels on the tops and rails
Spirit level, 60 cm or longer	setting the legs and feet
Calipers	top thickness, pad diameter, ALEX panel (the measurements on the last page of this chapter)

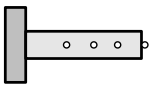
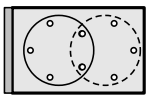
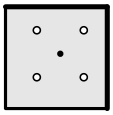
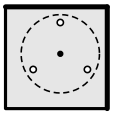
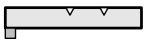
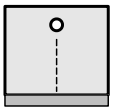
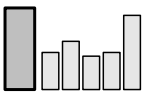
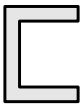
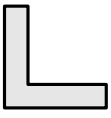
Protective equipment

- Safety glasses or goggles: every hole in steel, and opening the 7/16 in holes.
- Work gloves: deburring, filing and handling the cut rail ends.
- A blanket large enough for a whole top (2490 × 650).

Printed tools (P18): print these before you mark anything

All of them are in `parts/tools/`; `parts/README.md` has print settings. Print as saved, no supports. **Never scale a printed tool in the slicer: its distances are the tool.** The lipped and fenced tools print lip-up and are turned over to use.

> **EVERY JIG HOLE IS BIT-SIZED**
> Hold the jig on, start each pilot 3–4 mm through the jig's hole, lift the jig, then finish the hole to depth with the drill guide's puck. The small centre holes are for a scriber only; nothing is drilled through them.

Tool	What it does	Qty
	**Edge T-square.** Lip over any edge; punch holes at 50 (ring centres on the ALEX), 85 (feet, with the plate's four 1/8 in holes 46.8 apart round it), 100 (legs L2–L6) and 150 (leg L1).	1
	**Leg jig.** Lip over the wall edge; an OLOV's centre and its five 1/8 in holes in one placement: 100 pattern for L2–L6 (solid circle), 150 pattern for L1 (dashed circle, ringed holes).	1
	**Feet corner jig.** Fences on a far corner of a face-down top; a foot's centre 85/85 from both edges and its four 1/8 in holes 46.8 apart: F2, F4, F7, F8.	1
	**ALEX corner template.** Fences on an ALEX corner; ring centre 50/50, the ring's three 5/64 in holes (radius 33) and the three optional ALEX-foot holes (radius 14.5). Top and bottom corners.	1
	**Rail marking gauge.** Lip over the angle's edge; pencil notches at 19 (screw line on the flat leg) and 29 (bolt line on the vertical leg).	1
	**Rail drill jig.** Lip over the flat leg's free edge; a 1/8 in bore on the 19 line; sight grooves on the station line. PETG.	1
	**Slot plug** (fitting-centring-plug). Drops into a fitting's 21 × 12 slot; a 1/8 in bore dead centre, so the pilot cannot wander in the slot's play.	2
	**Pilot drill guide.** SET tower (45 for the 1/8 in bit, 35 for the 5/64 in) and five depth pucks: 12 PLATE, 15 LEG, 10 EDGE, 12 ALEX, 25 RAIL. Re-set the bit every time you change it.	1 set
	**Foot height gauge.** A C with 102 inside: all eight feet wound to the same height without a ruler.	1
	**Wall gauge.** An L: 65 (ALEX back to wall) and 30 (a top's raw edge to wall).	1

Also print the eight locating rings (P10) now: `parts/structure/foot-ring`, PETG or ABS, flange down. Their 51 mm bore depends on the pad diameter you measure (last page of this chapter).

How sure is each number?

Every dimension in this manual carries one of three confidence labels, or a fourth label that is an instruction to you:

VERIFIED	Measured on the purchased part, or taken from the part's own geometry file (the printed parts).
SOURCE-DEFINED	Defined by the project documents (the build manual): a design position, not a measurement. It is right by definition; you put the part there.
ASSUMPTION / FIELD CHECK	An engineering estimate or a check done on site. Confirm it before you rely on it.
MEASURE ON HARDWARE	Not yet measured on your parts. Measure it yourself before the step that uses it, and stop if it differs.

Confirm before you drill anything

> **MEASURE BEFORE DRILLING**
> These numbers are not yet confirmed on the parts you have. Each one is checked again where it is used; if one differs, stop and correct the layout before any hole is drilled.

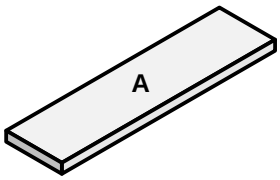
Measure	Expected	Label	Used in
EKBACKEN thickness, both tops (calipers)	28	MEASURE ON HARDWARE	D: every screw depth into the tops. At 27, shorten every screw into the top by 1 mm.
Delivered rail lengths, both pieces, and the angle thickness	2130 layout length (bought as 84 in = 2133.6); 4.8 thick	MEASURE ON HARDWARE	E: hole stations. 2130 is the layout length; the far end may carry the extra.
Strut fitting slots, positions from the heel	21 × 12 slots, centres 29.15 and 78.45 (legs 100.5 × 41)	VERIFIED	E, F: measured 23 Oct 2026; the fitting with the slot plug is still the drilling template.
Feet pad diameter (calipers)	50	MEASURE ON HARDWARE	G: the rings are bored 51 for a 50 pad; re-print them if yours differs.
ALEX top panel thickness, at an edge	16 or more for the 5/8 in screws	MEASURE ON HARDWARE	G: ring screws, 13 mm into the ALEX top.
ALEX height, with and without its stock feet	700	ASSUMPTION / FIELD CHECK	G: 700 + foot 102 = underside 802.
VIVO top arm reach; maximum clamp opening	about 89 ASSUMPTION / FIELD CHECK ; 31 or more	MEASURE ON HARDWARE	I: the clamps on the 28 top plus a pad plate.
Room: wall to wall along both walls; baseboard thickness	2520 and 3170	MEASURE ON HARDWARE	B (work area), G (placing the ALEX units by the walls).
Feet load rating	unknown; the design wants 50 kg or more each	ASSUMPTION / FIELD CHECK	C (P09); check the listing or the maker.

Already measured and printed as VERIFIED: the OLOV plate (105, five 5.5 holes on an 85 circle), the feet plate (59.5, four 5.2 holes 46.8 apart), the strut fittings (slots 21 × 12 at 29.15 and 78.45 from the heel), the VIVO crossbar (502 centre to centre), pole (35), pole inset (43.7), clamp back (80 × 17.5) and lower pad.

C · Complete parts inventory

Check every part against this list before you start. Part numbers P01–P21 and hardware numbers H01–H11 are used in every callout balloon in this manual. Dimensions in mm. REQUIRED parts are needed for the desk to stand; OPTIONAL parts can be left out.

thickness drawn ×4



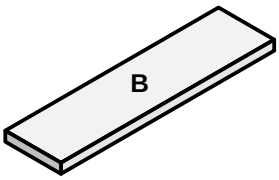
P01 **× 1**

Top A, IKEA EKBACKEN brown walnut, 2490 × 650 × 28. Front long edge rounded and laminated; back long edge square and unfinished (wall side).

MEASURE ON HARDWARE thickness

REQUIRED

thickness drawn ×4

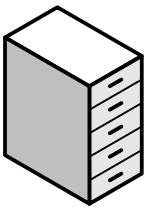


P02 **× 1**

Top B, identical to P01: 2490 × 650 × 28. Becomes B by being laid along the side wall (chapter D).

MEASURE ON HARDWARE thickness

REQUIRED



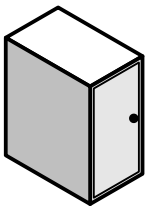
front

P03 **× 1**

ALEX 1, IKEA drawer unit, 360 × 580 × 700. Under top A's far end, drawers toward you.

ASSUMPTION / FIELD CHECK height 700

REQUIRED



front

P04 **× 1**

ALEX 2, IKEA storage unit with door, 360 × 580 × 700. Under top B's far end, door toward you.

ASSUMPTION / FIELD CHECK height 700

REQUIRED

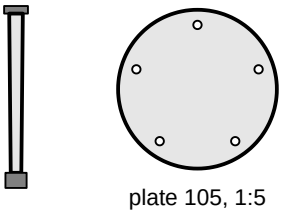


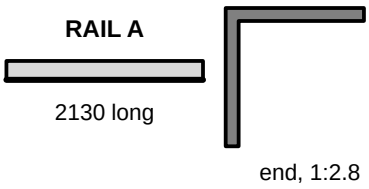
plate 105, 1:5

P05 **× 6**

IKEA OLOV leg, adjustable 600–900, set to 802. Plate ⌀105, five ⌀5.5 holes on an ⌀85 circle. Max 13 kg each.

VERIFIED plate

REQUIRED

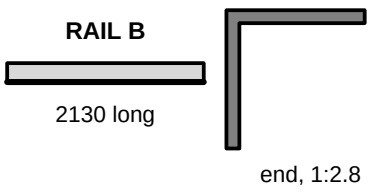


P06 **× 1**

Rail A, 2 × 2 × 3/16 in hot-rolled steel angle (50.8 × 50.8 × 4.76), 2130 layout length (bought as 84 in = 2133.6).

MEASURE ON HARDWARE length

REQUIRED

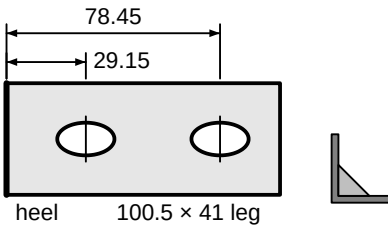


P07 **× 1**

Rail B, identical to P06 until its corner end is taped (chapter D).

MEASURE ON HARDWARE length

REQUIRED

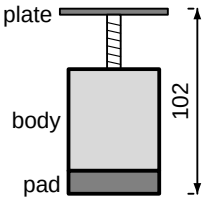


P08 **× 2**

Gusseted 90° strut fitting, 5 mm steel, for 1-5/8 in channel. Legs 100.5 × 41; two 21 × 12 slots per leg, centres 29.15 and 78.45 from the heel.

VERIFIED

REQUIRED



P09 **× 8 (+4)**

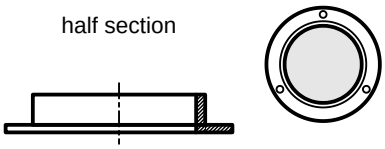
Round adjustable foot, 4 in, set to 102. Body and pad ⌀50; plate 59.5 square, four ⌀5.2 holes 46.8 apart. 12 on hand: 8 used, 4 SPARE.

VERIFIED plate ·

MEASURE ON HARDWARE pad · load

ASSUMPTION / FIELD CHECK

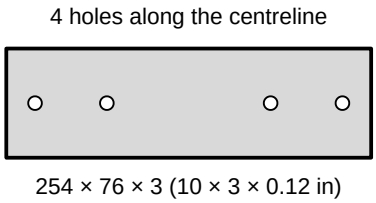
REQUIRED



P10 **× 8**

Locating ring, printed PETG or ABS: bore 51, OD 57, ring 10 high on a ⌀75 × 2.5 flange, three countersunk ⌀3.8 holes on radius 33. **VERIFIED** geometry file

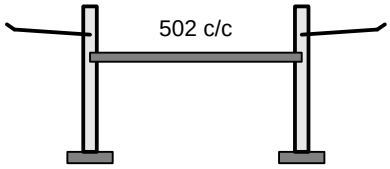
REQUIRED



P11 **× 4**

Mending plate, 10 × 3 × 0.12 in, holes at 20, 70, 184, 234 from one end. Two join the tops (P1, P2); two pad the VIVO clamp jaws. **SOURCE-DEFINED**

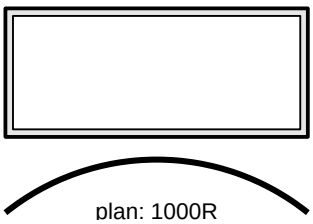
REQUIRED



P12 **× 1**

VIVO STAND-V300CU: two clamps, two ⌀35 poles 610 (24 in) tall, crossbar 502 centre to centre, two side arms, centre bracket. Clamp back 80 along the edge, 17.5 behind it. **VERIFIED**

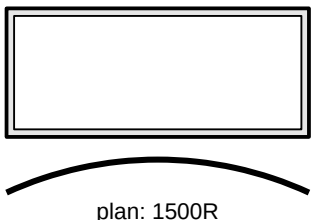
REQUIRED



P13 **× 1**

Samsung Odyssey G5 32 in, 1000R, VESA 75, about 5 kg. Centre screen, landscape, bottom edge 230 above the top. **SOURCE-DEFINED**

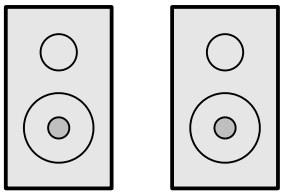
REQUIRED



P14 **× 2**

Acer XZ320QU 31.5 in, 1500R, VESA 100, 5.6 kg with stand (about 5 kg on the arm). Side screens, landscape. **SOURCE-DEFINED**

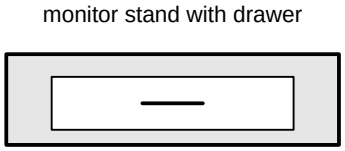
REQUIRED



P15 **× 2**

Edifier R1280DBs speakers, 234 tall, one under each Acer. **SOURCE-DEFINED**

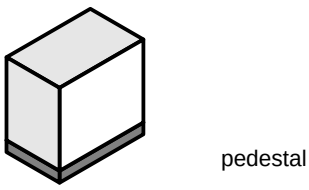
REQUIRED



P16 **× 1**

IKEA ELLOVEN monitor stand with drawer, 100 tall, under the G5. **SOURCE-DEFINED**

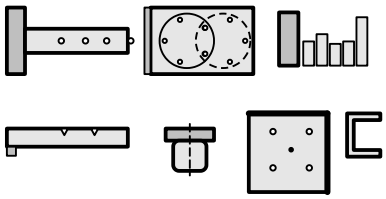
REQUIRED



P17 **× 1**

HYTE Y60 on its printed pedestal (MakerWorld 679012): footprint 490 × 310, 480 tall with the pedestal, over ALEX 1. **SOURCE-DEFINED**

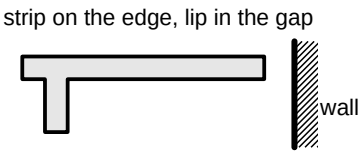
REQUIRED



P18 **1 set**

Printed tools: T-square, leg jig, feet corner jig, ALEX corner template, rail gauge, rail drill jig, slot plug × 2, drill guide (tower + 5 pucks), foot height gauge, wall gauge. List in chapter B.

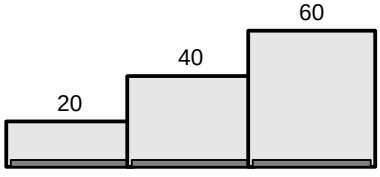
REQUIRED



P19 **1 set**

Wall-gap trim, printed: corner, fillers, clamp pieces, segments, connector tiles. Cut list in `parts/README.md`. Fixed with H07.

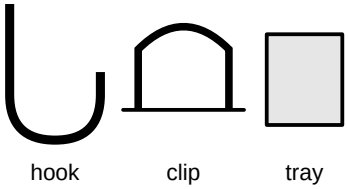
OPTIONAL



P20 **× 8**

ALEX feet, printed: ⌀56, one height for all (20, 40 or 60), with TPU pads. Raise the whole desk by their height.

OPTIONAL



P21 **1 set**

Cable parts, printed: rail cable hooks (magnets H11), cable clips, power-bar cradles, cable trays.

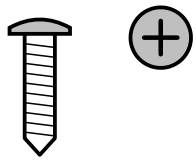
OPTIONAL

Drawings are silhouettes for recognition, not to scale unless a scale is written on them.

Hardware H01–H11

Screws and bolts are drawn **1:1** (true length, nominal diameter): lay a screw on the drawing to check it. Washers and the nut are symbols. All hardware was ordered 22 Oct 2026 SOURCE-DEFINED. Buy a few spares of each screw.

H01 × 30

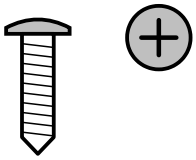


1:1 · head as supplied

M4 × 16 machine screw, fully threaded, IKEA 108443 (with the legs). OLOV plates, 5 per leg.

REQUIRED

H02 × 32

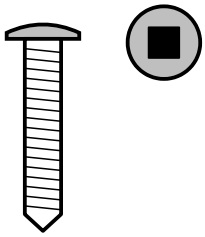


1:1 · pan head

#8 × 5/8 in pan head. Feet plates, 4 each.

REQUIRED

H03 × 19

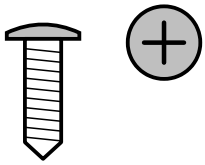


1:1 · pan head, Robertson

#10 × 1 in pan head, Robertson. Rails into the tops (9 + 10).

REQUIRED

H04 × 8

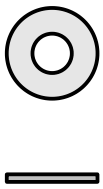


1:1 · pan head

#10 × 5/8 in pan head. Joining plates P1, P2, 4 each.

REQUIRED

H05 × 8

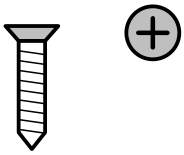


symbol, not to scale

#10 flat washer, under each H04.

REQUIRED

H06 × 24

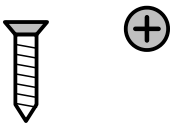


1:1 · flat head

#6 × 5/8 in flat head. Rings, 3 each. +24 for ALEX feet, more for cable parts.

REQUIRED

H07 × 48

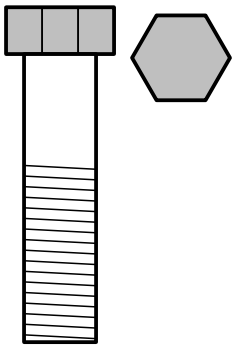


1:1 · flat head

#4 × 1/2 in flat head. Wall-gap trim, two per piece.

OPTIONAL

H08 × 6

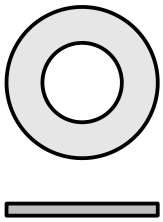


1:1 · hex head

3/8-16 × 1-1/2 in hex bolt, grade 5. Corner fittings.

REQUIRED

H09 × 12

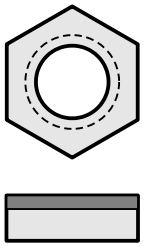


symbol, not to scale

3/8 flat washer, under each bolt head and each nut.

REQUIRED

H10 × 6

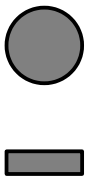


symbol, not to scale

3/8-16 nylon-insert lock nut. Stiff to turn; that is normal.

REQUIRED

H11 × 8



1:1

10 × 3 mm neodymium disc magnet, two per rail cable hook.

OPTIONAL

D · Component preparation

This chapter marks both tops face down and labels the two rails. Nothing is fixed yet. Every position is in mm from one **datum**: **top A's back-left corner**. **x** runs along the back wall, **y** along the side wall. Top A covers x 0–2490, y 0–650; top B covers x 0–650, y 650–3140.

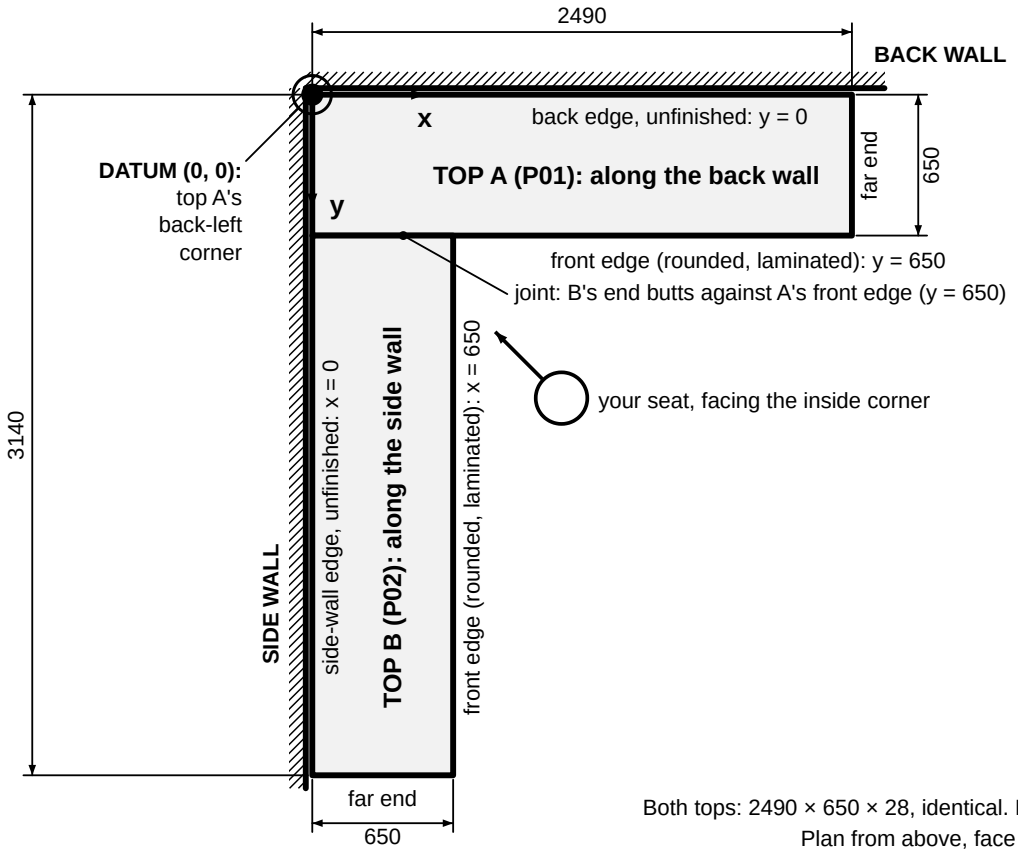


Fig. D-1 · The datum and the two tops, seen from above as installed. SOURCE-DEFINED

Which edge is which

Each top has one **rounded, laminated** long edge: the **front**. The other long edge is **square and unfinished**: it goes against the wall. Either short end can become the side-wall end.

Which top is A

The two tops (P01, P02) are identical. **A** is the one you lay along the back wall; its unfinished edge is y = 0. **B** lies along the side wall; its unfinished edge is x = 0 and one short end, the **joint end** (y = 650), butts against A's front edge. Top B is top A turned 90°, never mirrored.

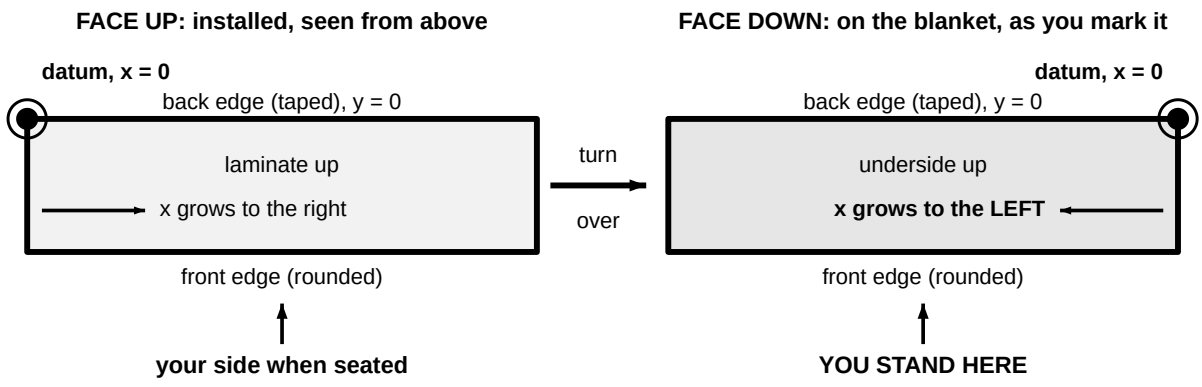
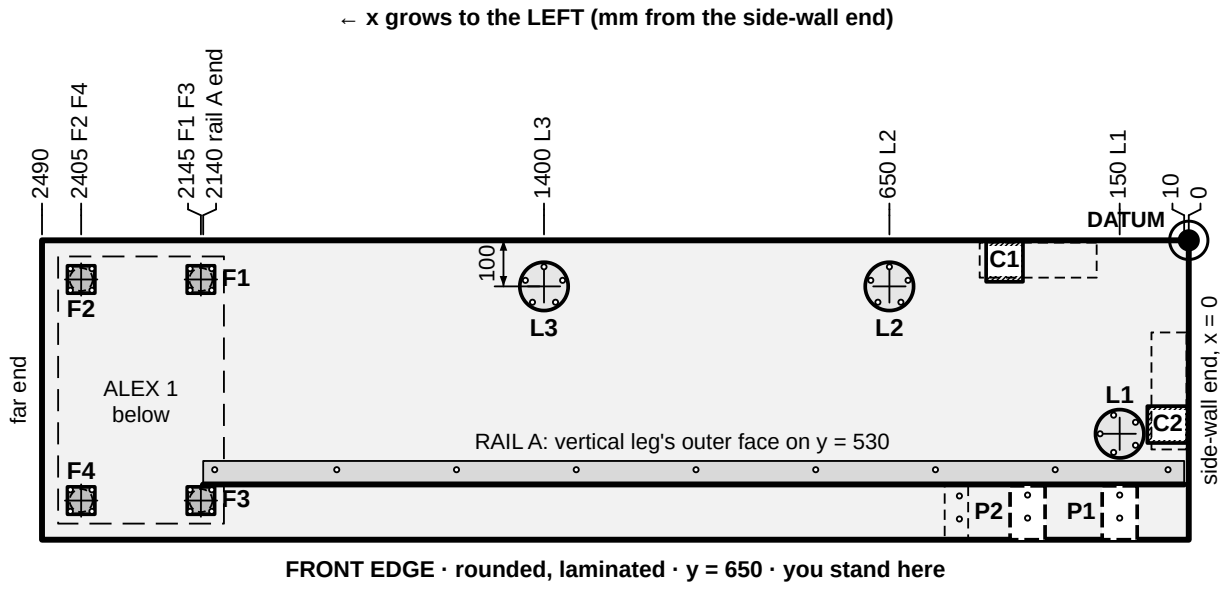


Fig. D-2 · Face-down convention used on every marking drawing. Stand opposite the taped reference edge (top A: at its front edge; top B: at its far end, joint end away from you): the underside is the plan mirrored left to right, so **x grows to the LEFT and the datum side (x = 0) is on your RIGHT**. Top B uses the same mapping.

Top A, face down: overview

1

Calipers on top A: 28 mm **MEASURE ON HARDWARE**. If it is 27, shorten every screw into it by 1 mm. Tape the **unfinished long edge** ("A · BACK EDGE") and the short end you choose as the **side-wall end** ("A · SIDE-WALL END"). Lay top A face down on the blanket and stand at the front edge, back edge away from you.



Scale 1:16. Corner and far end in Figs D-4 and D-5.

Fig. D-3 · Top A underside as you see it. Datum at the far right corner. Circles: OLOV plates (P05) with five pilots; squares: feet plates (P09) with four pilots; grey band: rail A (P06) with its screw positions; hatched C1, C2: clamp zones; dashed: pad plates, joining plates P1, P2 (P11), rail B's corner end and ALEX 1 (reference).

Mark	Centre (x, y)	Tool and placement	Note
L1	(150, 420)	leg jig, **150** pattern, lip over the **side-wall end**, station y = 420	under VIVO clamp 2; plate ends 6.5 short of rail A
L2	(650, 100)	leg jig, 100 pattern, lip over the back edge, station x = 650	nearest leg to clamp 1
L3	(1400, 100)	leg jig, 100 pattern, lip over the back edge, station x = 1400	
F1	(2145, 85)	T-square hole 85 and its four plate holes, lip over the back edge, station x = 2145	feet for ALEX 1, 50 in from each ALEX corner; plates 5 clear of rail A
F2	(2405, 85)	feet corner jig on the far back corner	
F3	(2145, 565)	T-square, lip over the **FRONT** (rounded) edge, station x = 2145	
F4	(2405, 565)	feet corner jig on the far front corner	
Rail A line	y = 530, x 10–2140	pencil line; the vertical leg's outer face goes on it	flat leg covers y 479–530
C1 / **C2**	x 360–440, y 0–90 / x 0–90, y 360–440	keep clear: VIVO clamp zones	nothing fixed here
Pad plates	x 200–454, y 5–81 / x 5–81, y 200–454	outline only	loose P11 plates under the clamp jaws, chapter I
P1 / **P2**	x 112–188 / x 312–388, y 533–650	outline only (each plate spans y 533–787 across the joint)	3 clear of rail A; screws later at y 553 and 603

All positions **SOURCE-DEFINED** (docs/build-manual.md step 2). Pilots: 1/8 in, started 3–4 mm through the jig; finishing them to depth and fitting the legs and feet follows in the next chapter.

Top A: the corner

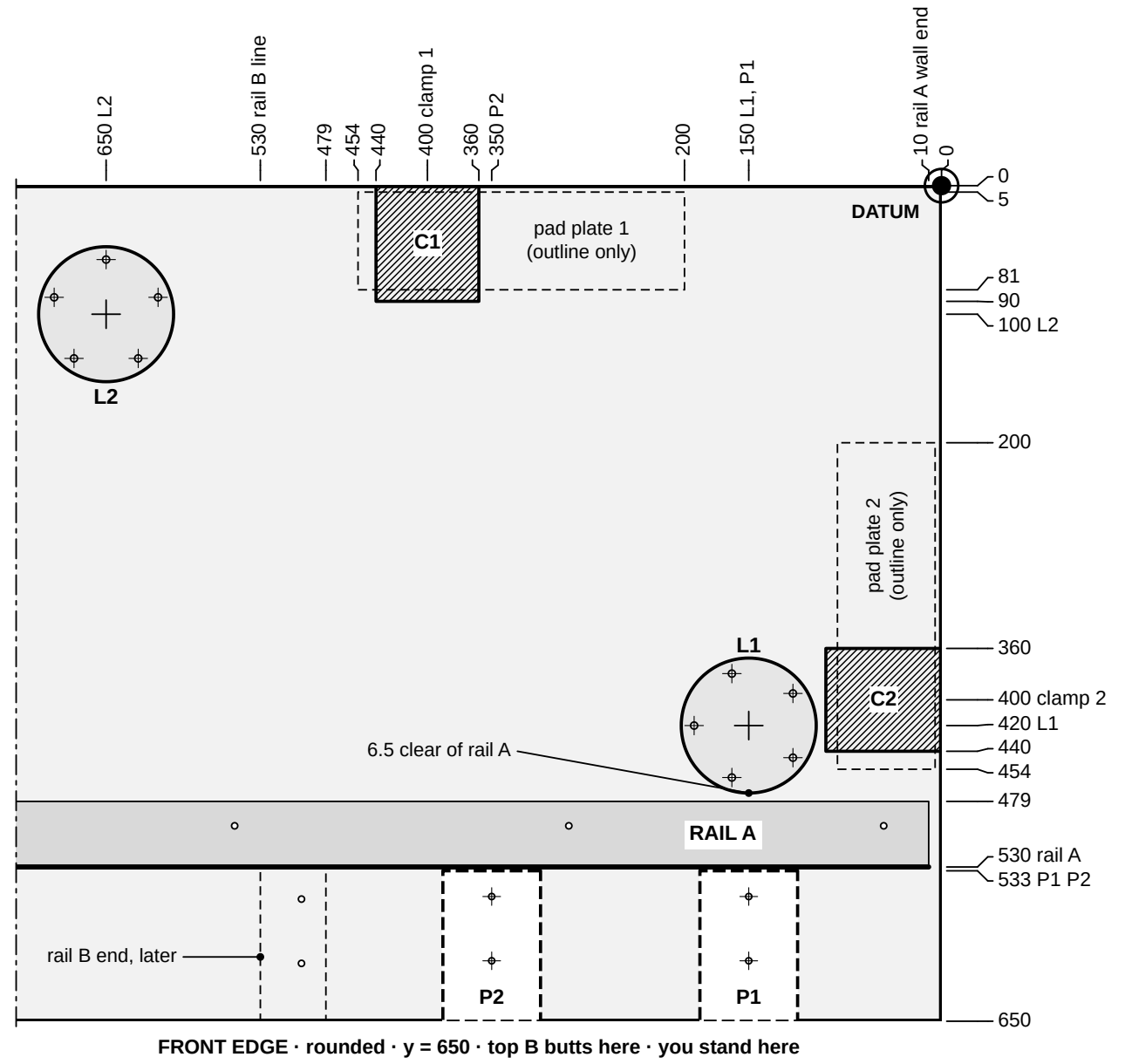
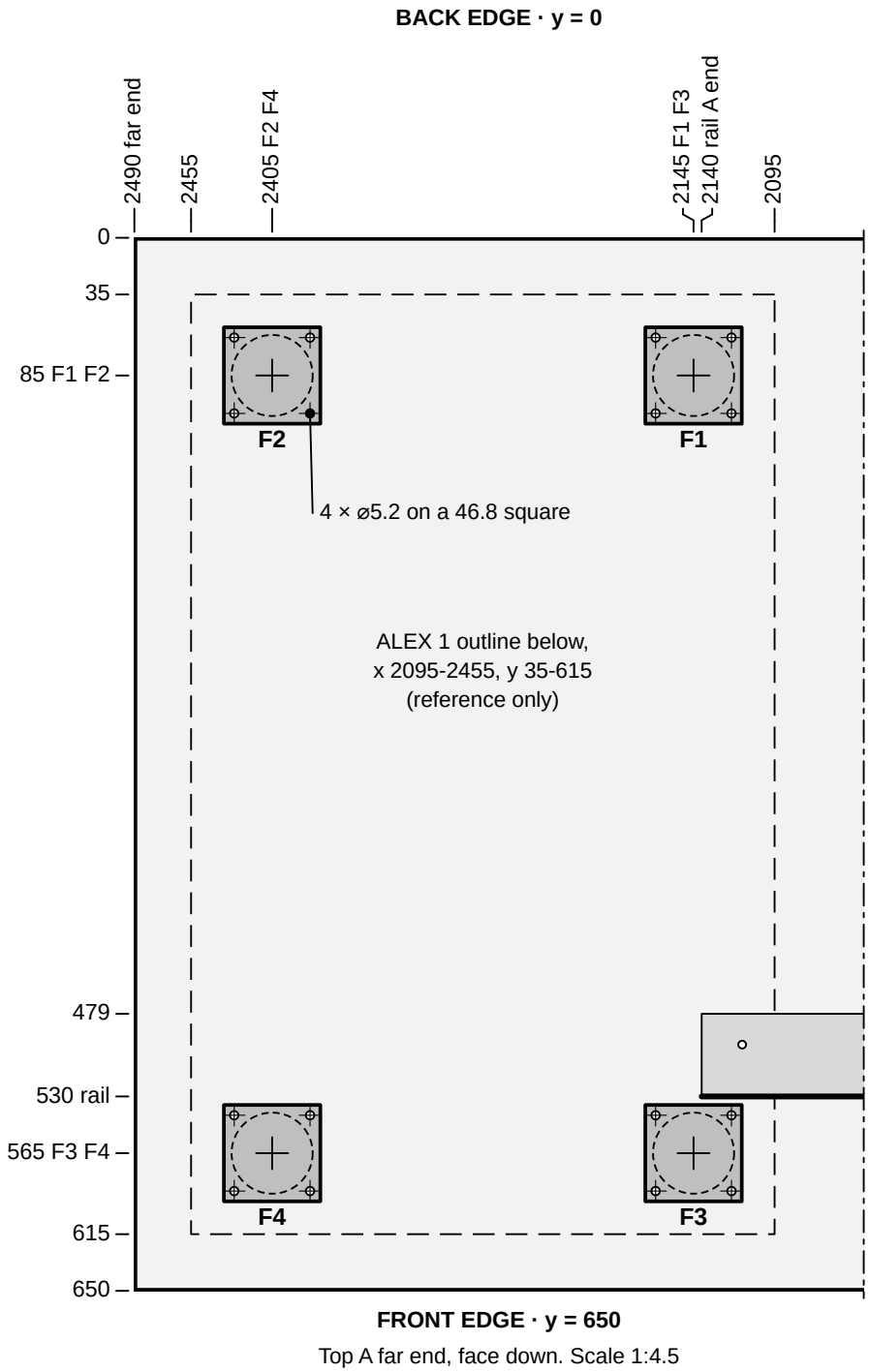


Fig. D-4 · Top A corner (x 0–720), face down, 1:5. Every station is measured from the datum: x along the back edge from the side-wall end, y in from the back edge.

2 **L1:** hook the leg jig's lip over the **side-wall end** (x = 0), slide it to station **y = 420**, and use its **150** pattern (dashed circle, ringed holes): start the five 1/8 in pilots. **L2:** lip over the back edge at x = 650, **100** pattern (solid circle).

3 Draw the rail A line at **y = 530** from x = 10 to 2140 (both ends and the middle measured from the back edge). Hatch C1 and C2 with pencil. Draw the two pad-plate outlines and the P1, P2 outlines. Nothing is drilled inside C1, C2 or the outlines.

Top A: the far end (feet for ALEX 1)



Top A far end, face down. Scale 1:4.5

Fig. D-5 · Top A far end (x 2040–2490), face down, 1:4.5.

4 **F2, F4:** drop the feet corner jig over the far back corner, then the far front corner, fences against both edges. Start the four 1/8 in pilots through it.

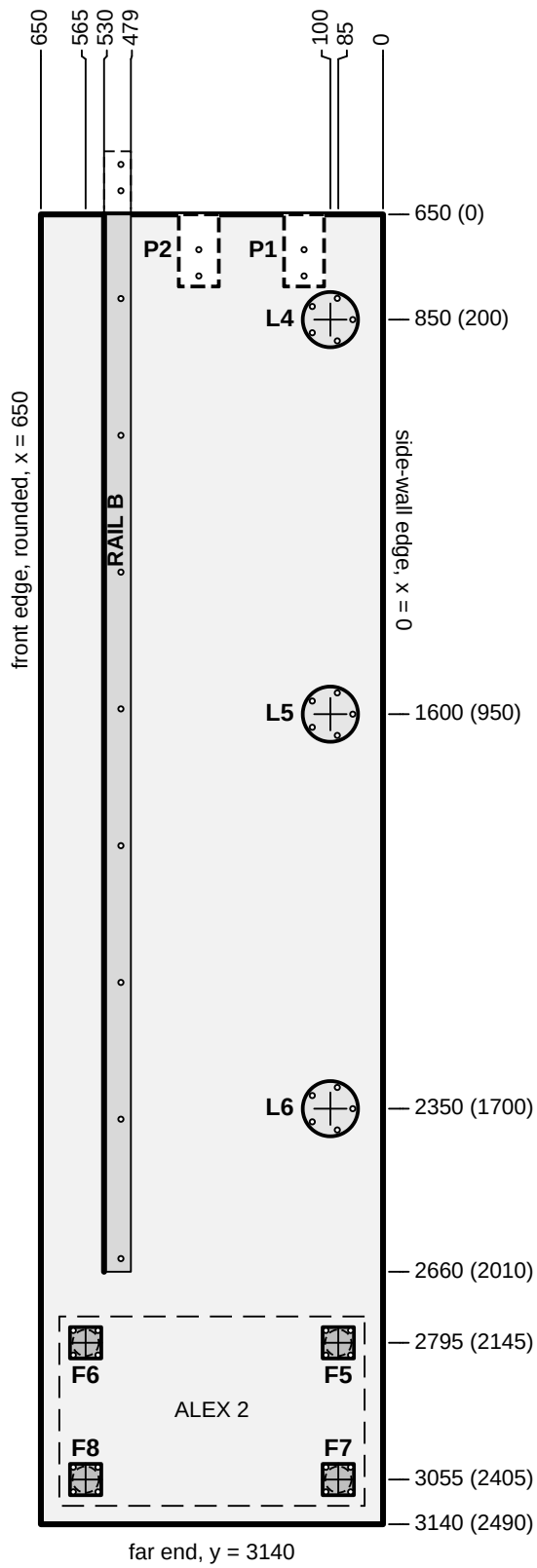
5 **F1:** T-square, lip over the back edge at station x = 2145: scriber through hole 85 to check, then start the four pilots round it. **F3:** the same with the lip over the **front** edge.

6 **L3:** leg jig, 100 pattern, lip over the back edge at x = 1400 (Fig. D-3).

Feet plate: 59.5 square, four ⌀5.2 holes 46.8 apart **VERIFIED**. Plates square with the top's edges.

The ALEX 1 outline (x 2095–2455, y 35–615) is for orientation only; nothing is marked for it.

Top B, face down: overview



y labels: desk y (tape from the joint end)
Scale 1:14

Fig. D-6 · Top B underside, joint end away from you, 1:14.

7

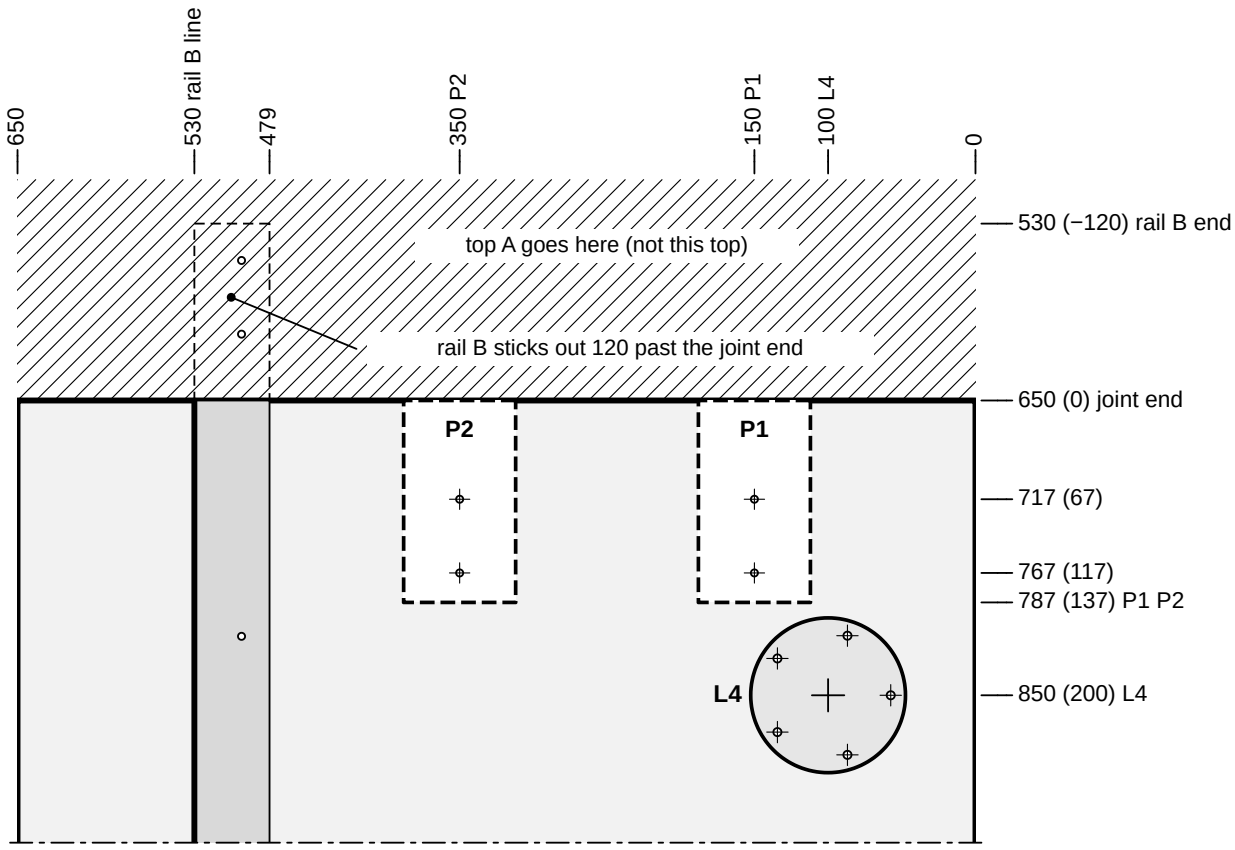
Calipers on top B: 28 mm MEASURE ON HARDWARE. Tape its **unfinished long edge** ("B · SIDE-WALL EDGE", x = 0) and the short end that will butt against top A ("B · JOINT END", y = 650). Lay it face down and stand at the far end, joint end away from you: the side-wall edge is on your right.

> Desk y on top B starts at 650. Hook the tape on the **joint end**: it reads **y − 650**, the number in brackets.

Mark	(x, y)	Tape from joint end	Tool and placement
L4	(100, 850)	200	leg jig, 100 pattern, lip over the side-wall edge
L5	(100, 1600)	950	
L6	(100, 2350)	1700	
F5	(85, 2795)	2145	T-square 85, lip over the side-wall edge
F6	(565, 2795)	2145	T-square 85, lip over the **FRONT** edge (x = 650)
F7	(85, 3055)	2405	feet corner jig, far wall-side corner
F8	(565, 3055)	2405	feet corner jig, far front corner
Rail B line	x = 530, y 650–2660	0–2010	vertical leg's outer face on it; flat leg x 479–530; corner end sticks out 120 past the joint end
P1 / **P2**	x 112–188 / 312–388, y 650–787	0–137	outline only; screws later at y 717 and 767 (67, 117)

All positions SOURCE-DEFINED (docs/build-manual.md step 3). L4–L6 have one pilot pointing at the side wall, like L2 and L3.

Top B: the joint end



Top B joint end, face down: joint end away from you, x grows to the left. Scale 1:5
(…) = tape reading from the joint end. P1, P2 holes: screw positions.

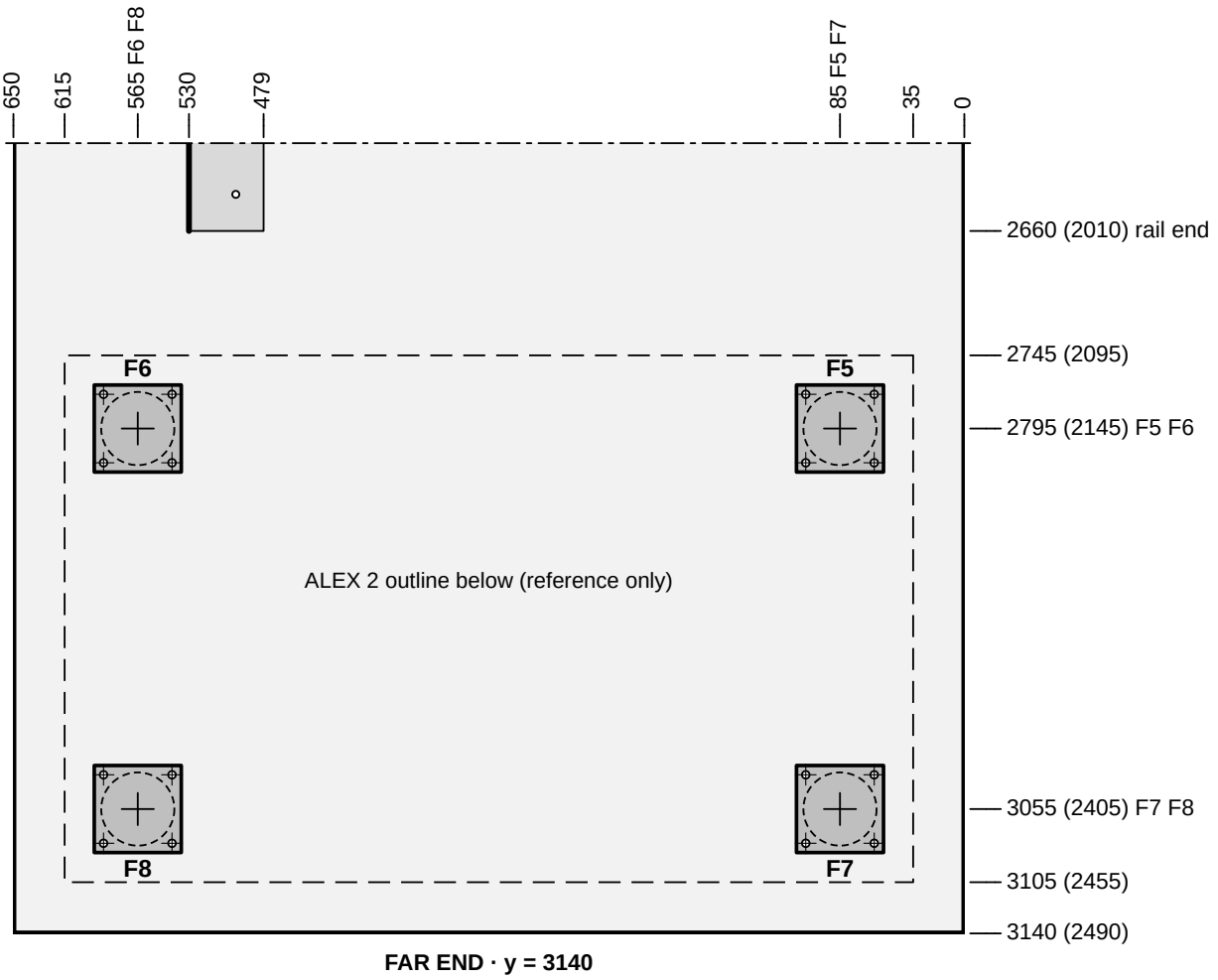
Fig. D-7 · Top B joint end (y 500–950), face down, 1:5. The hatched area beyond the joint end is where top A will be; rail B's corner end lies there.

8

L4: leg jig, 100 pattern, lip over the side-wall edge, tape 200 from the joint end. Draw the **rail B line at x = 530** from the joint end to tape 2010 (y 2660). Draw the P1 and P2 outlines from the joint end to tape 137 (y 787).

Rail B's holes 1 and 2 (shown beyond the joint end) go into top A when the tops are joined; on top B only its holes 3 to 10 are used.

Top B: the far end (feet for ALEX 2)



Top B far end, face down. Scale 1:5. (…) = tape from the joint end

Fig. D-8 · Top B far end (y 2600–3140), face down, 1:5.

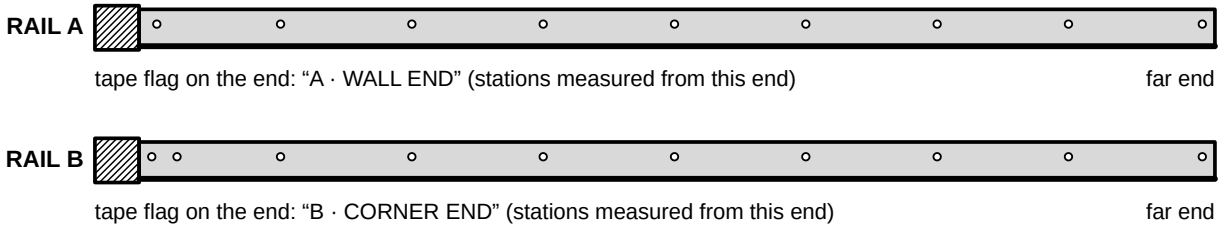
9

F7, F8: feet corner jig over the two far corners, fences against both edges; start the four 1/8 in pilots. **F5:** T-square, lip over the side-wall edge, tape 2145 from the joint end; **F6:** the same with the lip over the front edge. **L5, L6:** leg jig, 100 pattern, tape 950 and 1700 (Fig. D-6).

Rails: labels and orientation

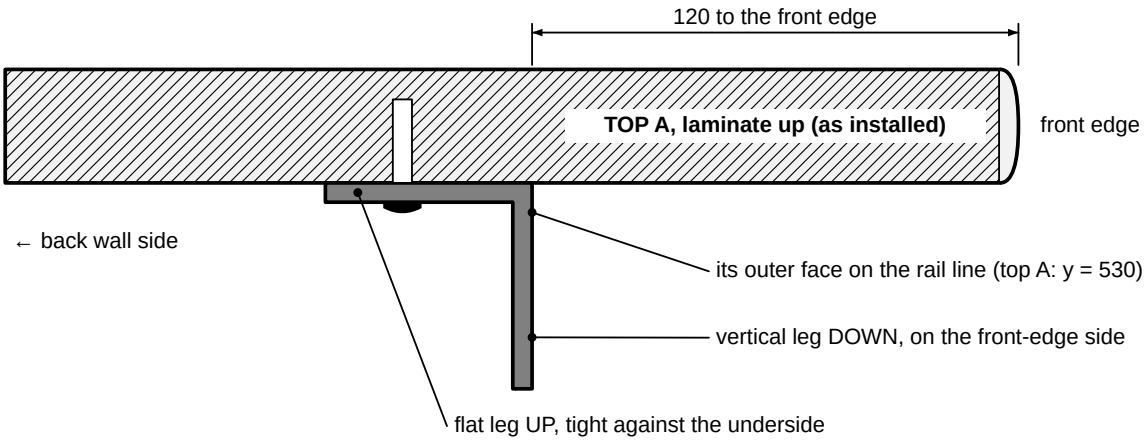
10

The two steel angles arrive identical; either can be rail A. Tape one end of each: **"A · WALL END"** and **"B · CORNER END"**. Every station on a rail is measured from its taped end. Measure both delivered lengths **MEASURE ON HARDWARE** (2130 layout length; the far end may carry the extra).



Bench view, flat leg up. Holes shown for orientation; positions are in chapter E.

Fig. D-9 · The two rails on the bench, flat legs up, labelled ends left.



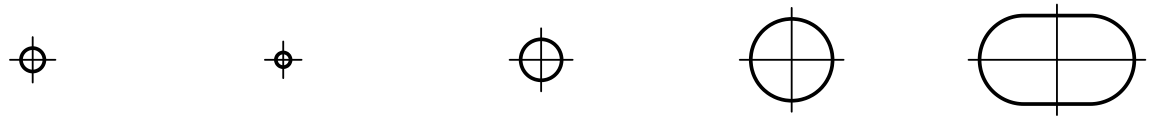
Section through top A and rail A, end-on, about 1:2. Rail B on top B: the same, with its outer face on x = 530.

Fig. D-10 · How a rail sits under its top: **flat leg UP** against the underside, all screws through it; **vertical leg DOWN** on the front-edge side, its outer face on the rail line, 120 from the front edge. Rail A: outer face on y = 530, wall end at x = 10. Rail B: outer face on x = 530, corner end 120 past top B's joint end.

The printed rail end caps are handed: cap A on rail A's far end and rail B's far end, cap B on rail A's wall end; rail B's corner end takes no cap.

Hole types and depths

The three hole types



pilot ⌀3.2 (1/8 in) into the tops: legs, feet, rails, joining plates

pilot ⌀2.0 (5/64 in) into the ALEX tops and the trim edge

clearance ⌀5.56 (7/32 in): rail flat legs, for #10 screws

through ⌀11.1 (7/16 in) rail vertical legs, 6 holes, 3/8 bolts

slot 21 × 12 in each fitting leg, bought, not drilled

Fig. D-11 · Pilot holes go into wood; clearance and through holes go through steel. The fittings come with 21 × 12 slots (not drilled); the slot plug centres the rail's 7/16 in pilot in them.

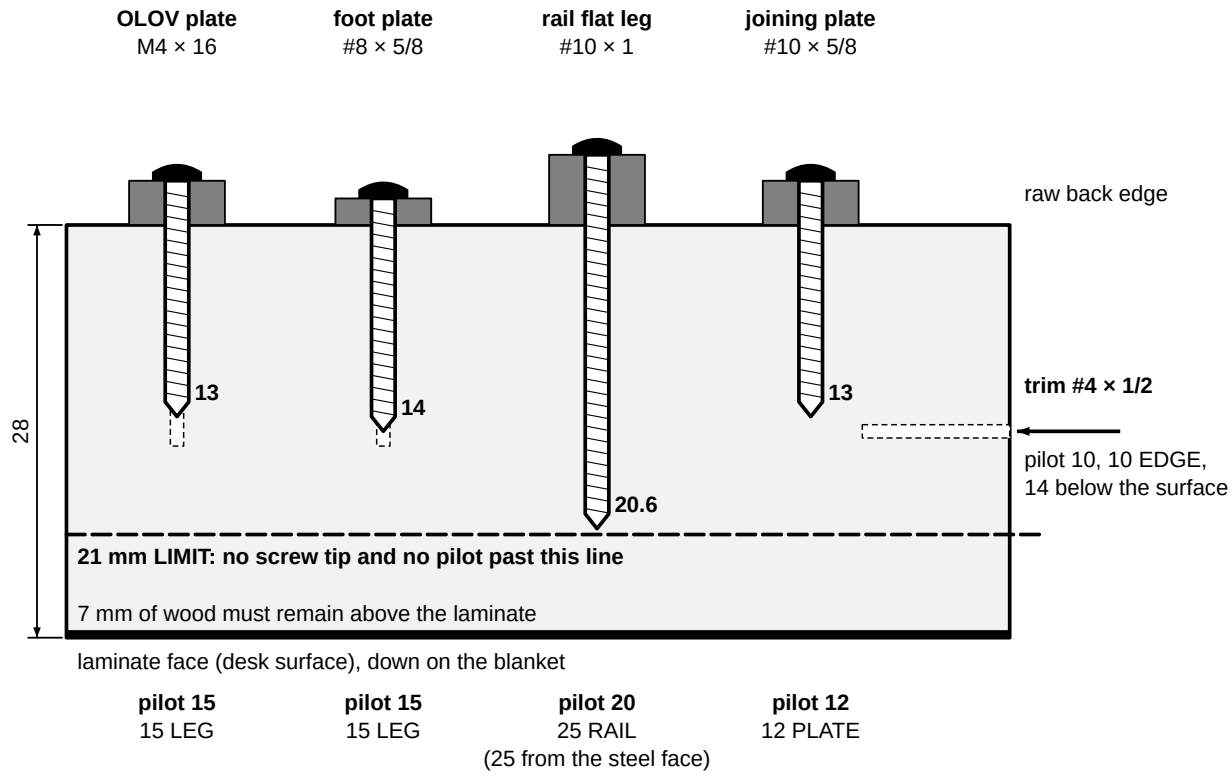
Every fastener, its pilot and its puck

Fastener	Qty	Goes through	Into the wood	Pilot	Puck
H01 · M4 × 16 (with the OLOV)	30	3 mm leg plate	13	1/8 in, 15 deep	15 LEG
H02 · #8 × 5/8 in pan head	32	feet plate (1.5–2 mm)	14	1/8 in, 15 deep	15 LEG
H03 · #10 × 1 in pan head, Robertson	19	4.8 mm angle	20.6	1/8 in, 25 from the flat leg's face = 20 in the wood	25 RAIL
H04 + H05 · #10 × 5/8 in with washer	8	3 mm joining plate	13	1/8 in, 12 deep	12 PLATE
H06 · #6 × 5/8 in flat head	24	2.5 mm ring flange	13 into the ALEX top	5/64 in, 12 deep	12 ALEX
H07 · #4 × 1/2 in flat head (optional trim)	48	trim lip	sideways into the raw back edge, 14 below the surface	5/64 in, 10 deep	10 EDGE
H08 + H09 + H10 · 3/8-16 bolts	6	fittings and rail vertical legs	none	7/16 in, bench-drilled	—

Depths SOURCE-DEFINED (guide.html fastener table, facts.json). The bolt holes and the 7/32 in screw holes in the steel are drilled on the bench (chapter E); everything into wood goes through a jig first and is finished with its puck.

Never through the laminate

> **NEVER THROUGH THE LAMINATE**
> No screw tip and no pilot goes deeper than **21 mm** into a 28 mm top: **28 − 21 = 7 mm** of wood must remain under the laminate face. Set every pilot depth with the drill guide's puck, never by eye. If a top measures 27 mm, shorten every screw into it by 1 mm.



Section through a 28 mm top, face down as you drill it (underside up), 2:1. Grey: the part being screwed on. Dashed: the pilot hole. Bold numbers: how far the screw goes into the wood, mm.

Fig. D-12 · Every fastener into the tops at its depth, 2:1.

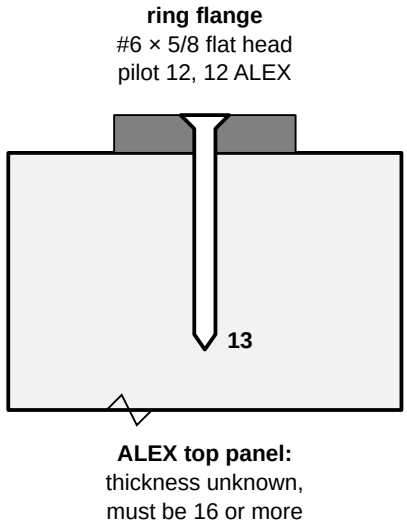


Fig. D-13 · Ring screws go into the ALEX top, not the desk. **MEASURE ON HARDWARE** panel thickness.

Setting a puck

11 Loosen the chuck. Drop the bit tip-first into the SET tower's hole for that bit (45 for 1/8 in, 35 for 5/64 in), rest the chuck nose on the tower, tighten. Re-set every time you change bits.

12 Stand the puck flat on the surface (on the rail's flat leg for 25 RAIL), bit down its bore, drill until the chuck nose lands on the puck. The hole is then the depth written on the puck.

If your chuck nose is domed, drill one test hole in scrap and measure it first.

E · Steel rail fabrication and drilling

All steel work happens on the bench, before either rail goes near a top. Under the desk you only drive screws and bolts; you never drill steel there. Two identical lengths of 2 × 2 × 3/16 in hot-rolled angle arrive; either can be rail A. All dimensions in mm.

MEASURE BEFORE MARKING OR DRILLING MEASURE ON HARDWARE
1 · Tape each rail from end to end and write the length on the tape label. Design length **2130**; bought as 84 in = **2133.6**. If a rail is under 2130, stop and check before drilling.
2 · Calipers on each angle: legs 50.8, thickness 4.76 (the docs expect about 4.8).
3 · Every station is measured from the **labelled end**: rail A's **wall end**, rail B's **corner end**. The **far end carries any extra** length. Never centre the hole pattern on the bar and never shift a hole to share out the extra.

1

Label the datum ends. Wrap tape round one end of each rail and write on it: **RAIL A · WALL END** on one, **RAIL B · CORNER END** on the other. Measure both rails from those ends (Fig. E-1). File both cut ends of each rail smooth.

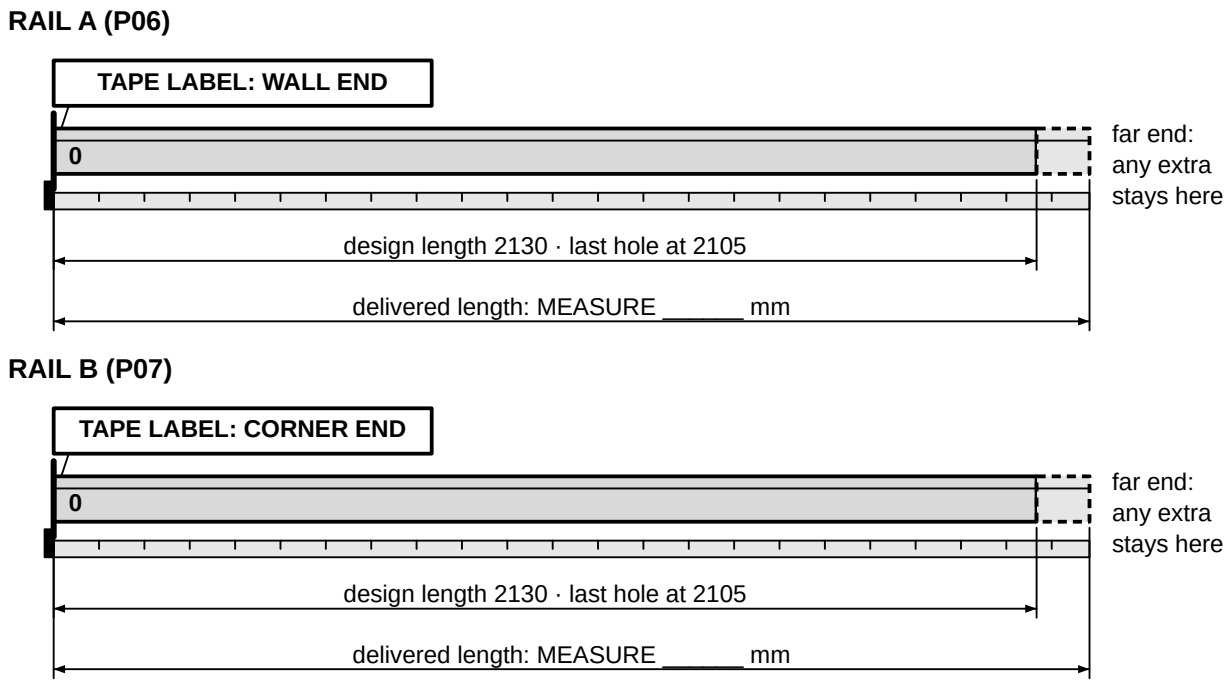


Fig. E-1 · The labelled end is 0 for every station. The last hole is at 2105 on both rails, so a longer rail only lengthens the far end beyond it. SOURCE-DEFINED design length; MEASURE ON HARDWARE delivered length.

You need	For
P06 rail A, P07 rail B; one P08 fitting as the template	the work pieces
P18 printed tools: edge T-square, rail marking gauge, rail drill jig (PETG), 2 fitting slot plugs	stations, the 19 and 29 lines, pilots, bolt-hole transfer
Bits 1/8, 7/32 and 7/16 in (the 7/16 cobalt), cutting oil, flat file, 2 clamps, tape, combination square, pencil	drilling and deburring; a drill press if you have one
Acetone or mineral spirits, 120 grit, flat black enamel	finish (step 9)

Which way up: the section and the two lines

FLAT LEG
lies against the top's underside

VERTICAL LEG
hangs down, toward you; outer face at
y = 530 (rail A)
x = 530 (rail B)

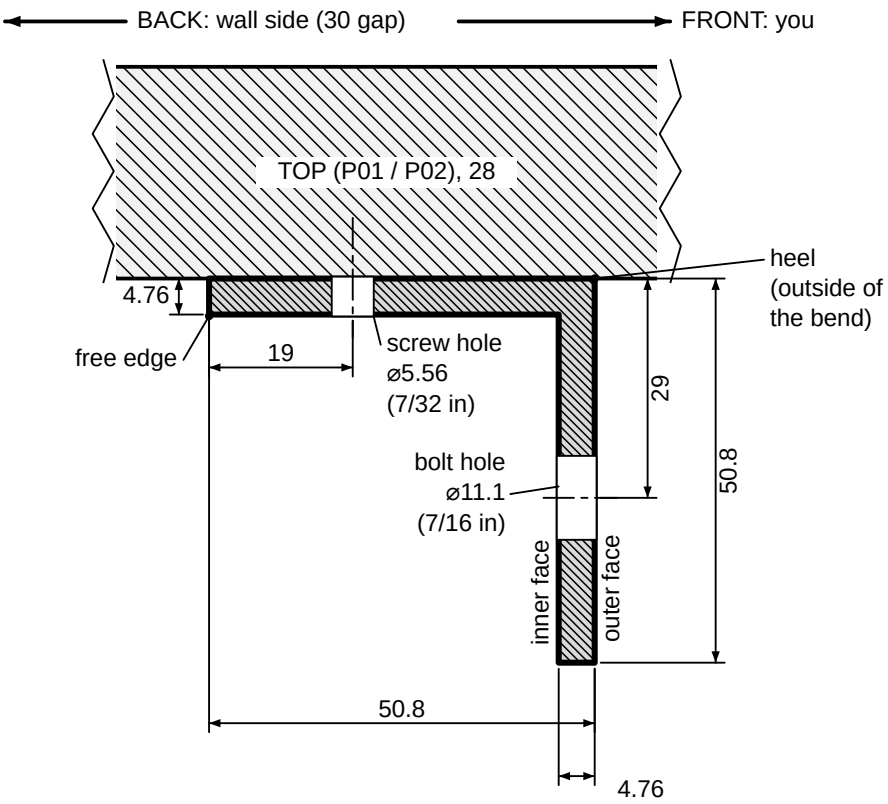
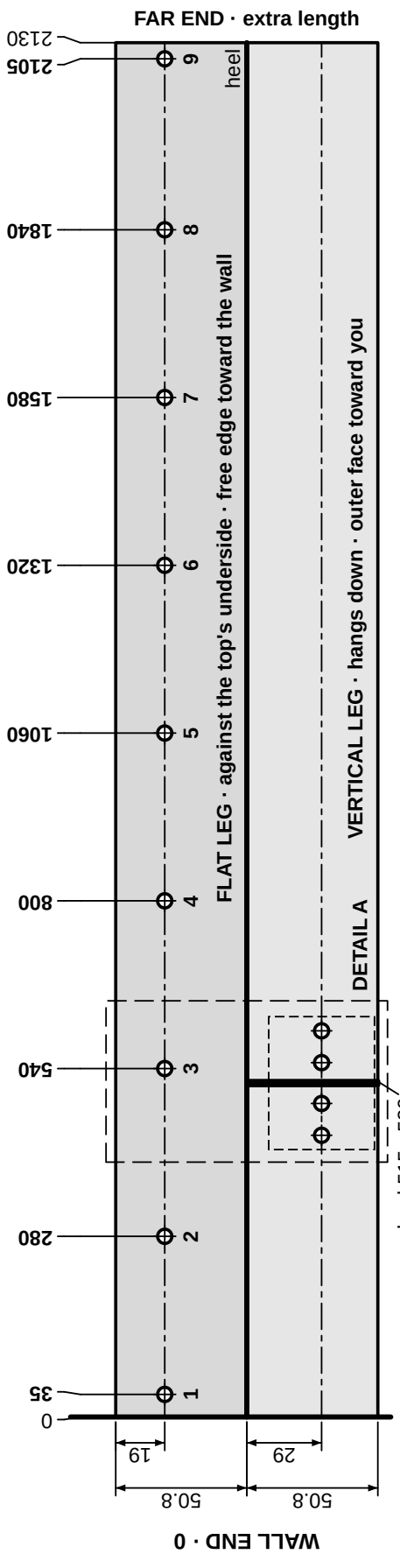


Fig. E-2 · Rail A seen end-on from its far end, as it hangs under top A (rail B is the same section turned 90°). Datum: top A's back-left corner, x along the back wall, y along the side wall. The vertical leg's outer face is at y = 530 on rail A and x = 530 on rail B, 120 back from the front edge (650).

Item	mm	What it means	Confidence
Section	50.8 × 50.8 × 4.76	2 × 2 × 3/16 in hot-rolled angle	SOURCE-DEFINED MEASURE ON HARDWARE thickness
Flat leg		lies against the top's underside; its free edge points to the wall	SOURCE-DEFINED
Vertical leg		hangs down on the side nearer you; outer face y = 530 (A), x = 530 (B)	SOURCE-DEFINED
Screw line	19	from the flat leg's free edge; centres of the ø5.56 (7/32 in) screw holes	SOURCE-DEFINED
Bolt line	29	below the heel (the top's underside); centres of the ø11.1 (7/16 in) bolt holes	SOURCE-DEFINED
Fitting slots	21 × 12	two per P08 leg, centres 29.15 and 78.45 from the heel; legs 100.5 × 41, 5 thick	VERIFIED 23 Oct 2026

Bench set-up. Clamp the rail with the flat leg's face up and the vertical leg hanging down over the bench edge. That is how it is marked and how the flat-leg holes are drilled. For the vertical leg (steps 6 and 7) turn it so the vertical leg's outer face is up and the flat leg hangs over the edge.

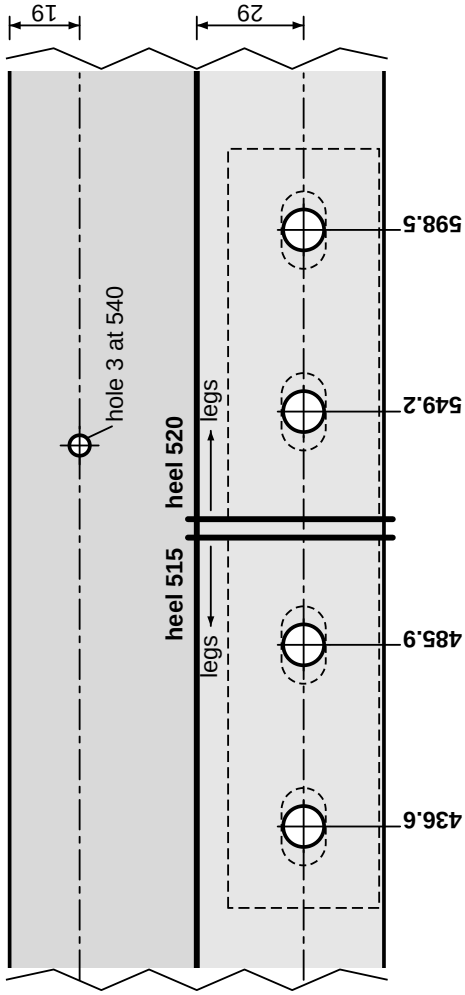
The two fabrication drawings that follow are full pages, turned sideways: turn the booklet a quarter turn clockwise to read them. Every hole is given as an ordinate from the labelled end, never from the hole before.



9× ø5.56 (7/32 in) THRU in the flat leg, centres on the screw line 19 from the free edge, at the ordinates above.
4× ø11.1 (7/16 in) THRU in the vertical leg on the bolt line 29 below the heel, at 436.6, 485.9, 549.2, 598.5 (Detail A).
Positions VERIFIED from the measured fitting; drill them THROUGH the clamped fitting with the slot plug (step 6).

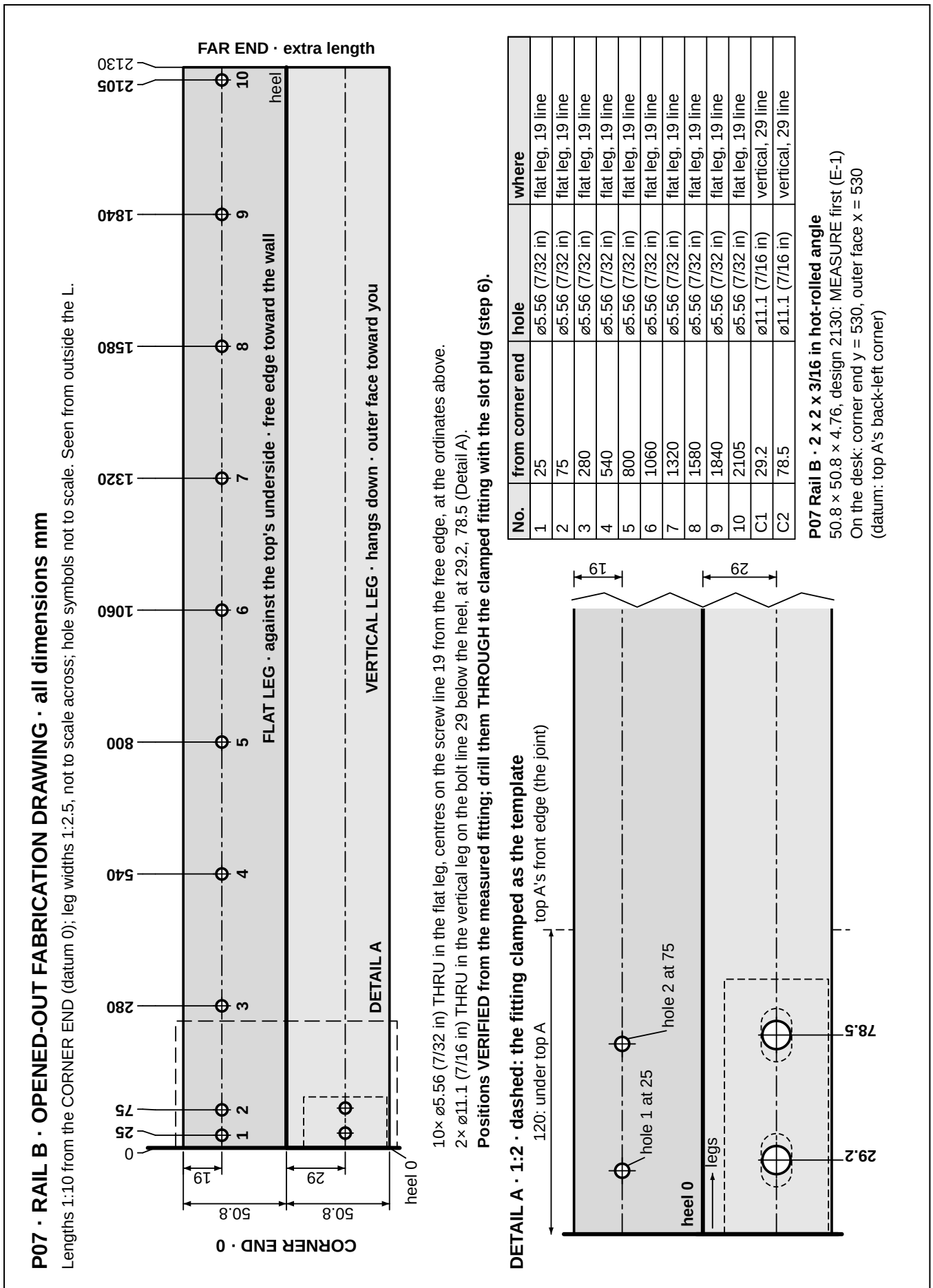
No.	from wall end	hole	where
1	35	ø5.56 (7/32 in)	flat leg, 19 line
2	280	ø5.56 (7/32 in)	flat leg, 19 line
3	540	ø5.56 (7/32 in)	flat leg, 19 line
4	800	ø5.56 (7/32 in)	flat leg, 19 line
5	1060	ø5.56 (7/32 in)	flat leg, 19 line
6	1320	ø5.56 (7/32 in)	flat leg, 19 line
7	1580	ø5.56 (7/32 in)	flat leg, 19 line
8	1840	ø5.56 (7/32 in)	flat leg, 19 line
9	2105	ø5.56 (7/32 in)	flat leg, 19 line
I2	436.6	ø11.1 (7/16 in)	vertical, 29 line
I1	485.9	ø11.1 (7/16 in)	vertical, 29 line
O1	549.2	ø11.1 (7/16 in)	vertical, 29 line
O2	598.5	ø11.1 (7/16 in)	vertical, 29 line

DETAIL A · 1:2 · dashed: the fitting clamped as the template



P06 Rail A · 2 x 2 x 3/16 in hot-rolled angle
50.8 × 50.8 × 4.76, design 2130: MEASURE first (E-1)
On the desk: wall end x = 10, outer face y = 530
(datum: top A's back-left corner)

Fig. E-3 · P06 rail A, opened out: the flat leg (upper strip) and the vertical leg (lower strip) unfolded about the heel. Nine screw holes from the wall end; four bolt holes at 436.6, 485.9, 549.2 and 598.5, drilled through the fitting (heels at 515 and 520). SOURCE-DEFINED stations; VERIFIED bolt holes.



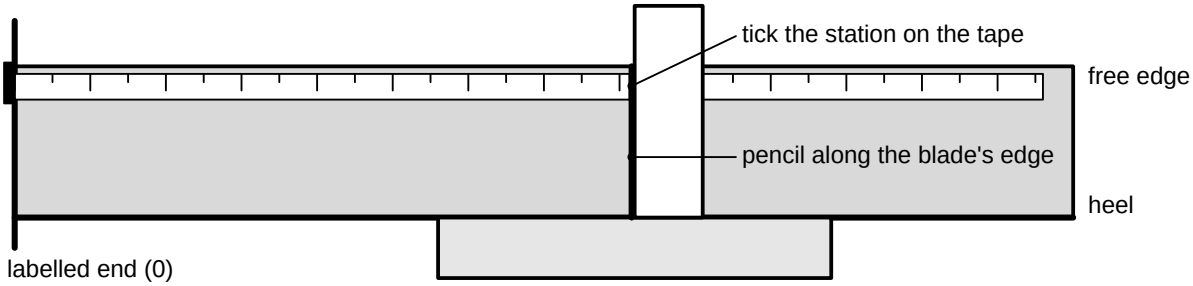
No.	from corner end	hole	where
1	25	ø5.56 (7/32 in)	flat leg, 19 line
2	75	ø5.56 (7/32 in)	flat leg, 19 line
3	280	ø5.56 (7/32 in)	flat leg, 19 line
4	540	ø5.56 (7/32 in)	flat leg, 19 line
5	800	ø5.56 (7/32 in)	flat leg, 19 line
6	1060	ø5.56 (7/32 in)	flat leg, 19 line
7	1320	ø5.56 (7/32 in)	flat leg, 19 line
8	1580	ø5.56 (7/32 in)	flat leg, 19 line
9	1840	ø5.56 (7/32 in)	flat leg, 19 line
10	2105	ø5.56 (7/32 in)	flat leg, 19 line
C1	29.2	ø11.1 (7/16 in)	vertical, 29 line
C2	78.5	ø11.1 (7/16 in)	vertical, 29 line

P07 Rail B · 2 x 2 x 3/16 in hot-rolled angle
50.8 × 50.8 × 4.76, design 2130: MEASURE first (E-1)
On the desk: corner end y = 530, outer face x = 530
(datum: top A's back-left corner)

Fig. E-4 · P07 rail B, opened out. Ten screw holes from the corner end: holes 1 and 2 go into top A (the first 120 sits under top A), holes 3 to 10 into top B. Two bolt holes at 29.2 and 78.5, shared by both fittings, drilled through the fitting with its heel flush with the corner end. SOURCE-DEFINED stations; VERIFIED bolt holes.

2 **Mark the stations.** Hook the tape over the labelled end and tick every station on the flat leg from that one hook: rail A 35, 280, 540, 800, 1060, 1320, 1580, 1840, 2105; rail B 25, 75, 280, 540, 800, 1060, 1320, 1580, 1840, 2105. Square each tick across the leg along the T-square's blade, its lip hooked over the heel edge.

Hook the tape over the labelled end once and tick every station from that one hook;
never measure from one hole to the next.



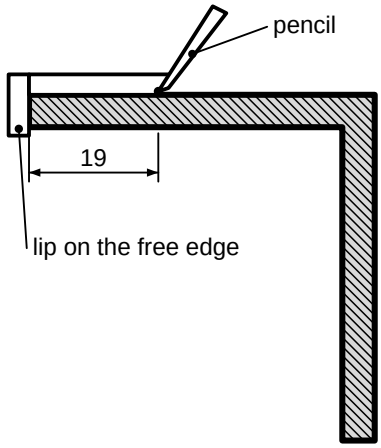
T-square (P18): lip over the heel edge, blade across the leg

Plan: flat leg face up, vertical leg over the bench edge, labelled end at the left.

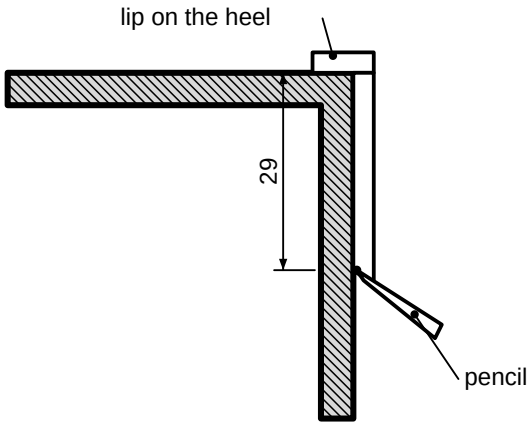
Fig. E-5 · One hook, every station. The station line only has to cross the leg; the jig finds the 19.

3 **Pencil the two lines with the rail marking gauge, full length.** Screw line: lip over the flat leg's free edge, pencil in the **19** notch, slide. Bolt line: gauge on the vertical leg's outer face, lip over the heel, pencil in the **29** notch, slide. The bolt line is only a check: the fitting's holes must sit on it in step 6.

a SCREW LINE on the flat leg
lip over the free edge,
pencil in the 19 notch, slide



b BOLT LINE on the vertical leg
gauge on the outer face, lip over the heel,
pencil in the 29 notch, slide



Sections, marking gauge (P18) in use. Run each line the full length of both rails.

Fig. E-6 · The gauge's lip is the zero; the V notches put the pencil point on 19.0 and 29.0. SOURCE-DEFINED

4 **Pilot the screw holes through the drill jig.** Jig lip down over the flat leg's free edge, sight grooves on the station line, hold it down and drill 1/8 in through its bore: square, on the 19 line, no punch needed. All 19 stations (9 on rail A, 10 on rail B). Use a drill press if you have one.

5 **Open and deburr.** Jig off. Open every pilot to **⌀5.56 (7/32 in)** at low speed with a drop of oil on each; the bit follows the pilot. Deburr both faces with the file or a larger bit turned by hand.

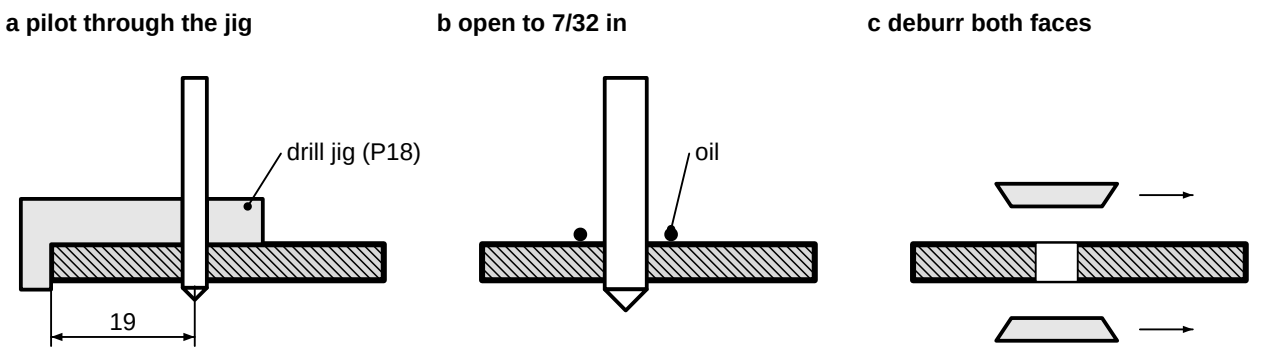


Fig. E-7 · Flat leg in section. The jig's bore is 1/8 in plus a running clearance; it keeps the bit square and on the 19 line.

THE FITTING IS THE TEMPLATE VERIFIED
The fittings were measured on 23 Oct 2026: legs 100.5 × 41, two 21 × 12 slots per leg with centres 29.15 and 78.45 from the heel. That puts the six bolt holes at the positions below. Even so, do not lay them out with the tape: clamp the fitting on its heel mark and drill through it with the printed **slot plug** (steps 6 and 7), so each hole lands on its slot's centre. Check first that a plug drops into each slot without rattle. The heel of a fitting is the **outside of its bend**.

Clamp the fitting	Heel on	Legs toward	Holes made (mm from the labelled end)
Rail A, outside fitting	520 from the wall end	the far end	O1 549.2, O2 598.5
Rail A, inside fitting	515 from the wall end	the wall end	I1 485.9, I2 436.6
Rail B, one fitting (both share the holes)	the corner end itself (0)	the far end	C1 29.2, C2 78.5

Heel stations SOURCE-DEFINED; hole positions from the measured slot centres VERIFIED. Both fittings are identical, so holes copied from one fit the other.

6 **Clamp the fitting and pilot through the plugs.** Rail with the vertical leg's outer face up. Clamp one fitting to the vertical leg with its heel exactly on the mark and its holes centred on the 29 line, legs the way Fig. E-8 shows. Drop a slot plug into each slot and drill the 1/8 in pilot through each plug's bore into the rail. Repeat for each of the three placements.

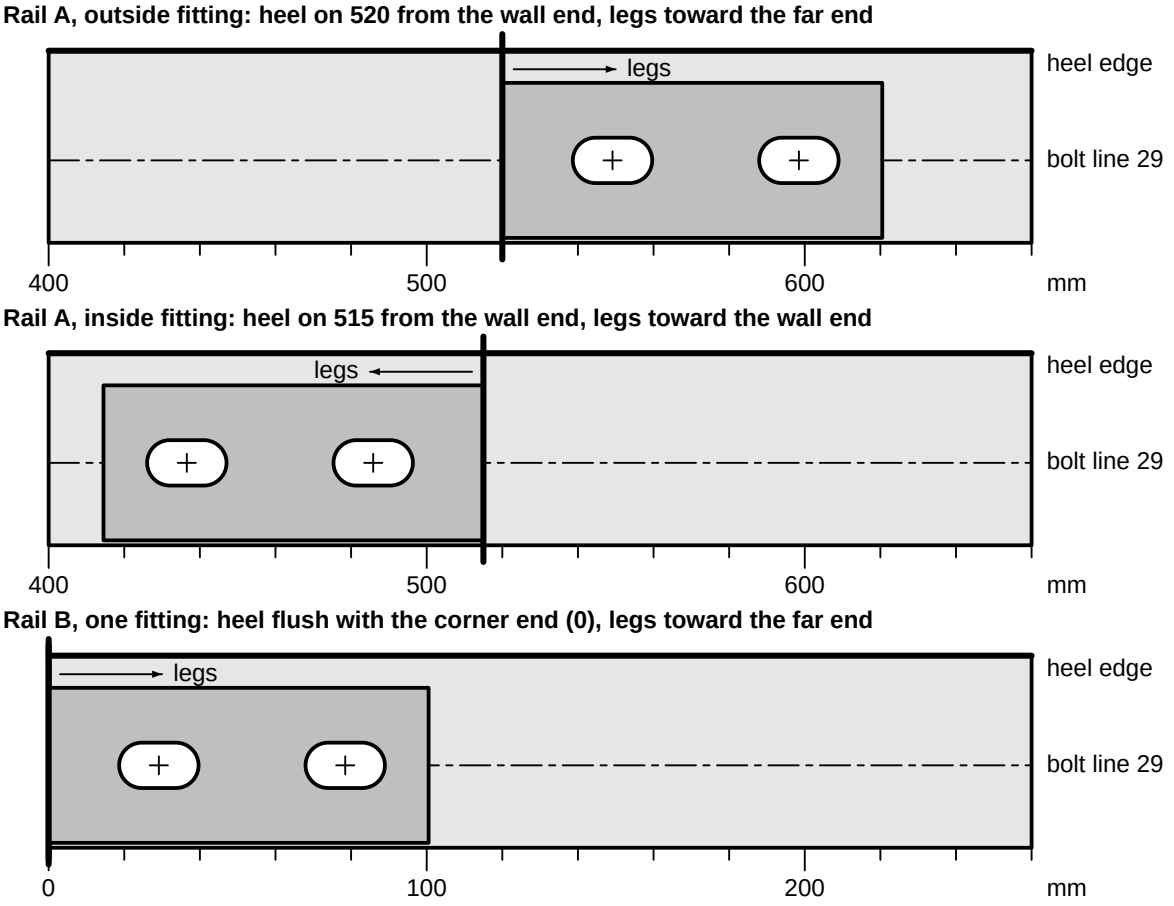


Fig. E-8 · The three clamp positions, vertical leg seen face on (1:2). Heavy line: the heel mark. Slots and their centres as measured. **VERIFIED**

7 **Fitting off, open to ⌀11.1 (7/16 in).** Cobalt bit, slow, oil on each. The bit follows the pilot, so the hole lands on the slot's centre, not anywhere along its 21 mm. Deburr. Six holes: four in rail A, two in rail B.

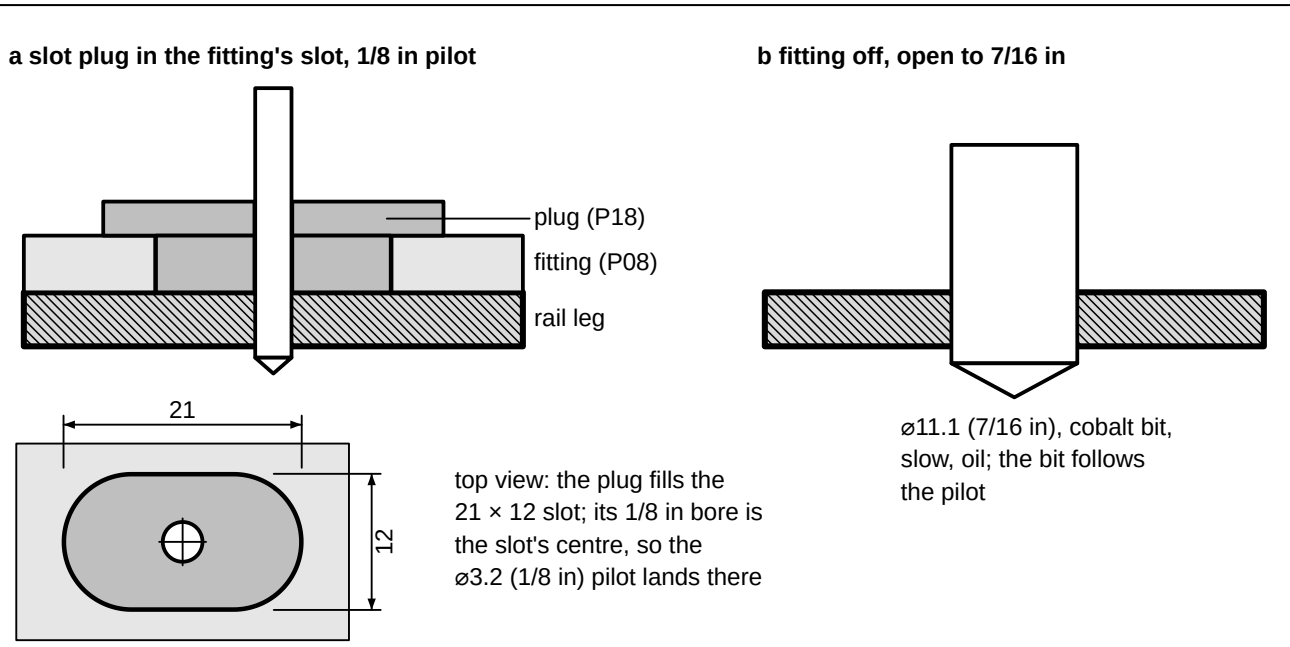
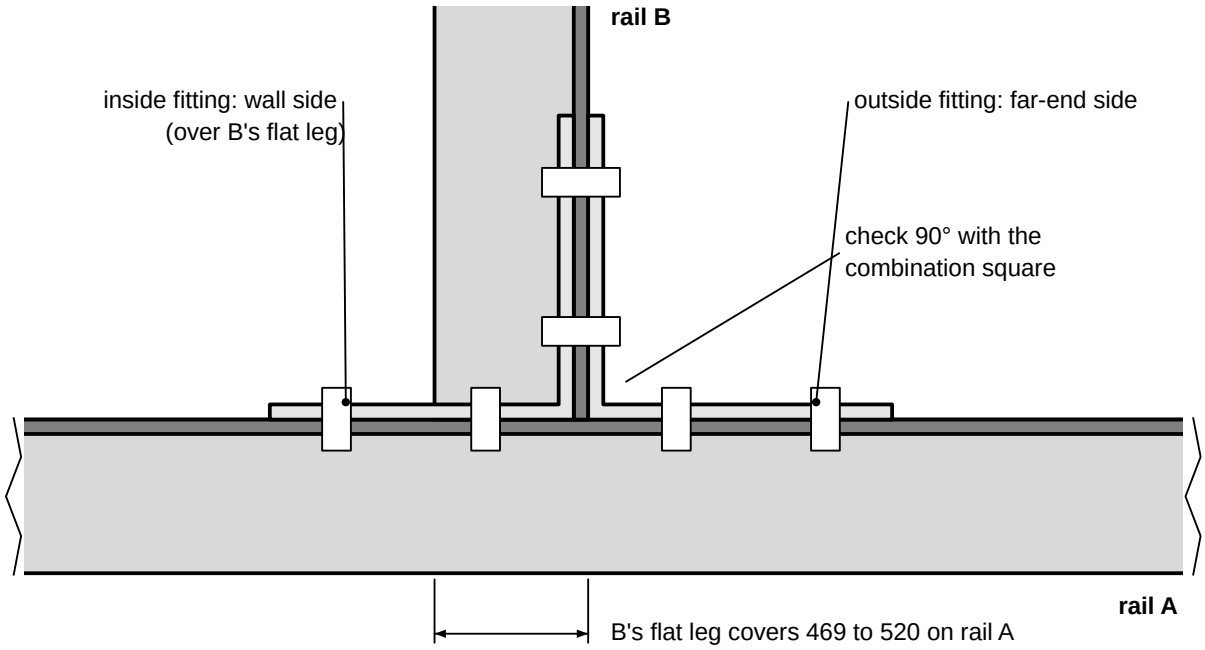


Fig. E-9 · The slot plug: a stadium that fills the 21 × 12 slot with a running clearance, 1/8 in bore at its centre. Without it a punch could land anywhere along the slot.

8 **Floor test.** Lay both rails upside down on the floor, flat legs down, vertical legs up. Butt rail B's corner end against the outer face of rail A's vertical leg so that rail B's outer face is on rail A's **520** mark; rail B's flat leg then covers 469 to 520 on rail A's scale. Bolt both fittings on loosely: the outside one on the far-end side of rail B, the inside one on the wall side. Every bolt must slide in by hand. Check the corner with the combination square. If a bolt will not go in, open that hole with a round file now, not under the desk.

Plan from above. Both rails upside down on the floor: flat legs down, vertical legs up.
Bolts in loosely; every one must slide in by hand.



B's outer face on rail A's 520 mark; B's end touching A: no gap

Fig. E-10 · Floor test, plan from above. Bolts drawn as boxes. 469 = rail B's flat-leg edge (x = 479) less rail A's wall-end offset (x = 10). SOURCE-DEFINED

9 **Finish.** Take the fittings off. Degrease both rails, scuff with 120 grit, two coats of flat black enamel, 24 h to cure. The holes need no masking.

The rails are now ready for the tops: rail A goes on top A with its wall end at x = 10 (1 cm in from the taped end), rail B on top B with its corner end projecting 120 past the joint end. Chapter F bolts the corner once both tops stand.

F · Steel corner assembly

The two rails are joined at the inside corner by two gusseted strut fittings, one each side of rail B's end, bolted to the vertical legs. This bolted joint, not the joining plates, carries the inside corner of top B. Both tops are standing (build manual step 7); you work from below.

> **SEQUENCE: THE FITTINGS GO ON LAST** SOURCE-DEFINED
> Before the first fitting goes on, all of these must be done:
> 1 · The two surfaces are flush at the joint, leveled with the feet (never packed).
> 2 · Rail B's end touches the outer face of rail A's vertical leg with no gap, and rail B's outer face is on the 520 mark on rail A.
> 3 · Joining plates P1 and P2 are screwed across the joint. They align the tops; they do not carry the corner.
> 4 · Rail B's screws 1 and 2 are driven into top A. The inside fitting sits right under them and would block the driver.

	Part	Qty		Hardware	Qty
P06	Rail A, angle 2130, drilled (E)	1	**H08**	3/8-16 × 1½ in hex bolt, grade 5	6
P07	Rail B, angle 2130, drilled (E)	1	**H09**	3/8 flat washer, under each head and nut	12
P08	Gusseted 90° strut fitting, 5 mm, legs 100.5 × 41, two 21 × 12 slots per leg VERIFIED	2	**H10**	3/8-16 nylon-insert lock nut	6

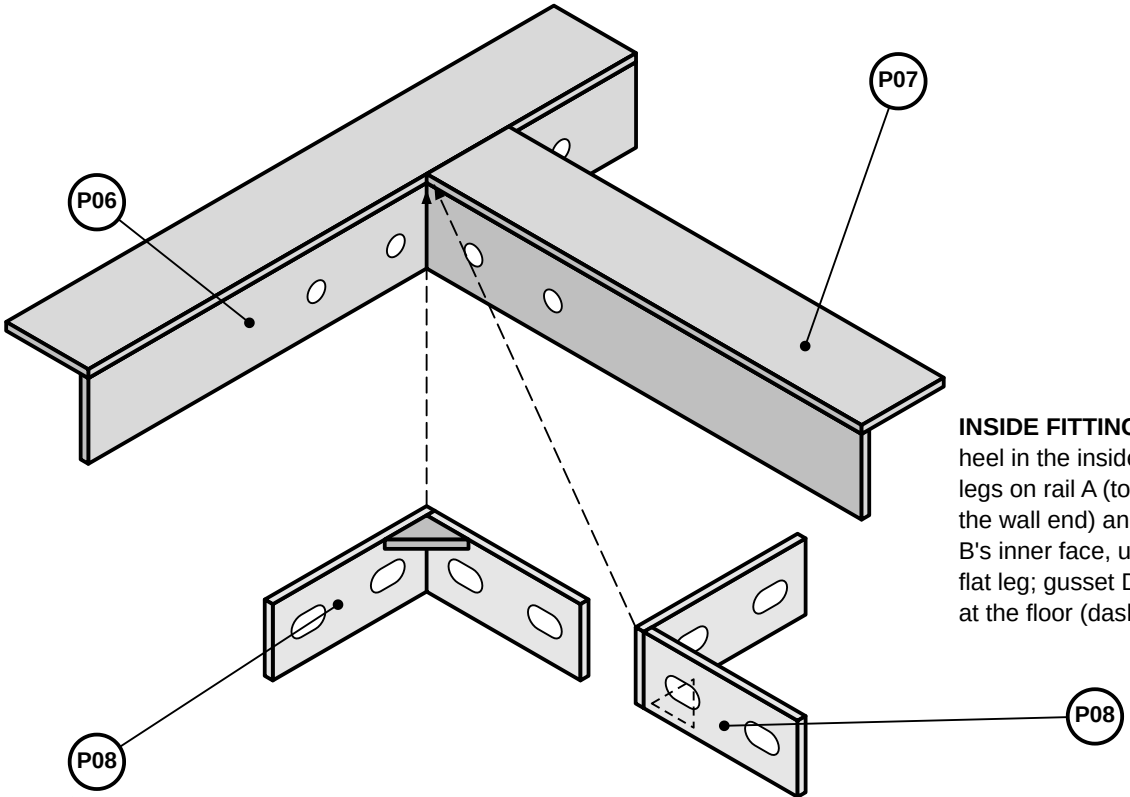


Fig. F-1 · The corner seen from the knee space with the tops lifted away. Both fittings are drawn pulled out of place; the dashed lines run from each heel to where it goes. The outside fitting fills the outside corner, the inside one the pocket under rail B's flat leg, gusset down. Neither is a mirror image: they are the same part, the inside one turned gusset-down. SOURCE-DEFINED placement.

Which bolt goes where

Bolts	Through	Head and washer	Washer and nut
O1, O2	outside fitting + rail A's vertical leg	fitting side, in the knee space	rail A's inner face, under its flat leg
I1, I2	inside fitting + rail A's vertical leg	fitting side, in the pocket under rail B	rail A's inner face, under its flat leg
C1, C2	outside fitting + rail B's vertical leg + inside fitting	outside fitting, knee side	inside fitting, in the pocket

Every bolt goes in **from the fitting side**: four through rail A's leg with one fitting each, two through rail B's leg with a fitting on each side.

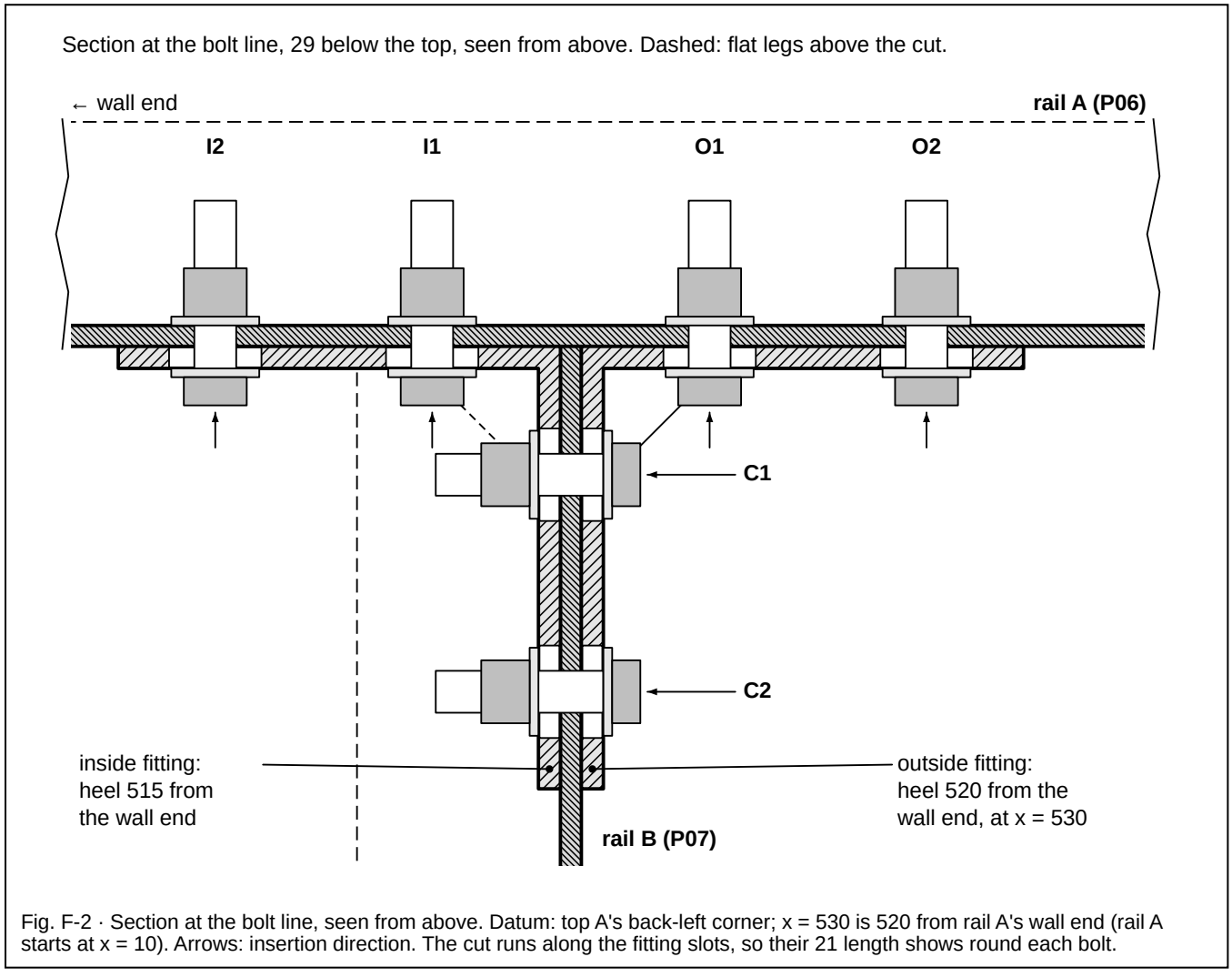


Fig. F-2 · Section at the bolt line, seen from above. Datum: top A's back-left corner; x = 530 is 520 from rail A's wall end (rail A starts at x = 10). Arrows: insertion direction. The cut runs along the fitting slots, so their 21 length shows round each bolt.

1

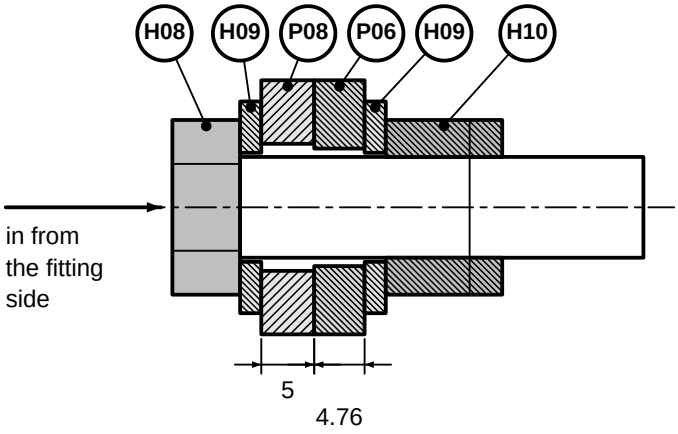
Outside fitting. Heel into the outside corner, one leg flat on rail A's vertical leg toward the far end, the other on rail B's outer face, gusset up, facing the knee space. Push O1 and O2 through from the fitting side with a washer under each head; washer and nut on rail A's inner face. Finger tight.

2

Inside fitting. Into the pocket: heel in the inside corner, one leg on rail A's vertical leg toward the wall end, the other on the inner face of rail B's vertical leg, **gusset pointing at the floor**. Gusset-up it would sit 1 mm under the head of rail B's screw 1. I1 and I2 from the fitting side, washers, nuts on rail A's inner face. Finger tight.

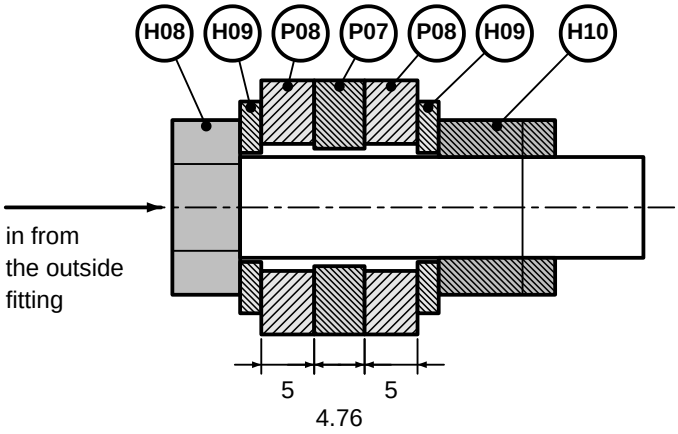
3 **Through bolts.** C1 and C2 from the outside fitting: through the outside fitting, rail B's vertical leg and the inside fitting, washer under the head, washer and nut in the pocket. Finger tight. All six bolts are now in; none is tight.

Rail A: 4 bolts, one fitting each (O1, O2, I1, I2)



fitting 21 × 12 slot (cut across): 2.5 mm play across, 11.5 mm along rail hole ⌀11.1 (7/16 in): 1.6 mm of play
so the joint holds by friction: tighten fully, 25 N·m

Rail B: 2 bolts through outside fitting, rail B and inside fitting (C1, C2)



head and washer on the outside fitting (knee side); washer and nut on the inside fitting, in the pocket under rail B's flat leg

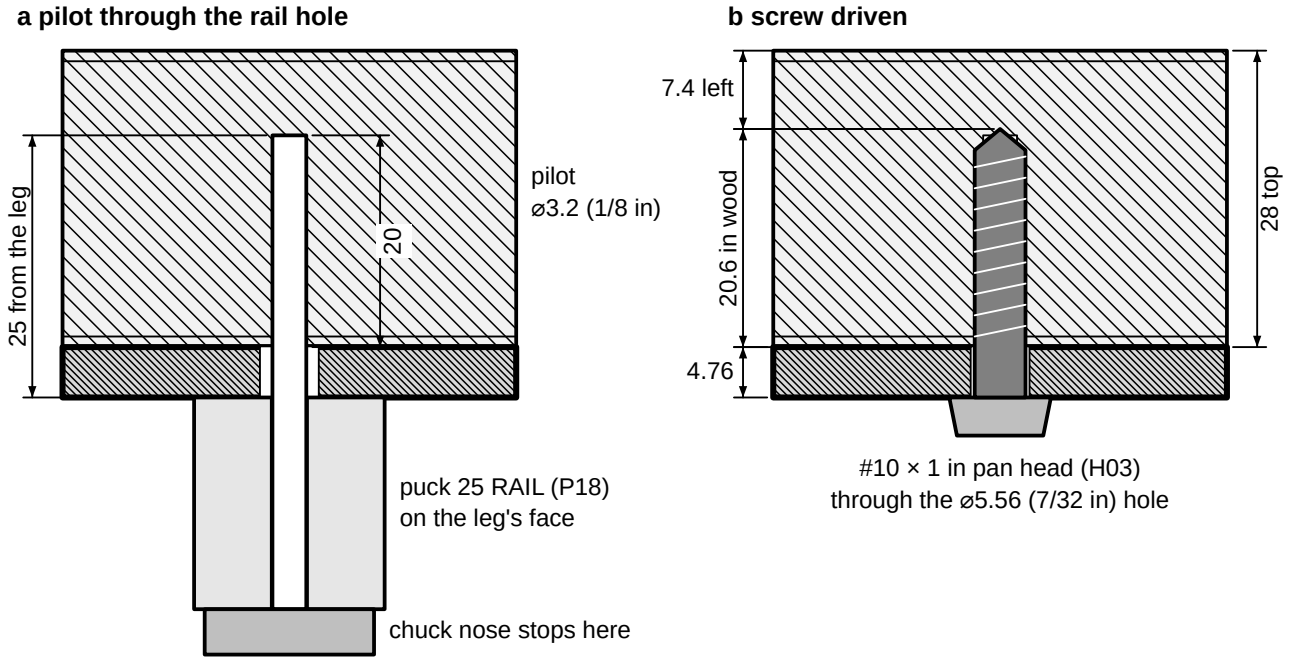
Fig. F-3 · Bolt stack-ups, cut along the bolt. Plates drawn cut off. The fitting's slots are 21 × 12; the cut shows their 12 width.
VERIFIED fitting; SOURCE-DEFINED bolts.

4 **Check again, then tighten all six.** Fingertip across the joint: no step. Rail B's end still touching rail A. Then tighten every bolt fully:

! **TIGHTENING: 25 N·m (18 ft·lb)** SOURCE-DEFINED
With a torque wrench, **25 N·m (18 ft·lb)**, under the standard dry torque even for a grade 2 bolt. Without one, pull as hard as one hand can on a 15 cm wrench, then stop.
Why tight, not snug: this is a friction joint. The fittings' 21 × 12 slots leave 2.5 mm of play across and 11.5 mm along each bolt, the 7/16 in rail holes 1.6 mm, so only the clamping force keeps rail B's corner from settling.
When: only after the two surfaces are flush and rail B's end touches rail A.
Later: if the corner ever drops a millimetre, retighten these six bolts.

How the rails hold the tops

Each rail is screwed to its top through the flat leg: nine #10 × 1 in pan-head Robertson screws (H03) in rail A, ten in rail B, 19 in all. Rail B's holes 1 and 2 land in top A (its first 120 lies under top A) and are driven in build-manual step 7.4, before the fittings; the rest go in with the tops face down.



Never more than 21 into the 28 top. Drawn upright, as for rail B's holes 1 and 2 (overhead).

Fig. F-4 · One rail screw in section. The pilot is 1/8 in, 25 from the flat leg's face (the face the puck stands on), which is 20 into the wood; the screw passes the 4.76 steel and goes 20.6 into the 28 top, leaving 7.4 under the laminate. SOURCE-DEFINED

Item	mm	Source
Top thickness	28	SOURCE-DEFINED MEASURE ON HARDWARE calipers (if 27, shorten every screw into the top by 1)
Flat leg	4.76	SOURCE-DEFINED
Screw into the top	20.6	SOURCE-DEFINED; limit 21 into any top
Pilot depth from the leg's face	25	printed drill guide, puck 25 RAIL SOURCE-DEFINED

Settling check. Lay a straightedge across the joint at the inside corner and write down any step. Check again after two weeks and after the first month; if the corner has settled, retighten the six fitting bolts.

G · Legs and cabinet support

Six OLOV legs and eight feet plates go on the tops while they lie face down on a blanket; a locating ring goes on each corner of the two ALEX tops; then the cabinets stand by the walls.

No.	Part	No.	Hardware	Qty
P01 **P02**	Top A and top B, EKBACKEN 2490 x 650 x 28	**H01**	M4 x 16 mm machine screw, IKEA 108443 (with the OLOV legs)	30
P03 **P04**	ALEX 1, drawer unit; ALEX 2, storage unit with door (360 x 580 x 700)	**H02**	#8 x 5/8 in pan head screw (feet plates, 4 each)	32
P05	OLOV leg x 6, 105 mm plate	**H06**	#6 x 5/8 in flat head (rings, 3 each)	24
P09 **P10**	Round adjustable foot 4 in x 8; printed locating ring x 8	**P18** printed tools: leg jig, feet corner jig, edge T-square, ALEX corner template, drill guide (tower + pucks), foot height gauge, wall gauge		

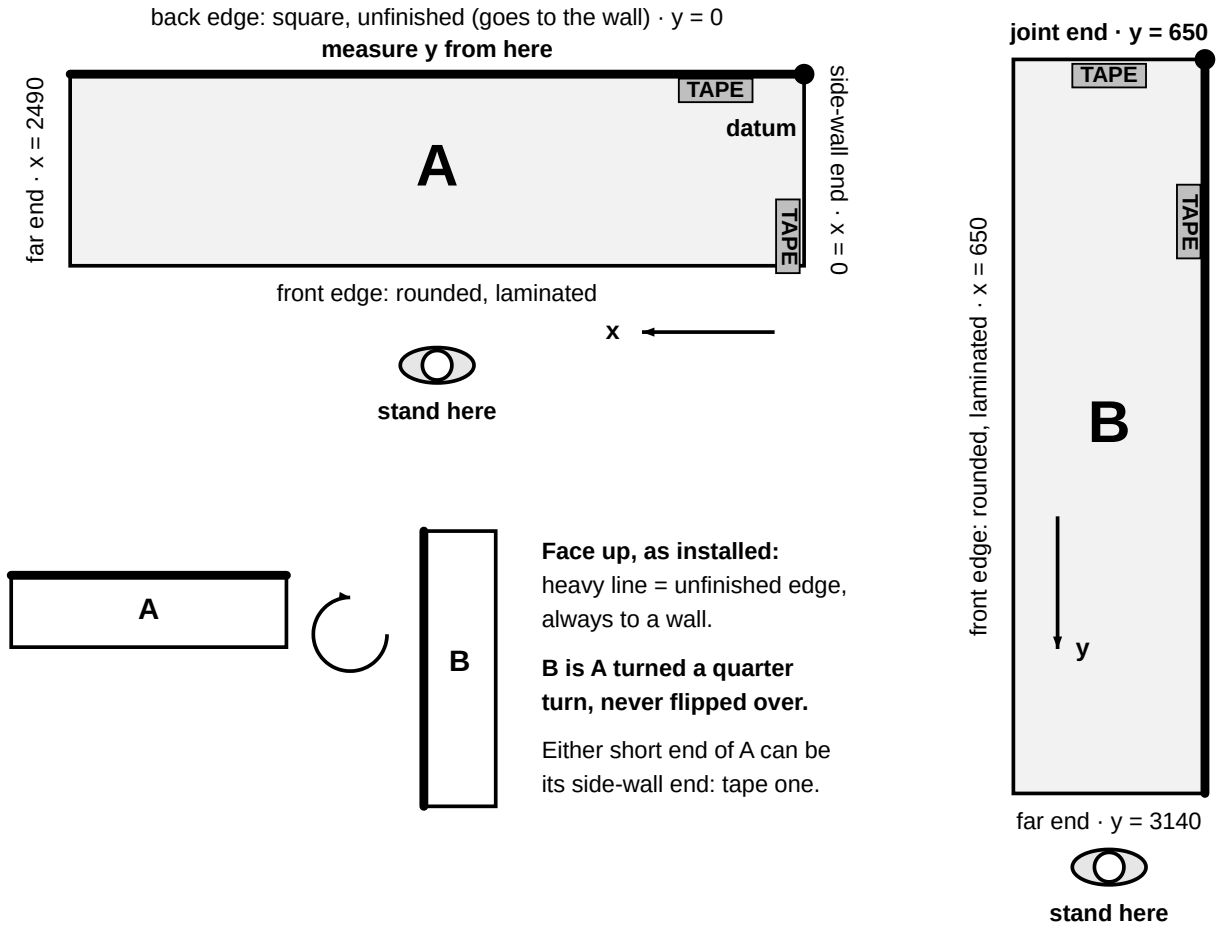


Fig. G-1 How each top lies while you mark it: face down, laminate on the blanket. Heavy line: the square, unfinished long edge, which always goes to a wall (y = 0 on A, x = 0 on B). Tape the unfinished edge and the end you measure from, and stand where the person symbol is; the drawings in this chapter then match what you see, with x growing to the left. Datum: top A's back-left corner (0, 0) as installed.

Decide the desk height first

The legs are pre-set in step 1, so choose now. The desk is built at 830 as planned. Optional printed ALEX feet (2, 4 or 6 cm, `parts/accessories/alex-foot.scad`, ⌀56) raise everything by their height; nothing on the desk moves.

SOURCE-DEFINED

ALEX feet	Underside of the tops	Top surface	Every OLOV set to	Rail lower edge
none (as planned)	802	830	802	751
20 (2 cm)	822	850	822	771
40 (4 cm)	842	870	842	791
60 (6 cm)	862	890	862 (OLOV max 900)	811

G · Underside support layout

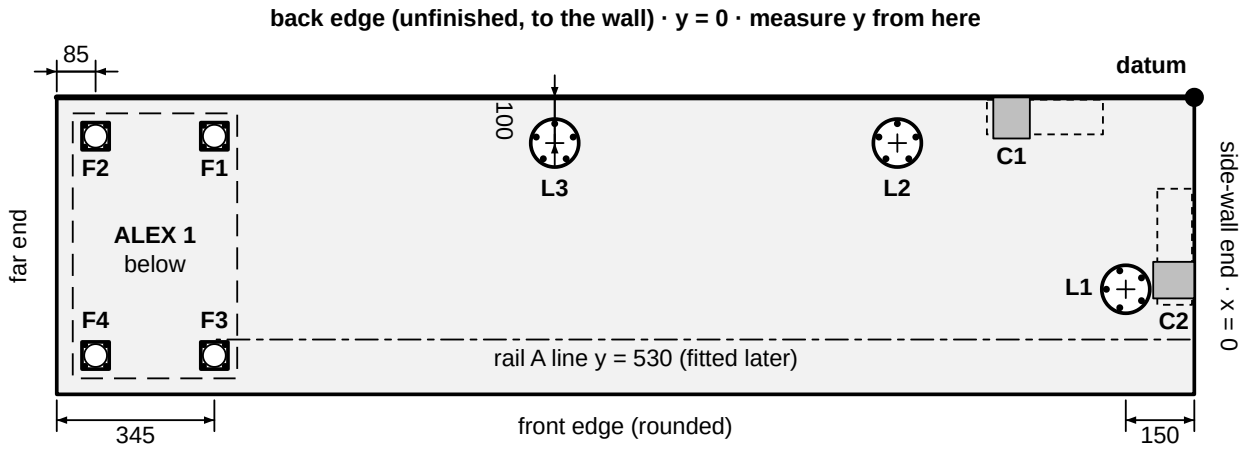


Fig. G-2 Top A face down, back edge away from you: x grows to the left from the side-wall end. Circles: OLOV plates with their five holes, one pointing at the wall (L1's pattern is turned). Squares: feet plates. Grey: clamp zones C1, C2, keep clear; thin dashed: the clamp pad outlines; dash-dot: rail A's line. Datum (0, 0) at the top-right corner of this view.

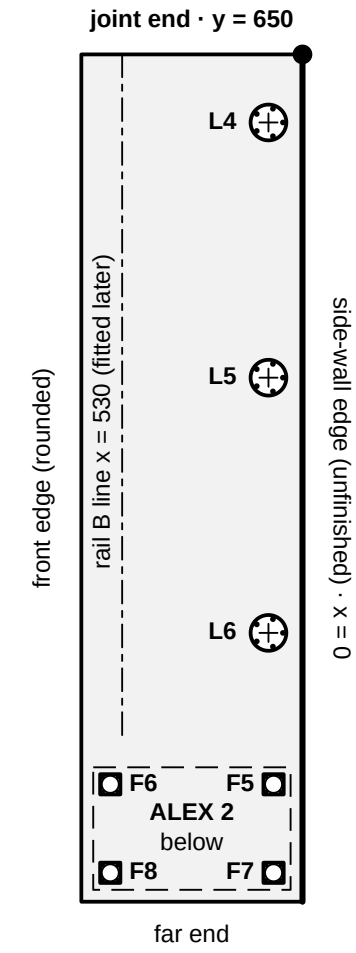


Fig. G-3 Top B face down, joint end away from you. Coordinates are y from top A's back edge: the tape from B's joint end reads y − 650.

Mark	x	y	Note
L1	150	420	A, under VIVO clamp 2
L2	650	100	A, nearest clamp 1
L3	1400	100	A
L4	100	850	B, 200 past the joint
L5	100	1600	B
L6	100	2350	B
F1	2145	85	A, ALEX 1
F2	2405	85	A, far back corner
F3	2145	565	A, ALEX 1
F4	2405	565	A, far front corner
F5	85	2795	B, ALEX 2
F6	565	2795	B, ALEX 2
F7	85	3055	B, far wall corner
F8	565	3055	B, far front corner

mm from the datum SOURCE-DEFINED. Feet are 50 in from each ALEX corner, which is 85 in from the top's edges. F2 F4 F7 F8 are 85 from the far end, F1 F3 F5 F6 are 345 from it.

Leg plates are ⌀105 and feet plates 59.5 square. L1's plate ends short of rail A and clear of the clamp 2 jaw; feet plates F3 and F6 stay 5 clear of the rails.

> Nothing here goes in the clamp zones C1 (x 360–440, the 90 next to the back edge) or C2 (y 360–440, the 90 next to the side-wall end).

G · Step 1: legs

> **MEASURE ON HARDWARE** MEASURE ON HARDWARE
> Calipers on both tops before the first hole: expect 28. If a top is 27, shorten every screw into it by 1 mm. No screw may go deeper than 21 into a top.

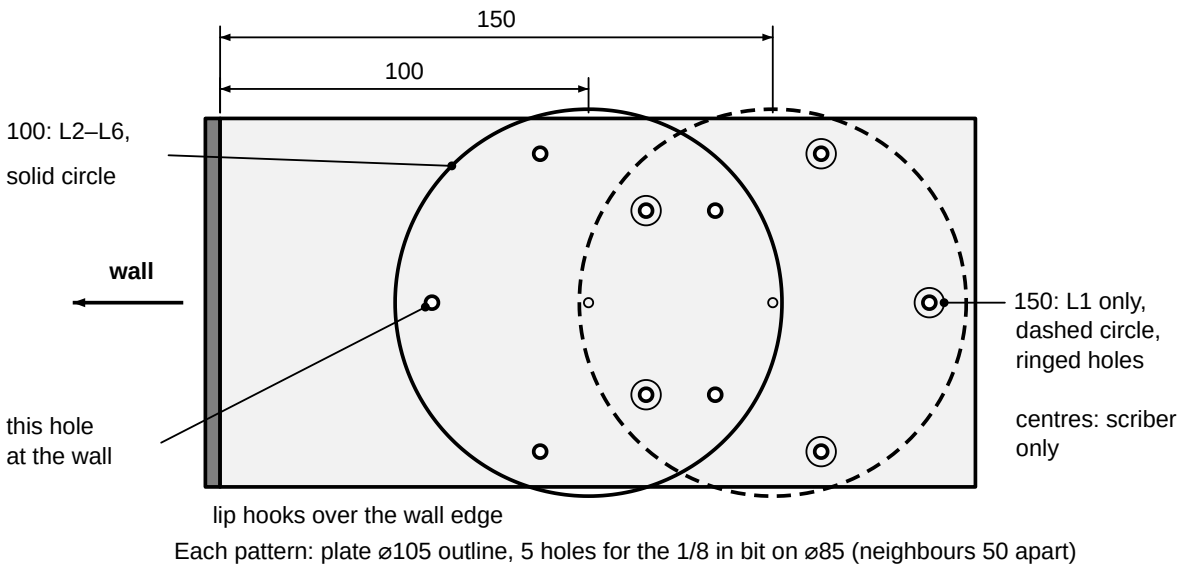


Fig. G-4 Leg jig (P18), seen from above in use, lip hooked down over the wall edge. The 100 pattern (solid plate circle, plain holes) is for L2 to L6 and puts one hole at the wall; the 150 pattern (dashed circle, ringed holes) is for L1 only.

Leg	Top	Hook the lip over	Tape station	Pattern
L1	A	side-wall end (x = 0)	y 420 from the back edge	150, ringed holes
L2 L3	A	back edge (y = 0)	x 650 and x 1400 from the side-wall end	100
L4 L5 L6	B	side-wall edge (x = 0)	y 850, 1600, 2350 (tape from the joint end reads y − 650; L4 is 200)	100

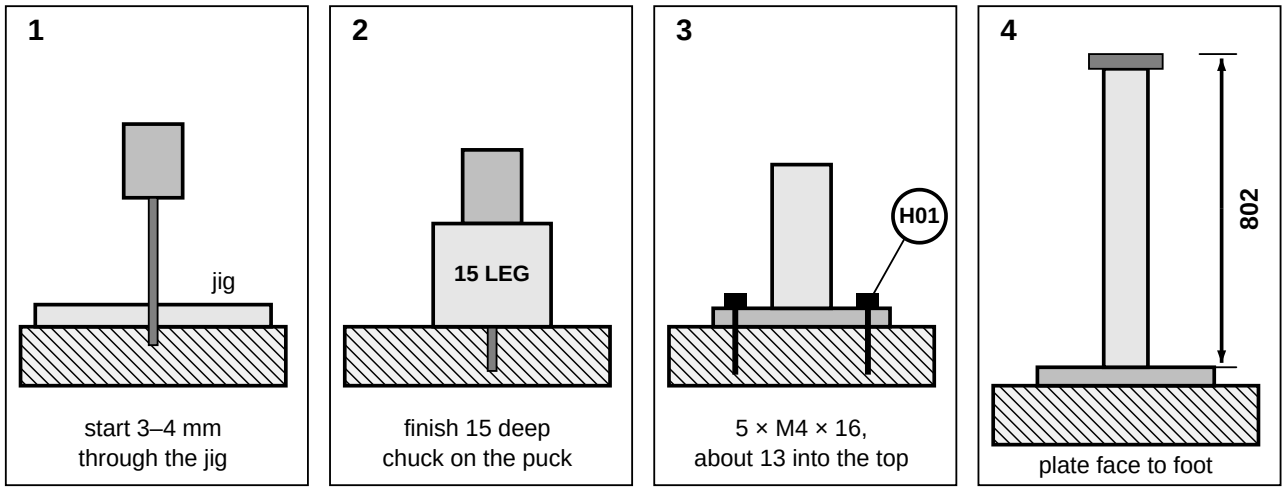


Fig. G-5 One leg, four operations (steps 1a–1d below). The drill guide's SET tower sets the 1/8 in bit first.

1 **a** Jig on the station, start all five pilots 3–4 mm through its holes with the 1/8 in bit. **b** Lift the jig; finish each pilot 15 deep with the **15 LEG** puck. **c** Screw the OLOV plate on with five **H01** M4 × 16 screws (about 13 into the top). **d** Pre-set the leg to **802** from plate face to foot (or the table's number on page G-1 if you fit ALEX feet). Six legs, 30 screws.

OLOV plate ⌀105, five ⌀5.5 holes on an ⌀85 circle, neighbouring holes 50 apart; range 600–900. VERIFIED 105 plate measured 22 Oct 2026; hole pattern from a shared OLOV template, printed and test-fitted.

G · Step 2: feet plates

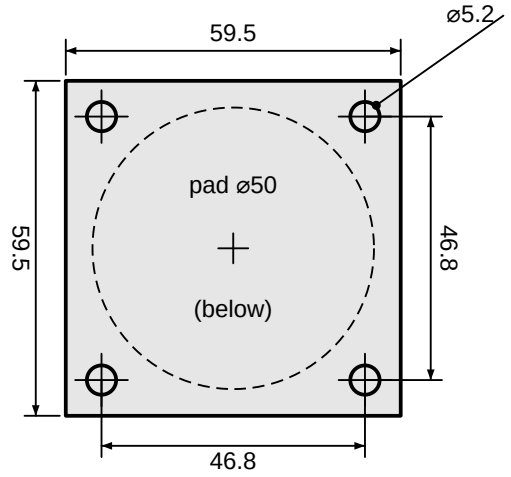


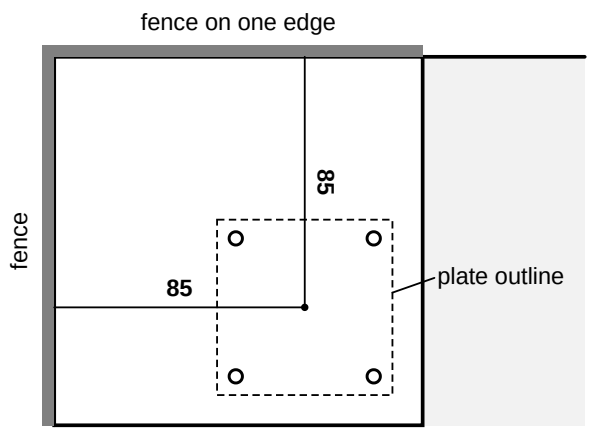
Fig. G-6 Feet plate (P09) from the screw side. VERIFIED on the purchased plate, 23 Oct 2026: 41.6 between two neighbouring holes (inside jaws), 52.0 across them (outside jaws).

Feet	Tool	Placement
F2 **F4**	feet corner jig	top A's two far corners (x 2490 end), fences on both edges
F1 **F3**	T-square, 85 hole	lip over A's back edge (F1) or front edge (F3), station x 2145
F7 **F8**	feet corner jig	top B's two far corners (y 3140 end)
F5 **F6**	T-square, 85 hole	lip over B's side-wall edge (F5) or front edge (F6), station y 2795

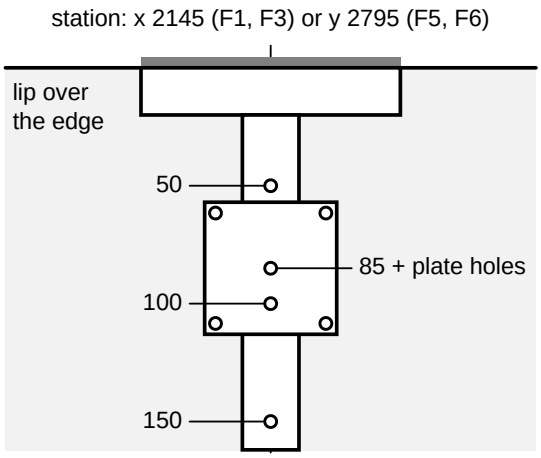
ASSUMPTION / FIELD CHECK

ASSUMPTION / FIELD CHECK

The feet's load rating is unknown; the design wants 50 kg or more each.



(a) corner jig on a far corner: F2 F4 F7 F8



(b) T-square, 85 hole: F1 F3 F5 F6

Fig. G-7 Both tools put the foot's four plate holes (1/8 in, on the 46.8 square) square to the edges in one placement, seen face down. Centre holes are for a scriber only.

2 Start the four pilots 3–4 mm through the tool's plate holes with the 1/8 in bit, lift the tool, finish 15 deep with the **15 LEG** puck. Screw the plate on with four **H02** #8 × 5/8 pan heads (14 into the top), pad pointing away from the top. Eight plates, 32 screws.

3 Wind each pad out until the printed foot height gauge (102.0 inside) just slides over the foot from the side with no rock. All eight then match without a ruler.

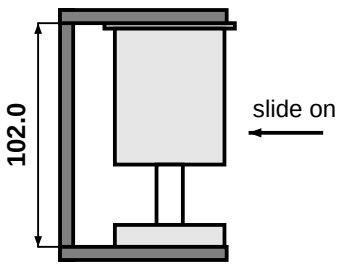


Fig. G-8 Foot height gauge.

G · Step 4: rings on the ALEX tops

> **MEASURE ON HARDWARE** MEASURE ON HARDWARE
> **Pad diameter:** calipers on one pad before you print the rings. The ring is bored 51 for a 50 pad. If yours differs, change `pad_d` in `parts/structure/foot-ring.scad` and reprint.
> **ALEX top panel:** measure its thickness at an edge. It must be 16 or more for the 5/8 in screws.

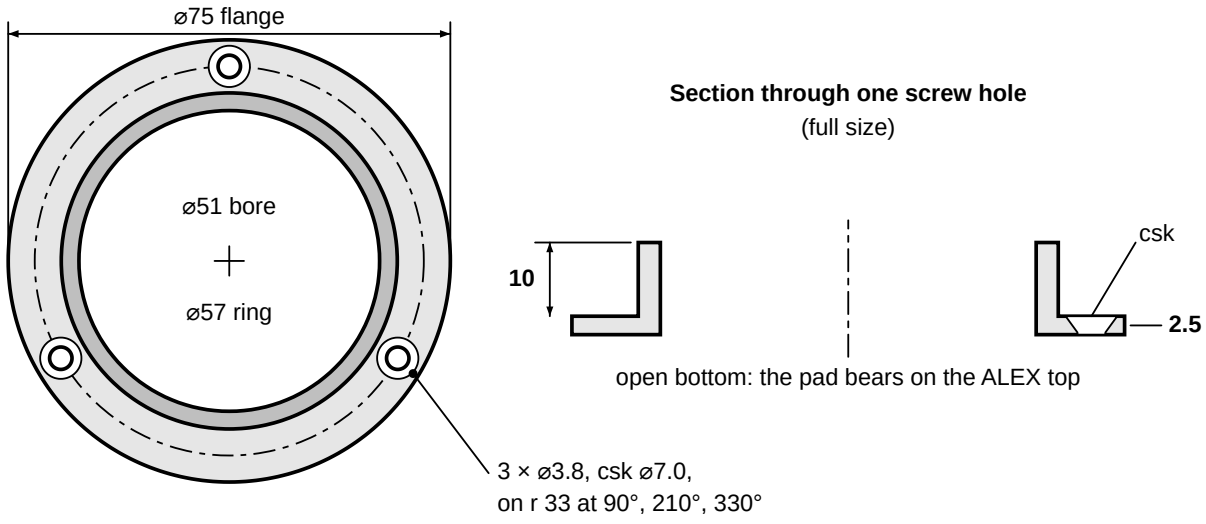


Fig. G-9 Printed locating ring (P10), from `parts/structure/foot-ring.scad`. VERIFIED from its own geometry file: bore ø51, ring ø57 (3 wall) and 10 tall on a ø75 × 2.5 flange, three ø3.8 holes countersunk ø7.0 on r 33. Print flange-down, PETG or ABS.

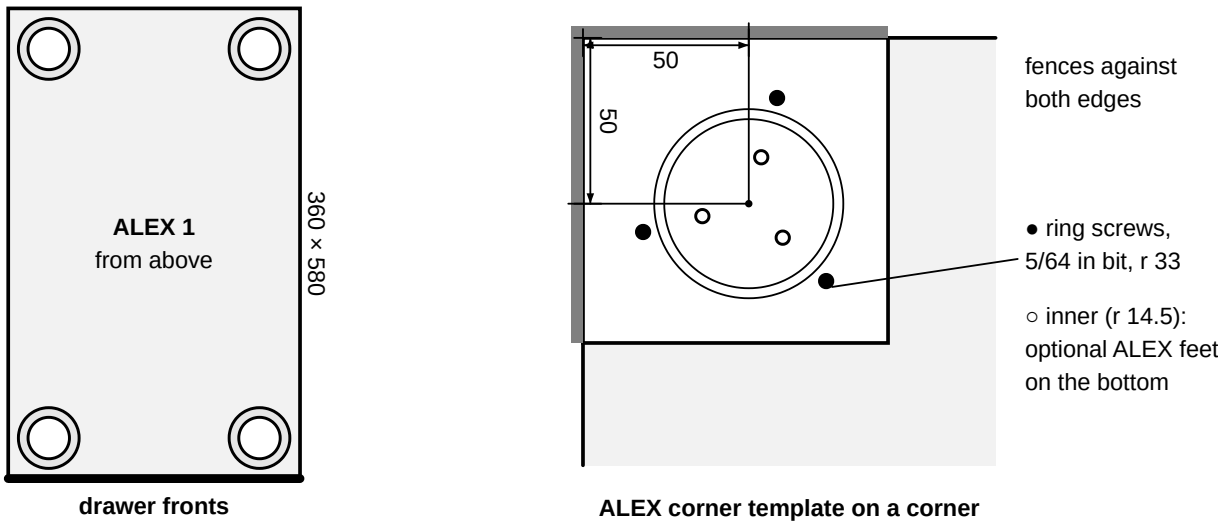


Fig. G-10 Left: ALEX 1's top with its four rings, centres 50 in from each corner. ALEX 2 is the same at all four corners. Right: the ALEX corner template dropped over a corner; the ring then fits over its three pilots only one way round.

4

Drop the corner template over each ALEX corner, fences against both edges. Start the three ring pilots 3–4 mm through it with the 5/64 in bit, lift it, finish 12 deep with the **12 ALEX** puck. Set a ring over the three pilots and drive three **H06** #6 × 5/8 flat heads (13 into the ALEX top). Eight rings, 24 screws.

G · Step 5: place the ALEX units

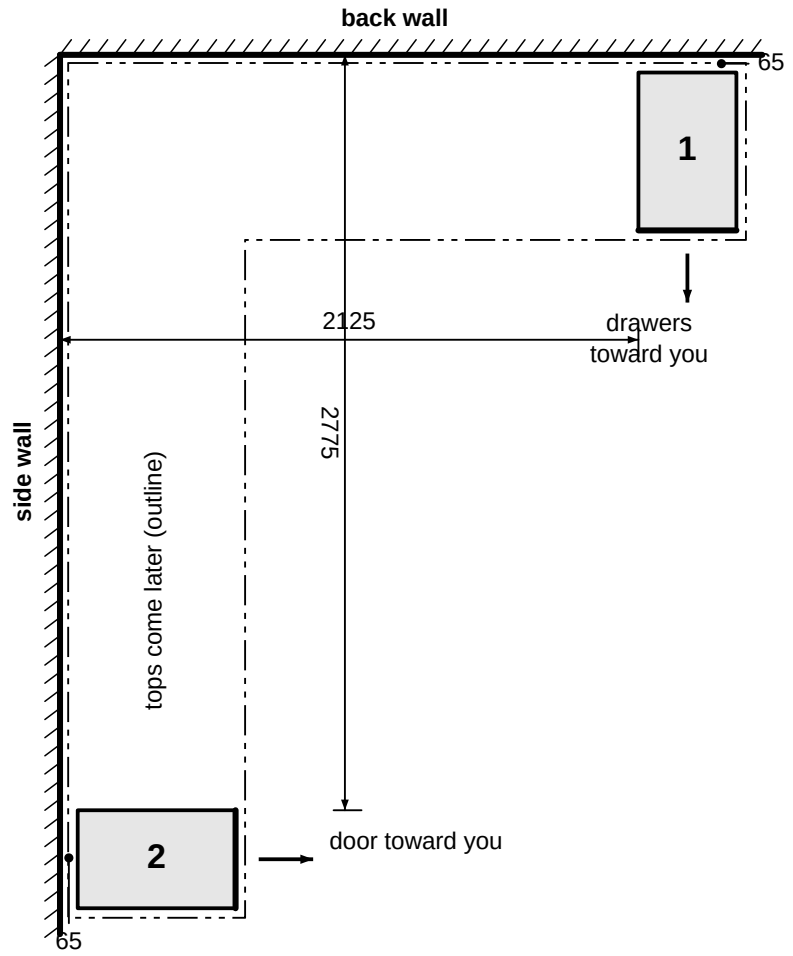


Fig. G-11 Plan from above. The tops are not up yet, so the cabinets are set from the walls (mm). Heavy line: the drawer fronts of ALEX 1 face you (+y); the door of ALEX 2 faces you (+x).

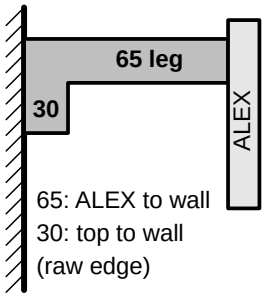


Fig. G-12 Wall gauge (P18): the 65 leg just fits between the wall and a cabinet.

5 Stand **ALEX 1** (P03) with its back 65 off the back wall and its near side 2125 from the side wall, drawers toward you. Stand **ALEX 2** (P04) with its side 65 off the side wall and its near end 2775 from the back wall, door toward you. Painter's tape on the floor round each one saves guessing.

Once the tops are on, that leaves 35 of top showing all round each cabinet and the tops 30 off the walls. The feet pads drop into the rings when the tops go on; that is what stops the desk sliding on the cabinets, and the cabinets then hold the whole desk against racking.

Optional ALEX feet (decided on page G-1): fit them now, before the cabinets stand. Unscrew the stock feet, lay the unit on its back, hook the corner template over each bottom corner and use its inner three holes (r 14.5). Pilot 5/64 in, 12 deep with the 12 ALEX puck; three #6 × 5/8 flat heads per foot down the wells, four feet per unit, all the same height.

! **ASSUMPTION / FIELD CHECK** ASSUMPTION / FIELD CHECK
The drawings take an ALEX as 700 tall, which is what makes the feet 102 between the ALEX top and the underside at 802. Measure your units, with and without their stock feet, before you wind the feet in step 3.

G · Foot, pad and ring in section

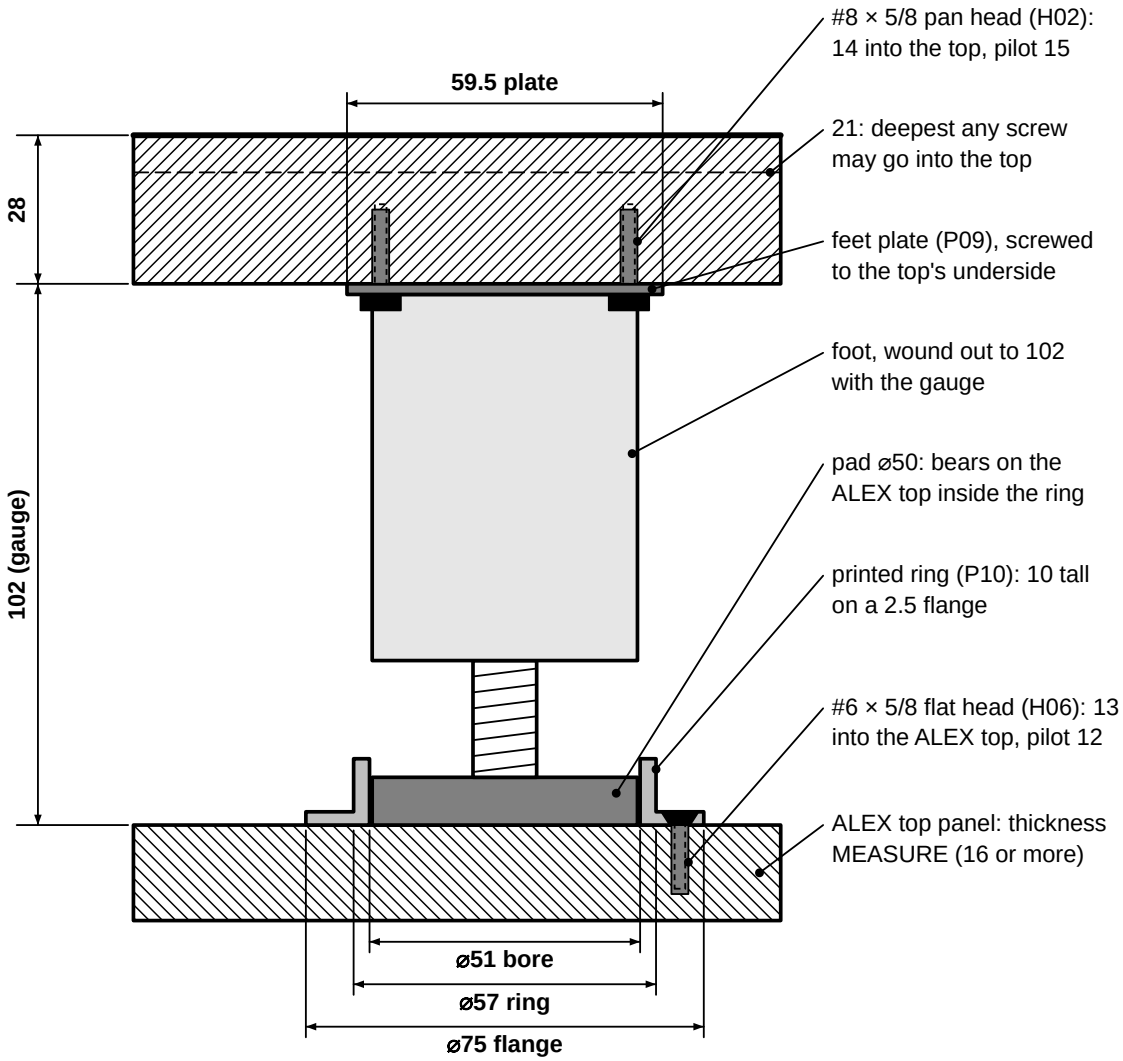


Fig. G-13 Section through one foot, its ring and one ring screw (mm). Hatched: the top above and the ALEX top panel below, both cut. The foot's internal shape is drawn schematically: only its 102 setting and ⌀50 body and pad are known. Heights above the floor: surface 830, underside 802, ALEX top 700 ASSUMPTION / FIELD CHECK.

Dimension	mm	Confidence
Top thickness	28	MEASURE ON HARDWARE expect 28
Foot, plate face to pad (gauge)	102	SOURCE-DEFINED
Feet plate, hole spacing	59.5 / 46.8	VERIFIED
Pad diameter	50	MEASURE ON HARDWARE before relying on the 51 bore
Ring bore / ring / flange	51 / 57 / 75 × 2.5	VERIFIED geometry file
ALEX top panel thickness	16 or more	MEASURE ON HARDWARE

G · Load path

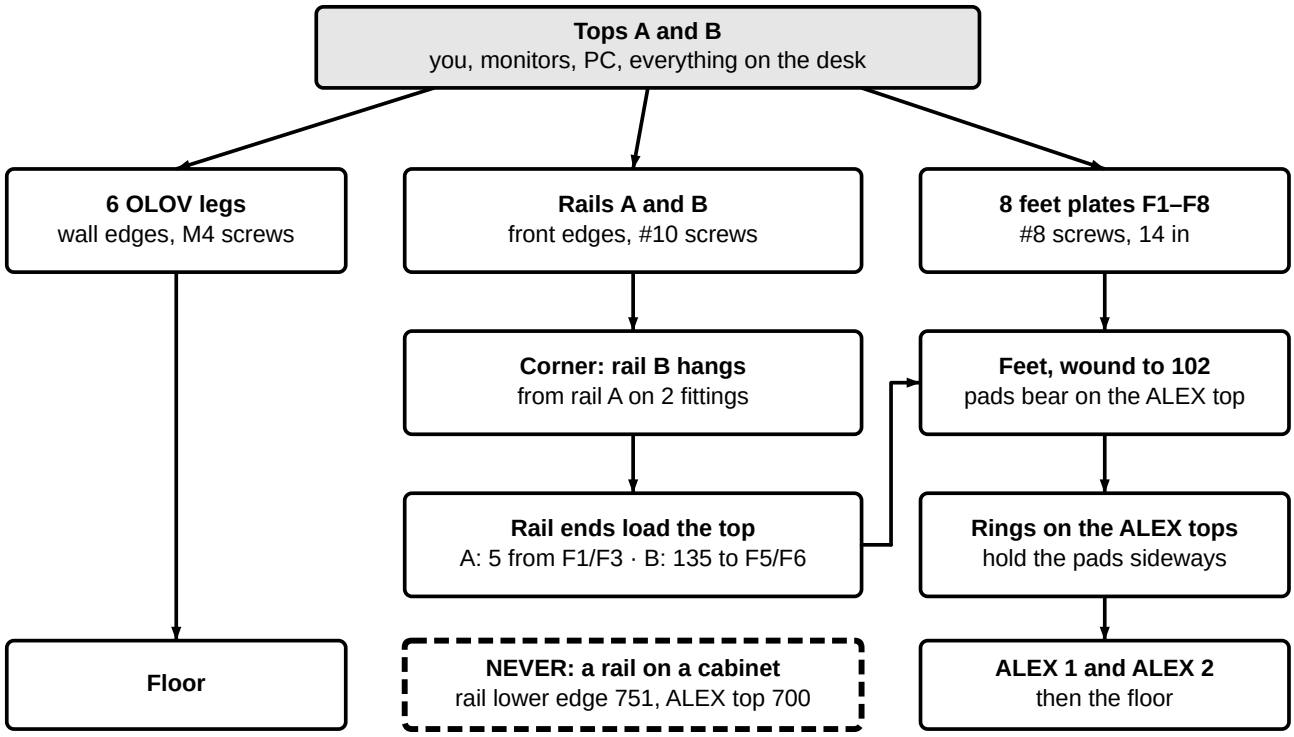
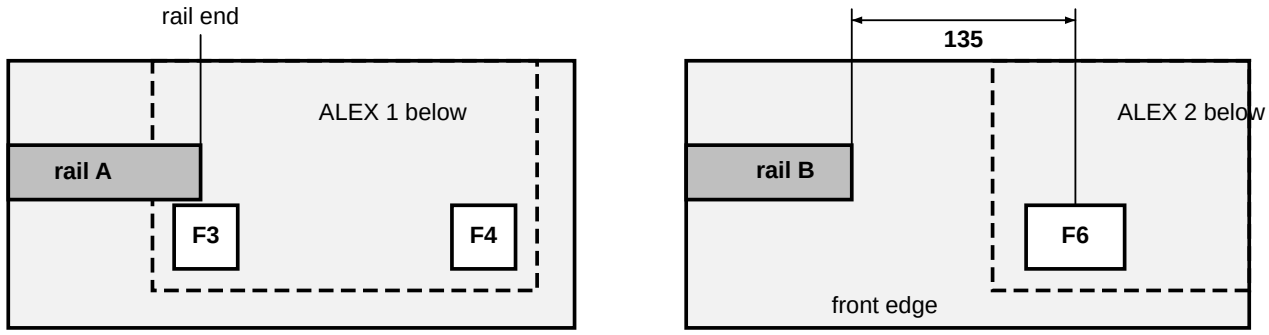


Fig. G-14 What carries what. The legs and the feet carry the tops to the floor. The rails carry the front edges; their ends load the top, and the top loads the nearest feet plates. The pads bear on the ALEX tops; the rings only stop them sliding.



(a) rail A's far end, plan
rail end x 2140 · F1/F3 centres x 2145

(b) rail B's end, plan (y to the right)
rail end y 2660 · ALEX 2 face 2745 · F5/F6 y 2795

Fig. G-15 Where the rails stop (mm from the datum; front edges at the bottom). (a) Rail A ends at x 2140, 5 short of the F1/F3 centres at 2145, beside F3's plate. (b) Rail B ends at y 2660; the top bridges the 135 to F5/F6.

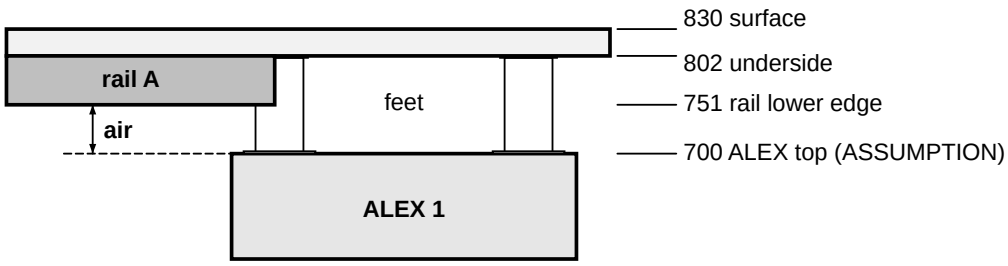


Fig. G-16 Elevation at ALEX 1, seen from the room. Rail A's end passes over the cabinet in plan but stays in the air above it: its lower edge is 751, the ALEX top 700.

G · The assembled underside

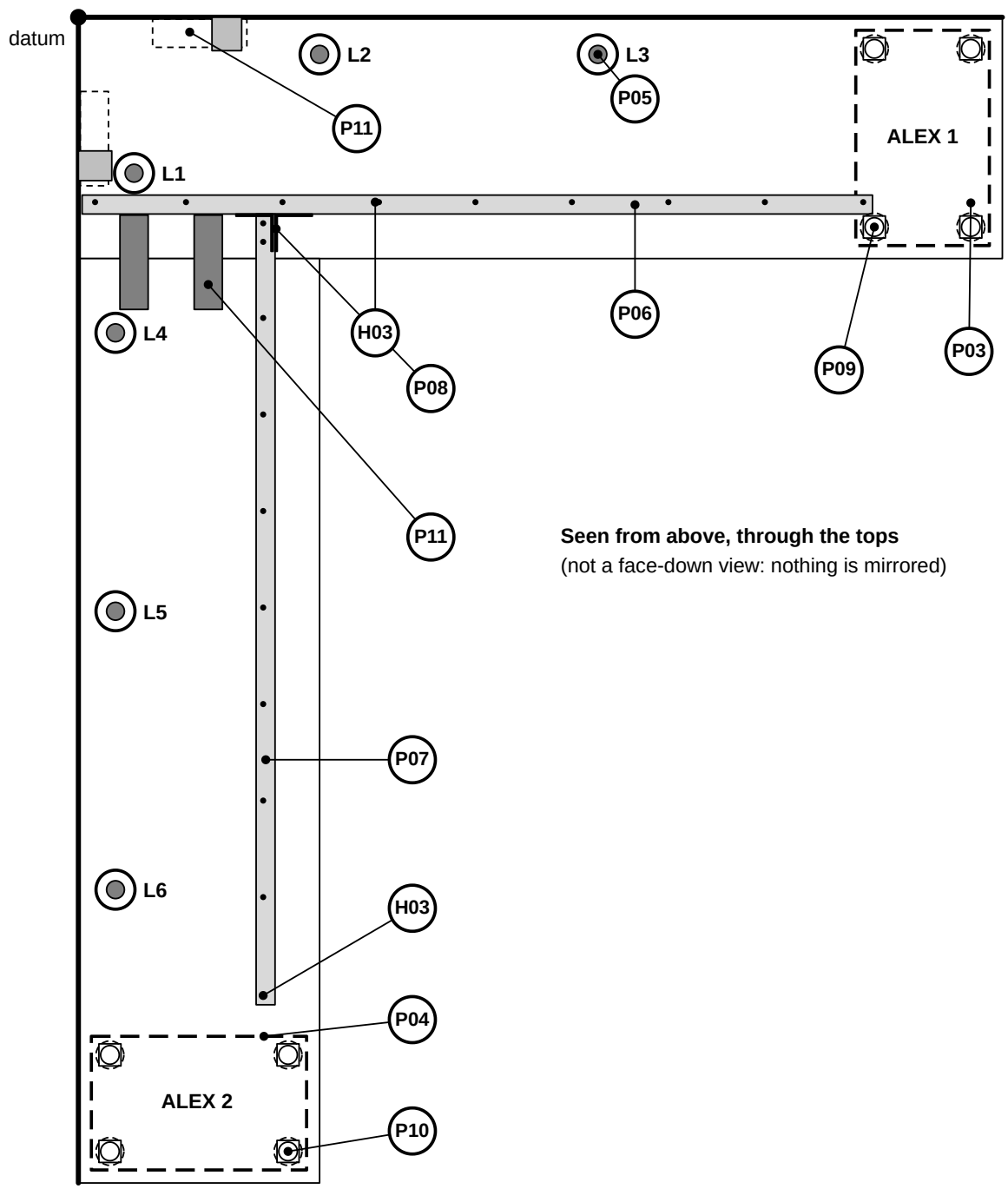


Fig. G-17 Everything under the tops once the desk is built, seen from above through the tops. Datum (0, 0) at top A's back-left corner. Rails, corner fittings and joining plates are fitted in later chapters; they are shown here to check against.

No.	Part	No.	Part
P03	ALEX 1, drawer unit 360 x 580 x 700	**P08**	Gusseted 90-degree strut fitting x 2
P04	ALEX 2, storage unit with door 360 x 580 x 700	**P09**	Round adjustable foot 4 in x 8
P05	OLOV leg x 6, 105 mm plate	**P10**	Printed locating ring x 8
P06	Rail A, steel angle 2130	**P11**	Mending plate 10 x 3 x 0.12 in x 4 (two joining, two clamp pads)
P07	Rail B, steel angle 2130	**H03**	#10 x 1 in pan head, Robertson (rails into the tops), 19

H · Joining and positioning the countertops

Two people. Both tops are face down with legs, feet and rails fitted, every OLOV pre-set to 802 (80.2 cm), every pad set with the foot height gauge (102); the ALEX units stand in place with their rings on.

Parts	Hardware	Tools
P01 top A · P02 top B · P03 ALEX 1 · P04 ALEX 2 · P05 OLOV leg · P06 rail A · P07 rail B · P08 strut fitting × 2 · P09 foot · P10 ring · P11 mending plate × 2 (joining)	H03 #10 × 1 in × 2 (rail B holes 1, 2) · H04 #10 × 5/8 in × 8 · H05 #10 washer × 8 · H08 3/8-16 × 1-1/2 in bolt × 6 · H09 3/8 washer × 12 · H10 3/8-16 nylon nut × 6	Drill guide (SET tower, pucks 25 RAIL and 12 PLATE), 1/8 in bit, Robertson #2 on a 3 in extension, level, straightedge, tape, torque wrench or 15 cm wrench

1 **Stand top A.** With a helper, turn top A over and stand it on its legs: lower the ALEX end first so the four pads drop into their rings on ALEX 1, then the leg end. Level along x and along y by turning the OLOV feet and the round feet until it reads flat and all four pads bear with no rock. Target: surface 830 (83.0 cm) at every corner. SOURCE-DEFINED

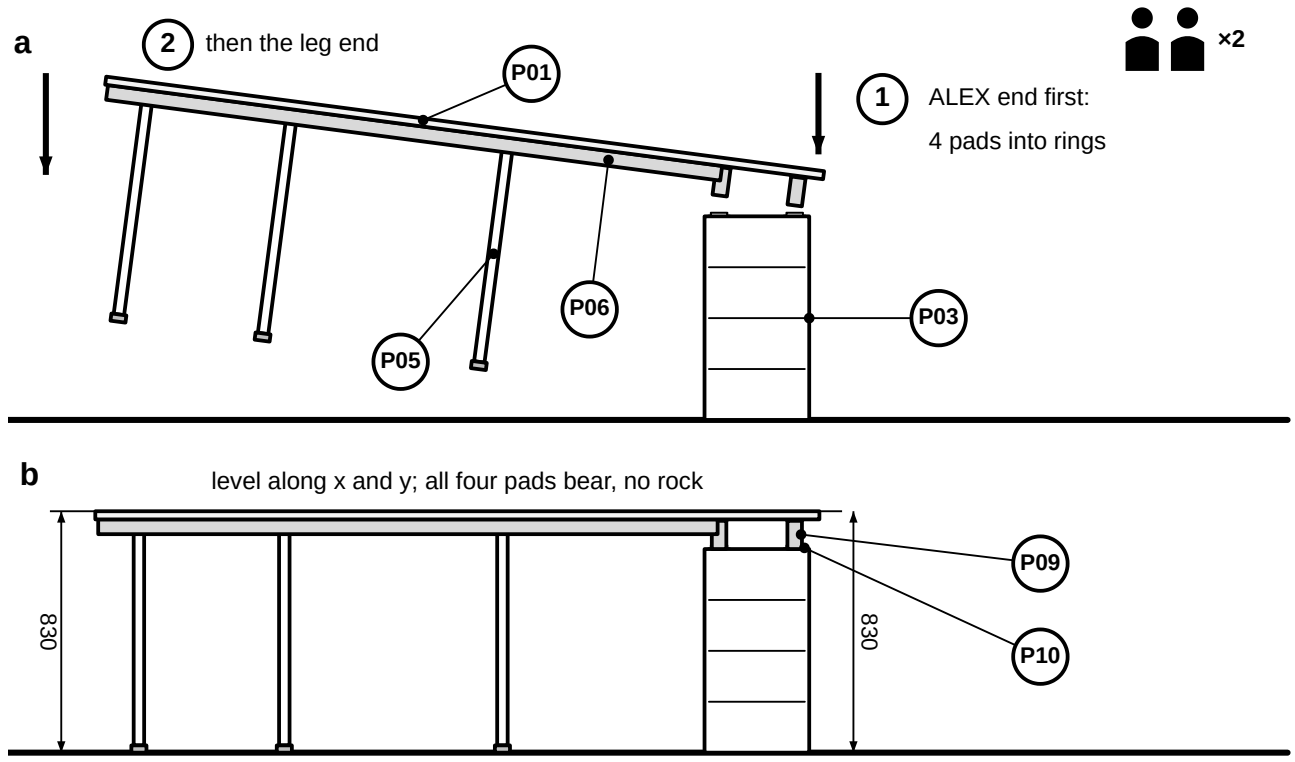


Fig. H-1 · Top A from the front (1:26). a: ALEX end first, the pads drop into the rings; b: leg end down, level, 830 at both ends. Rail A hangs under the top; it never touches ALEX 1.

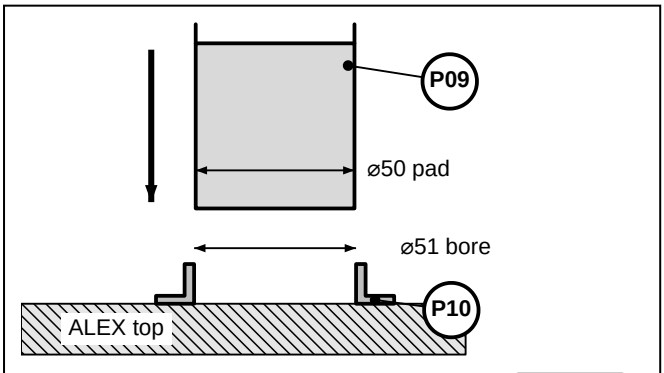


Fig. H-2 · Pad into ring (section, 1:2.4). Bore 51 VERIFIED; pad 50 MEASURE ON HARDWARE

! **Once the pads are in the rings, top A cannot be pushed.**
If the pads miss, or the top is not 30 (3 cm) off a wall, lift the ALEX end a few millimetres and slide the ALEX unit, not the top. The wall gauge's 30 leg, or a 3 cm offcut, against the wall is the easiest gauge.

830 assumes ALEX units 700 tall ASSUMPTION / FIELD CHECK until measured. With the optional ALEX feet the surface rises by their height (20, 40 or 60).

2

Stand top B and slide it home.

1. Wind top B's OLOV legs 2–3 mm short, so rail B's corner end (it projects 120 past B's end) can slide under top A.
2. Turn top B over onto its legs with its pads in the rings on ALEX 2.
3. Push top B (ALEX 2 slides with it) until rail B's corner end touches the outer face of rail A's vertical leg and B's end meets A's front edge at y 650 (65 cm), 30 off the side wall.
4. Wind B's legs up until the two surfaces are flush.

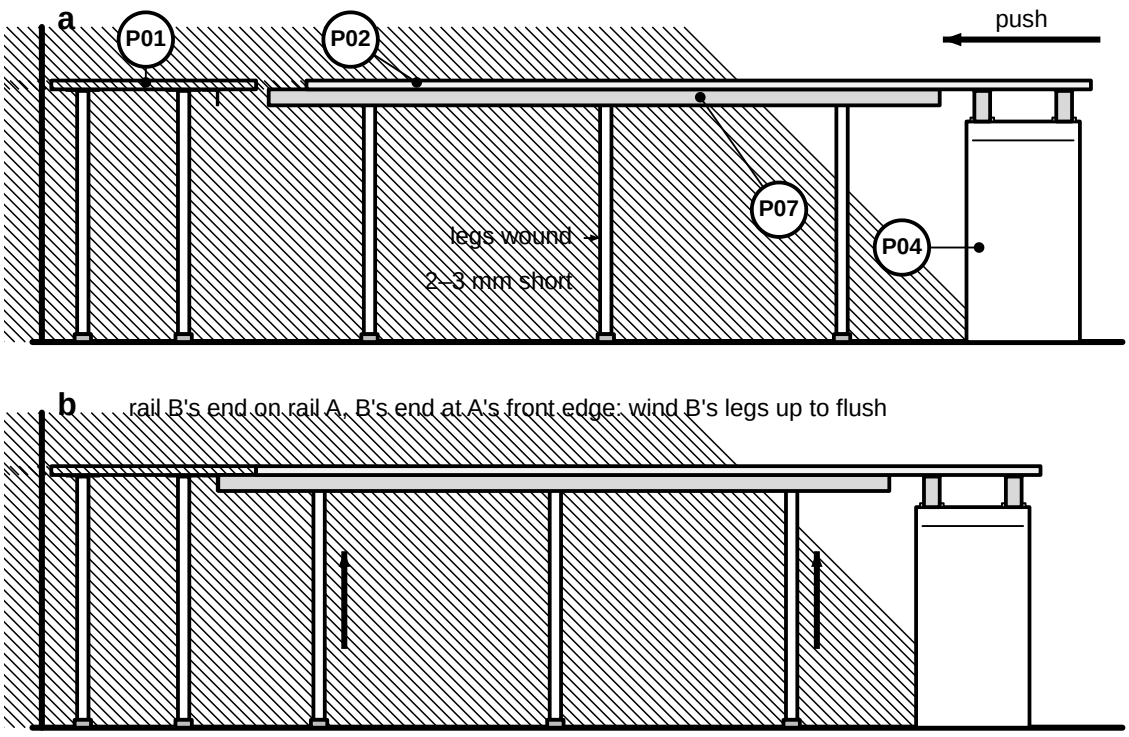


Fig. H-3 · Top B seen from the knee space, looking at the side wall (1:24); top A cut beyond B's inside edge, ALEX 1 not shown. a: legs short, push; b: home, wind the legs up. The rails rest only on the tops; neither touches an ALEX.

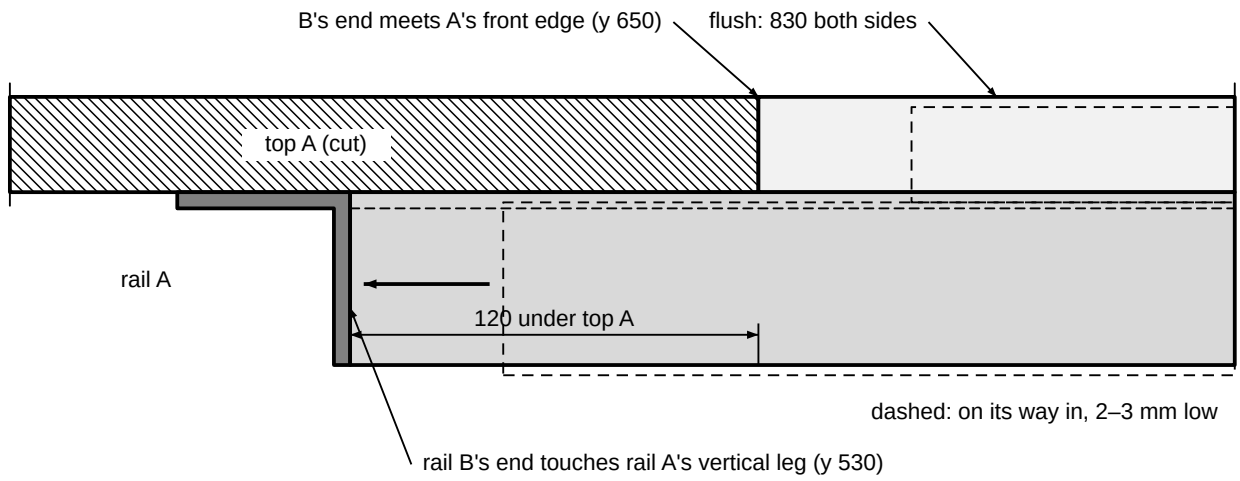


Fig. H-4 · The joint in section, same view (1:2.2). Dashed: top B on its way in, 2–3 mm low. Solid: home. Rail B's flat leg bears on top A's underside for its first 120; its holes 1 and 2 are drilled in step 5.

!

Never force the corner.
If the corner will not meet, or the surfaces will not come flush, adjust leg height, not the joint. Do not hammer, clamp or lever the tops together.

3 **Check under the corner, then level the joint.** Rail B's end must touch the face of rail A's vertical leg with no gap, and rail B's outer face must line up with the 52.0 cm (520) mark on rail A. If not, lift top B a little and move it. Then bring the two surfaces level at the joint by turning the OLOV feet and the round feet, never by packing anything under the joining plates. A fingertip across the joint finds no step.

4 **Joining plates P1 and P2, from below.** Screw each plate on the outline you drew: y 533–787 (53.3–78.7 cm), long axis along y, centred x 150 and 350. They are not centred on the joint: each starts 3 clear of rail A. Pilot every hole 1/8 in, 12 deep, with the drill guide's **12 PLATE** puck held up overhead; drive H04 #10 × 5/8 in with an H05 washer. Two screws in each top. SOURCE-DEFINED

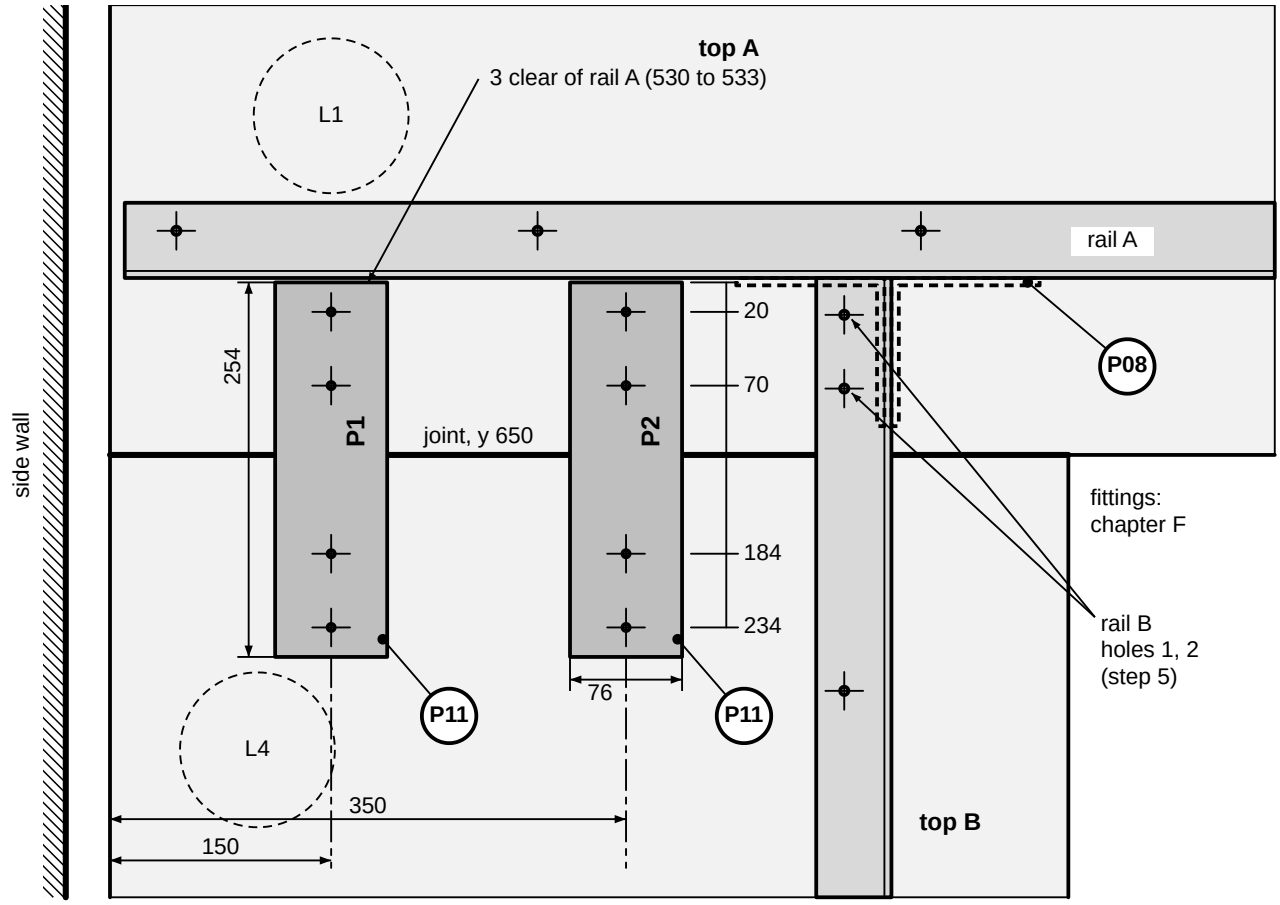


Fig. H-5 · The corner from above, looking through the tops (1:5). Datum: top A's back-left corner, x along the back wall, y along the side wall. Plate holes at 20, 70, 184 and 234 from the plate end nearest rail A. Dashed: OLOV plates L1 and L4, and the two fittings (orientation in chapter F).

Plate	Centre x	Span y	Screws in top A, y	Screws in top B, y	Pilot
P1	150	533–787	553, 603	717, 767	⌀3.2 (1/8 in) × 12, puck 12 PLATE
P2	350	533–787	553, 603	717, 767	same

The plates align the tops; they do not carry the corner. Every plate hole is more than 40 (4 cm) from an edge, so all four get a screw.

5 **Rail B meets top A.** Through rail B's holes 1 and 2 (25 and 75 from its corner end, both under top A) drill a 1/8 in pilot with the drill guide's **25 RAIL** puck on the flat leg: 4.8 of steel plus 20 into the wood. Drive H03 #10 × 1 in with the Robertson #2 bit on the 3 in extension until the head seats. SOURCE-DEFINED

6

Bolt on the two fittings, then tighten. Orientation is in chapter F: outside fitting heel at the outside corner, inside fitting heel at the inside corner with its gusset pointing at the floor.

1. Rail A: four H08 bolts from the fitting side, H09 washer under each head, washer and H10 nylon-insert nut on the inside of the leg. Finger-tight.
2. Rail B: two H08 bolts through outside fitting, leg and inside fitting, washer each end, H10 nut. Finger-tight.
3. Only when the surfaces are flush (step 3) and rail B's end touches rail A: tighten all six to **25 N·m (18 ft·lb)**; without a torque wrench, one hand's full pull on a 15 cm wrench, then stop. SOURCE-DEFINED

Tight, not snug: the fittings' 21 × 12 slots leave 2.5 of play across and 11.5 along each bolt, and the 7/16 in rail holes 1.6, so only the clamping force holds the corner. Fitting legs 100.5 × 41, 5 thick; slot centres 29.15 and 78.45 from the heel VERIFIED. Rail bolt holes: rail A 549.2 and 598.5 (outside fitting) and 485.9 and 436.6 (inside fitting) from the wall end; rail B 29.2 and 78.5 from the corner end VERIFIED.

7

Checks. Do every one; if one fails, fix it with the feet or the legs, never at the joint.

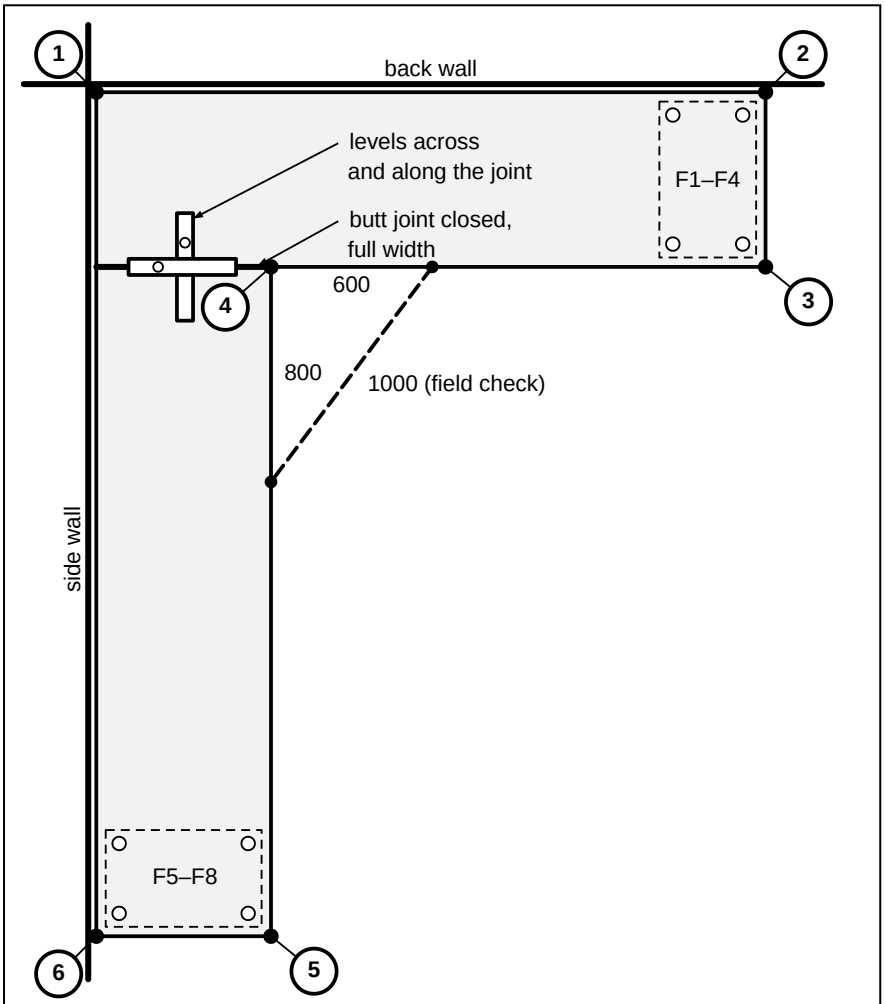


Fig. H-6 · The checks in plan (1:28). Datum: top A's back-left corner.

- **Square:** the butt joint is closed across its full 650 width (no light), rail B's end touches rail A, rail B's outer face is on rail A's 520 mark.
- **Level:** a level across the joint and along it reads flat both ways; a fingertip finds no step.
- **Height:** surface 830 ± 3 at all six outer corners, 1–6. SOURCE-DEFINED
- **Pads:** all eight bear. Turn each of F1–F8 by hand in its ring: none spins free or rocks.

! **FIELD CHECK, not in the build manual** ASSUMPTION / FIELD CHECK
Optional 3-4-5 triangle at the inside corner: mark 600 along A's front edge and 800 along B's inside edge; the diagonal between the marks is 1000 when the L is square. Write down what it reads. If it is off, check the butt joint for a gap first, then move ALEX 2 with top B on it. Never force the corner.

The corner load test (20 kg on top B's inside corner), the sideways push and the settling record are in the final checks. Never stand or sit on the corner.

8b · Wall-gap trim (optional)

> **Do this after chapter I.**
> The clamp pieces wrap round the VIVO clamp backs, so both clamps must already be fitted and tight. The trim goes on before the final push to the walls.

Parts P19 (corner, two 80 fillers, two clamp pieces, 240 segments, connector tiles), H07 #4 × 1/2 in flat head, the drill guide's **10 EDGE** puck with a 5/64 in bit, the wall gauge.

8 **Measure, set, print.** With the desk in place against both walls, measure the gap between each top's raw back edge and the wall at five or six points along each wall. Set `gap` in `parts/accessories/wall-gap-common.scad` to the smallest value minus 1 and print the trim. MEASURE ON HARDWARE (design gap 30, so `gap` 29 as saved).

Wall	From the wall corner outward (desk positions)	Slotted segments
back	corner (to 200), 80 filler (200–280), clamp piece (280–520), 8 × 240, 1 × 50 (to 2490)	3rd (1000–1240), 8th (2200–2440)
side	corner (to 200), 80 filler (200–280), clamp piece (280–520), 10 × 240, 1 × 220 (to 3140)	3rd (1000–1240)

The corner piece covers each wall's first 200 of desk. 24 connector tiles: 11 joints on the back wall, 13 on the side wall. Cut list from parts/README.md.

9 **Pull the desk out, 400 (40 cm) from both walls.** Two people, one at each ALEX unit: slide the whole desk away from both walls. The ALEX units slide on their feet and the pads stay in their rings. Nothing comes apart.

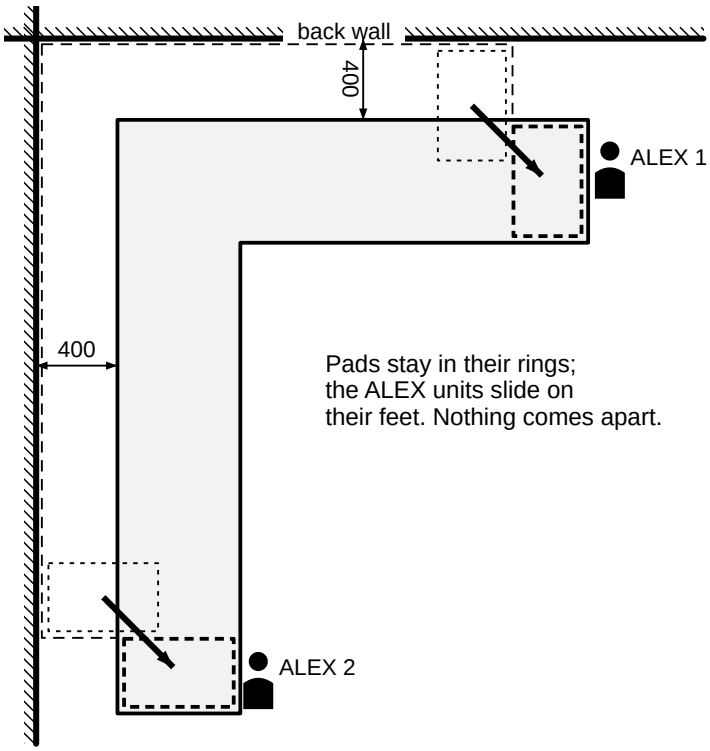
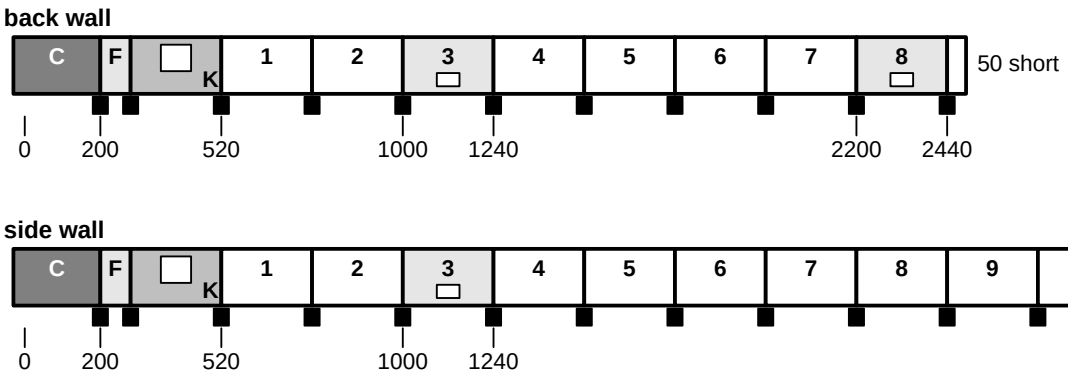


Fig. H-7 · Plan (1:40). Dashed: the desk in place, 30 off both walls. Grey: pulled 400 off both, so the raw edges are open to work on.

10 **Build each chain from the corner outward.** Corner piece, 80 filler, clamp piece with its notch round the clamp back, then the 240 segments (slotted ones where the cut list says), ending with the short piece. Butt every joint and push a connector tile up into the pockets.



C corner · F 80 filler · K clamp piece (white: notch round the clamp back)
1, 2 … 240 segments, grey with a slot = slotted · ■ connector tile under every joint

Fig. H-8 · Chain order along both walls, desk positions in mm from the corner (1:20). The notch (white, in K) wraps the clamp back, 80 long and 17.5 proud VERIFIED.

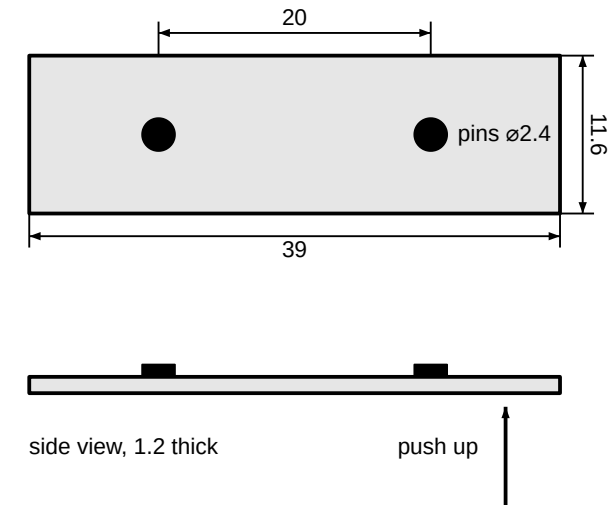


Fig. H-9 · Connector tile (1.8:1), one under every joint, pins up. It bridges the 20 × 12 pockets of two butted pieces.

Every piece, the corner included, has a pocket on its underside at each end, over the gap. Butt two pieces, then push a tile up into both pockets until its pins seat: the joint cannot step, bow or drift, and nothing shows on top.

If a wall is out and the notch misses the clamp, change the 80 filler's `seg` and reprint it, nothing else.

11 **Pilot and screw.** Through each lip hole, pilot 5/64 in, 10 deep, drill horizontal, with the drill guide's **10 EDGE** puck held against the raw edge. Drive an H07 #4 × 1/2 in flat head into each: two per segment, one per corner leg, one each side of a clamp notch.

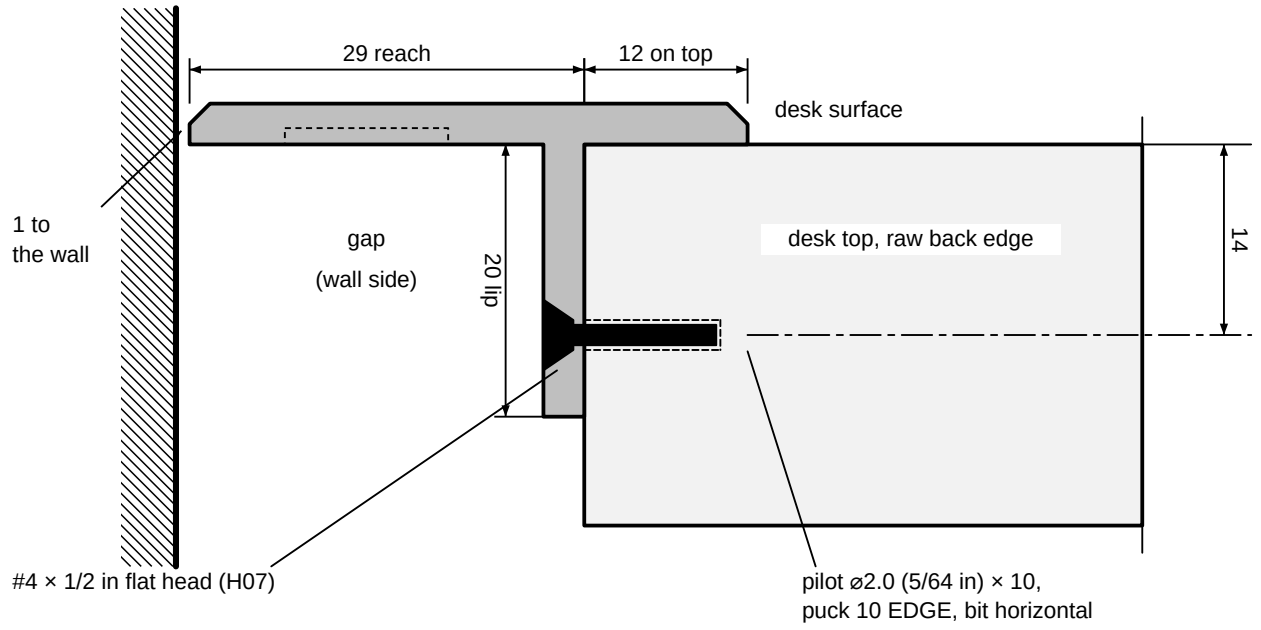


Fig. H-10 · Trim on the raw back edge, section (1.8:1). 12 rests on the surface, the 20 lip drops into the gap, the screw sits 14 below the surface: nothing shows on top and nothing goes into the top face. Profile from parts/accessories/wall-gap-common.scad SOURCE-DEFINED; top 28 thick MEASURE ON HARDWARE.

12 **Slide the desk back.** Two people again, one at each ALEX, until the wall gauge's 65 leg just fits between the wall and each ALEX. The trim's outer edge stops 1 short of the wall.

Retighten every screw after two weeks.

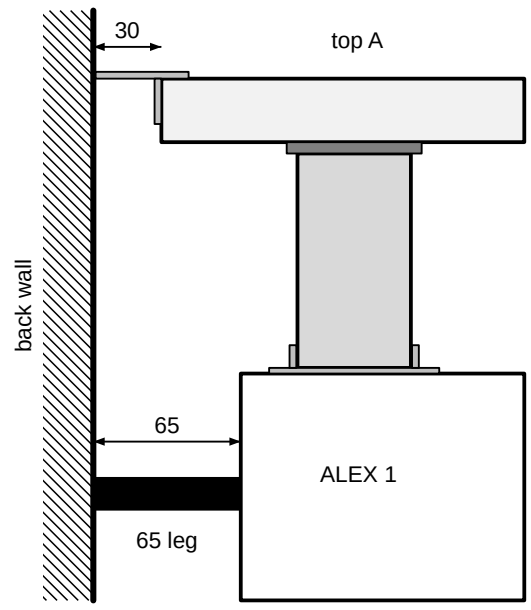


Fig. H-11 · Wall gauge at ALEX 1, section along the back wall (1:3.3).

I · Monitor mount and accessories

The VIVO STAND-V300CU (P12) has two independent C-clamps and no base bar. Both clamp onto top A's own wall edges, never on the joint: clamp 1 centred at **x = 400 (40.0 cm)** on the back edge, clamp 2 centred at **y = 400 (40.0 cm)** on the side-wall end. The crossbar runs on the diagonal between the poles. Each lower jaw bears on a spare steel mending plate (P11), because particleboard crushes under a bare steel jaw over months. Datum: top A's back-left corner (0, 0); x along the back wall, y along the side wall; mm.

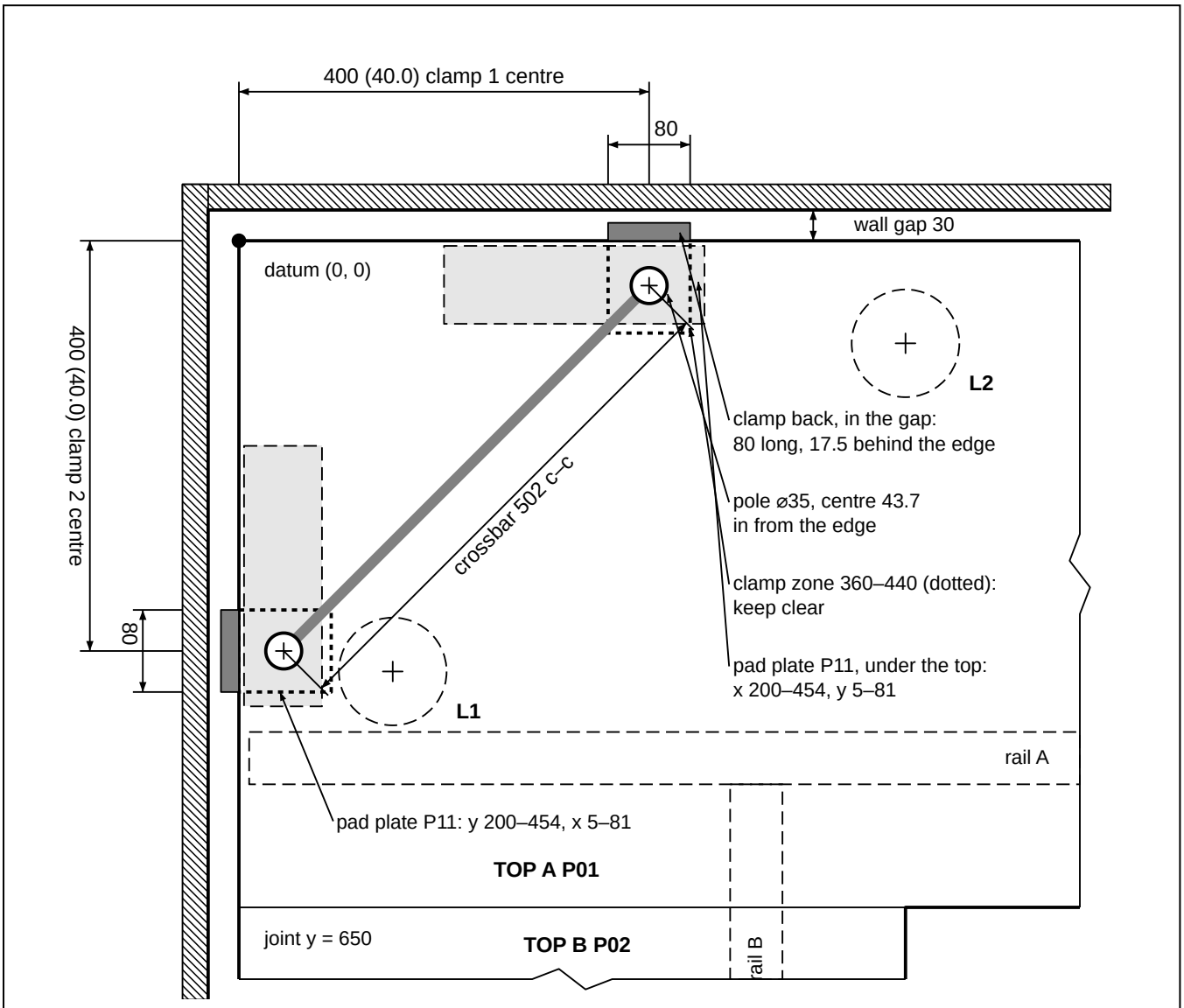


Fig. I-1 Plan of the corner from above. Dashed: under the top (pad plates P11, legs L1 and L2, rails). Dotted: the clamp zones, 360–440 along the edge, keep clear. Dark: the clamp backs in the 30 mm wall gap. Black circles: poles, 43.7 in from the edge.

Balloon	Part	Balloon	Part
P01	Top A, EKBACKEN 2490 × 650 × 28	P11	Mending plate 10 × 3 × 0.12 in, two used here as clamp pads
P02	Top B, EKBACKEN 2490 × 650 × 28	P12	VIVO STAND-V300CU (clamps, poles, crossbar, arms)

What is measured at the clamp

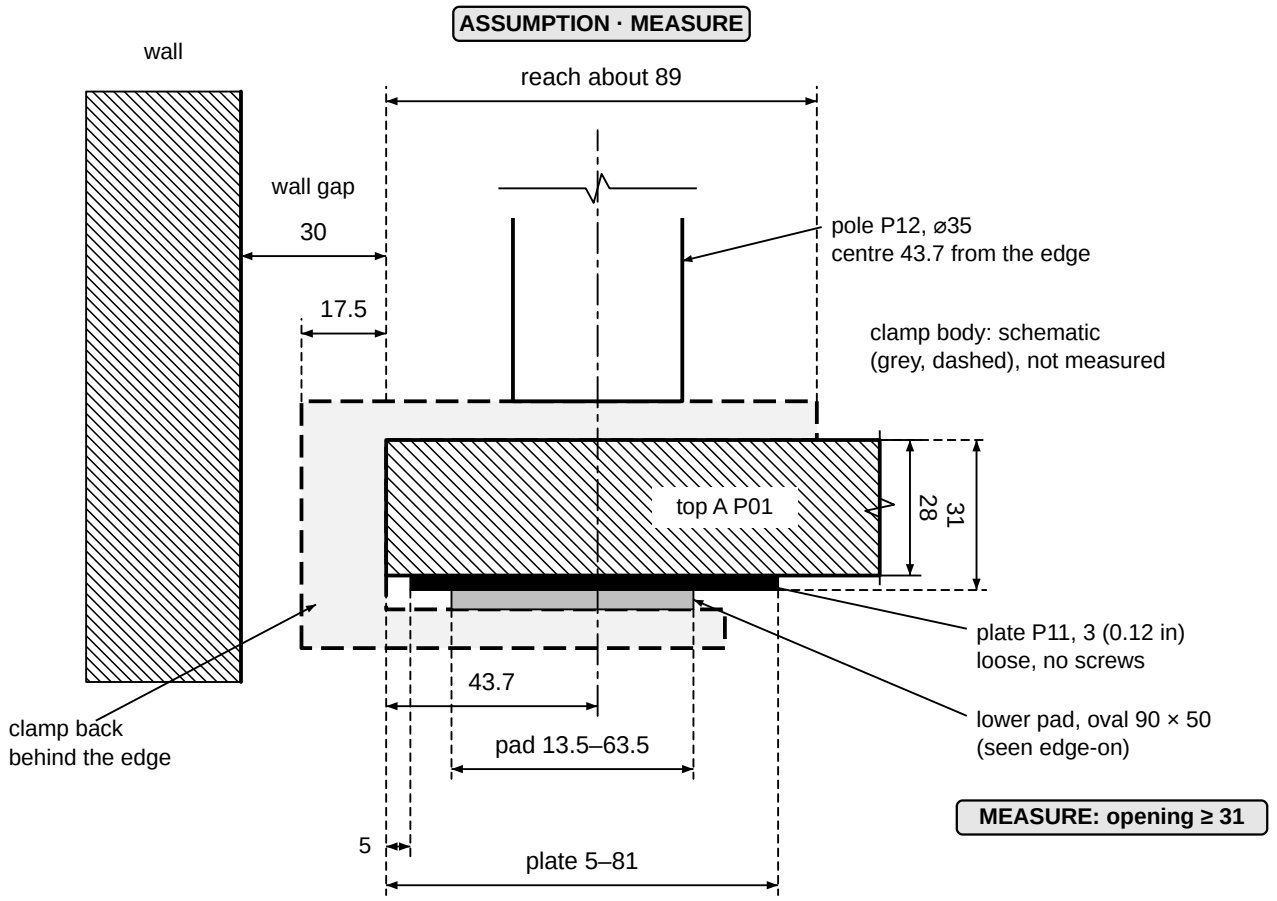


Fig. I-2 Section through clamp 1 at x = 400, looking along the back wall. The clamp body is drawn as a plain C (grey, dashed) because its shape is not measured; only the dimensioned values are. The section through clamp 2 at y = 400 is the same, turned 90°.

Value	mm	Confidence
Crossbar, centre to centre of its end collars (19¾ in)	502	VERIFIED
Pole diameter; pole centre in from the desk edge	35; 43.7	VERIFIED
Clamp back: length along the edge; projection behind the edge (sits in the 30 wall gap)	80; 17.5	VERIFIED
Lower pad: oval 90 × 50, from 13.5 to 63.5 in from the edge, so it lands on the plate (5–81)	13.5–63.5	VERIFIED
Ideal clamp centre from the corner: 43.7 + 502 ÷ 1.414; built at 400, the crossbar sets the last millimetre	399 → 400	SOURCE-DEFINED
Clamp zones (x 360–440, y 0–90 at clamp 1; y 360–440, x 0–90 at clamp 2); pad plate outlines x 200–454 / y 5–81 and y 200–454 / x 5–81	see Fig. I-1	SOURCE-DEFINED
Top plus plate: 28 + 3 (0.12 in), under 60, so VIVO's Option 1 holes (2 and 4 on the lower clamp)	31	SOURCE-DEFINED
Top arm reach from the edge	about 89	ASSUMPTION / FIELD CHECK
Maximum jaw opening (must be 31 or more)	≥ 31	MEASURE ON HARDWARE

Hang the mount

1 **Lay the pad plates.** Lay a spare mending plate P11 flat on the underside, on each pad outline drawn when top A was marked, long side along the edge: x 200–454, y 5–81 at clamp 1; y 200–454, x 5–81 at clamp 2. Loose, no screws: the jaw holds it. SOURCE-DEFINED

> ! **MEASURE ON HARDWARE before hanging the clamps**
> Top arm reach (about 89) and maximum opening (needs 31 or more: 28 top + 3 plate) are not measured. Open the lower clamp fully on the Option 1 holes and measure the gap; if it is under 31, stop. The clamp zones are drawn 90 deep from the edge; if the reach measures more, keep the zone clear to the measured reach.

2 **Hang both lower clamps, finger-tight only.** Clamp 1 centred at x = 400 on A's back edge, clamp 2 centred at y = 400 on A's side-wall end (Fig. I-1). Use the VIVO sheet's Option 1 holes (2 and 4), since 28 + 3 = 31 is under 60. The clamp backs sit in the 30 mm wall gap; the lower pad lands on the plate, 13.5–63.5 from the edge (Fig. I-2).

3 **Fit the poles and the crossbar as the spacing guide** (VIVO sheet step 3). The crossbar measured 502 centre to centre, so it fixes the pole spacing. **Only now tighten both clamps**, with the crossbar on. VERIFIED

4 **Refit the crossbar at monitor height** (VIVO sheet step 6): 400–430 above the top (1230–1260 from the floor).

5 **Hang the three screens, all landscape**, and set the heights in Fig. I-3. The G5 goes on the centre bracket (VESA 75), screen centred on the diagonal at (365, 365), 200 (20 cm) in front of the crossbar. The Acers go on the side arms, toed in 15°; each Acer's near end ends up about 120 (12 cm) from the G5's curved end. The side arms run along the wall behind each panel, then forward into the VESA plate.

Weights against the mount's ratings

Screen	On the mount	Rating	Confidence
Samsung Odyssey G5 32 in P13 (1000R, VESA 75), centre	about 5 kg	centre bracket 48.4 lb curved (59.4 lb flat); top edge limit 26.5 in (673) above the top	SOURCE-DEFINED
Acer XZ320QU 31.5 in P14 (1500R, VESA 100), each side	about 5 kg (5.6 kg with its stand)	17.6 lb (8 kg) per side arm	SOURCE-DEFINED

The clamps load top A's edges near legs L1 and L2, not the corner tip. Do not add a soft pad under a jaw: the steel plate is the pad.

Heights above the top

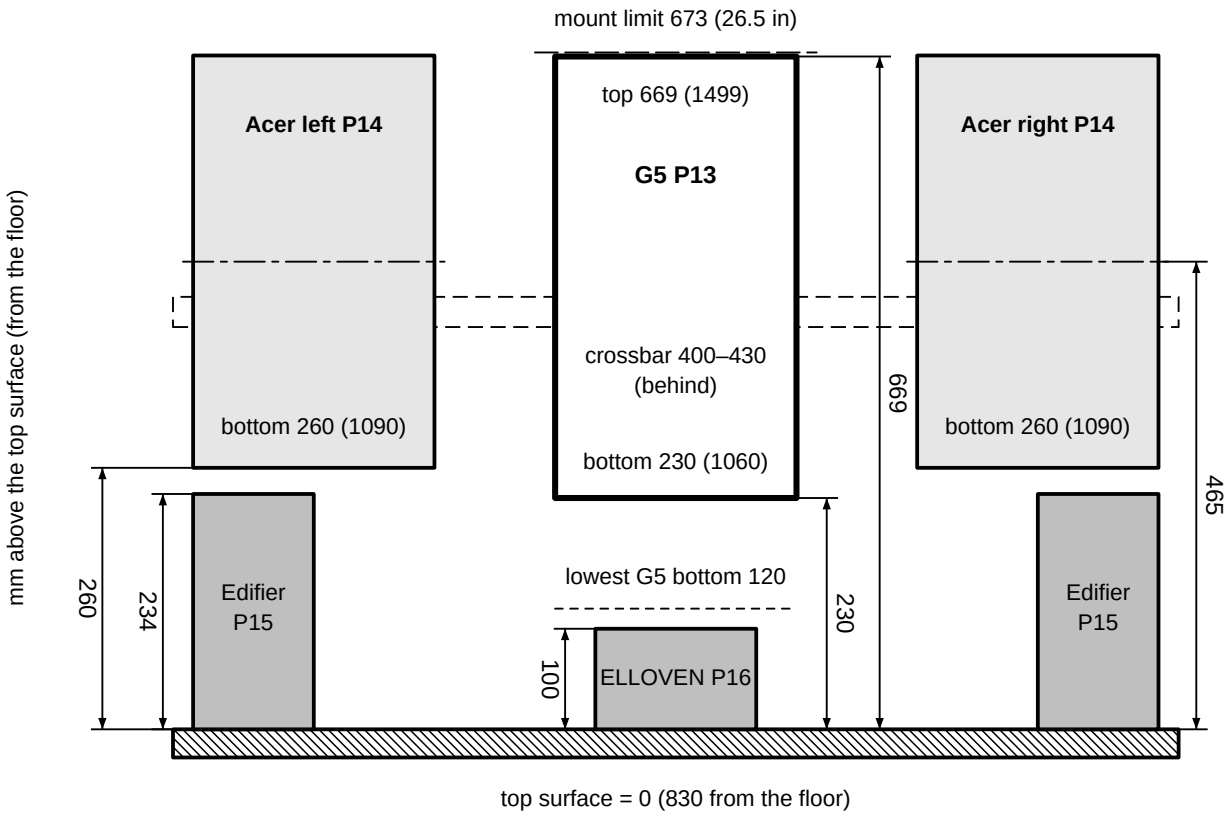


Fig. I-3 The three screens seen from the seat, unfolded flat. Heights to scale, in mm above the top surface (in brackets, from the floor); widths are schematic. The Edifiers stand under the outer half of each Acer.

Item	Above the top	From the floor	Note
Top surface	0	830	28 top on an 802 underside
ELLOVEN P16 top	100	930	130 under the G5
G5 bottom edge	230	1060	SOURCE-DEFINED
Edifier P15 top	234	1064	26 under each Acer
Acer bottom edges	260	1090	centre 465 above the top
VIVO crossbar	400–430	1230–1260	refitted at monitor height
G5 top edge	669	1499	4 under the mount's 673 (26.5 in) limit

IF THE ASSEMBLED MOUNT IS LOWER
The G5's top edge at 669 leaves only 4 below the 673 limit. If the assembled mount will not reach, lower the G5: a bottom edge anywhere down to 120 above the top still clears the ELLOVEN (100 tall).

Everything else on the desk

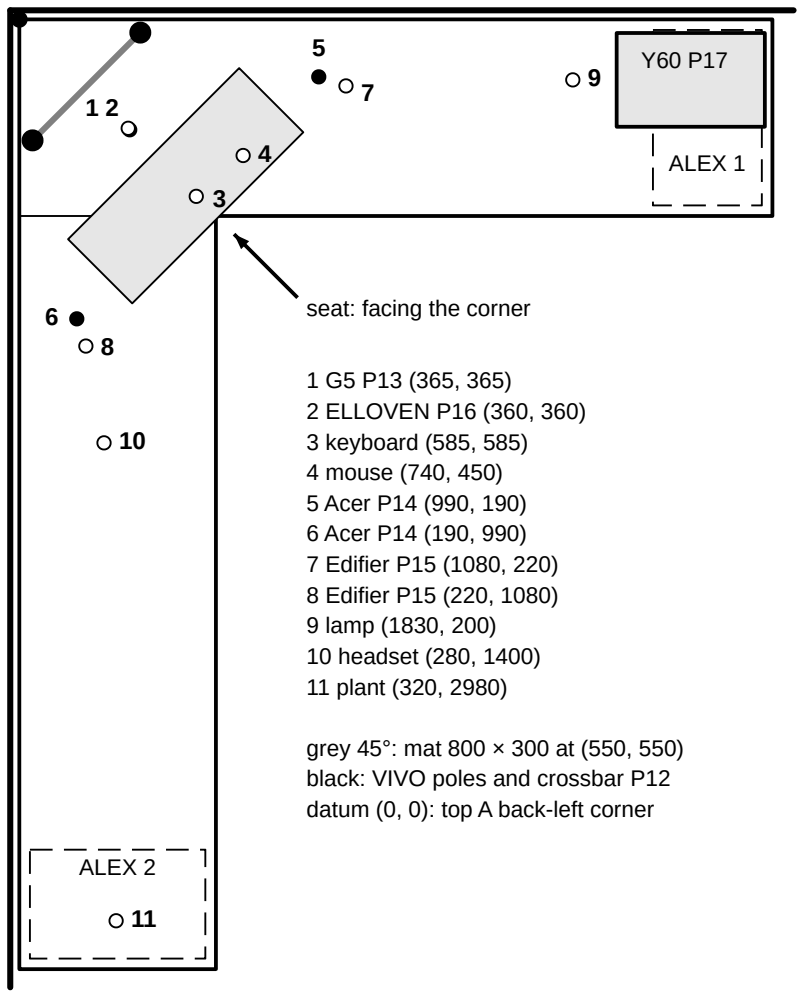


Fig. I-4 Plan of the desk, centres in mm from the datum, top A's back-left corner. Dashed: ALEX 1 and ALEX 2.

ELLOVEN at (360, 360), turned 45°, drawer toward you: its back corners clear the clamp plates and poles by over 20. At (330, 330) it would sit on the clamps.

The mat's back edge touches the ELLOVEN's front feet; its front edge reaches the inside corner of the L.

The Y60 (P17) stands on its I/O end on the printed pedestal over ALEX 1: long axis along the back wall, side glass toward you, front glass up. Keep it there, not mid-wing.

Item	Centre (x, y) mm	How
G5 P13	(365, 365)	on the diagonal, 200 in front of the crossbar
ELLOVEN P16	(360, 360)	turned 45°, drawer toward you
Deskmat 800 × 300	(550, 550)	turned 45°
NuPhy Air75 V2; mouse	(585, 585); (740, 450)	keyboard turned 45°, on the mat
Acer P14, right; left	(990, 190); (190, 990)	along each wall, toed 15° toward you
Edifier P15 ×2	(1080, 220); (220, 1080)	grilles toward you, toed with the Acers
HYTE Y60 P17	x 1975–2465, y 45–355	on its pedestal, over ALEX 1
ELYONA lamp	(1830, 200)	beside the Y60, arm toward the keyboard
Headset stand; plant	(280, 1400); (320, 2980)	on top B, hook toward you; plant over ALEX 2

All positions **SOURCE-DEFINED** (docs/build-manual.md step 8).

J · Final inspection and load testing

Ten checks, in the build manual's order, before you call the desk done. Each is an ACCEPTANCE CHECK: a pass or fail you can see or measure. Several of them test a STRUCTURAL ESTIMATE from the design review that has not been measured on this desk; the check says which. Write the results in the boxes and in the sign-off table at the end of this chapter. Datum for every drawing: top A's back-left corner (0, 0); mm.

Check 1 · Surface height at the six corners ACCEPTANCE CHECK

1

Measure from the floor to the top face at each of the six corners of the L (Fig. J-1). Every reading **830 ± 3** (83.0 cm). With the optional ALEX feet fitted: 830 plus their height. If a corner is out, turn that corner's OLOV foot or round foot, then measure all six again.

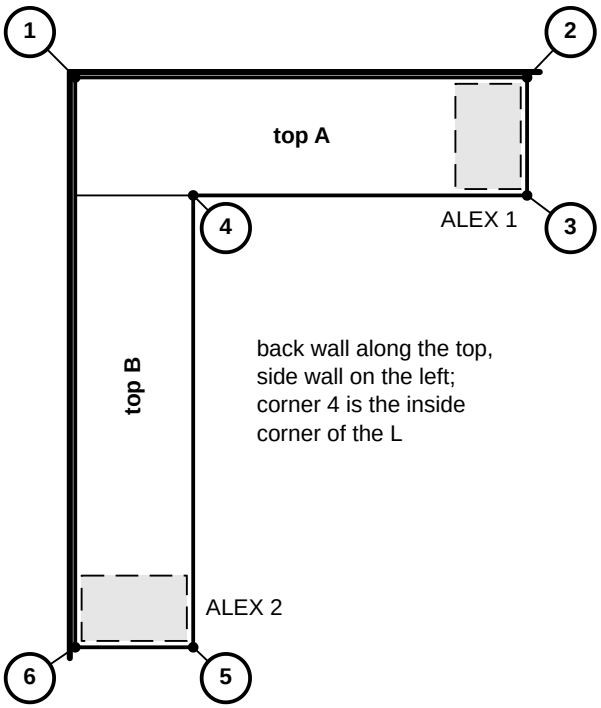


Fig. J-1a The six corners, numbered for the record.

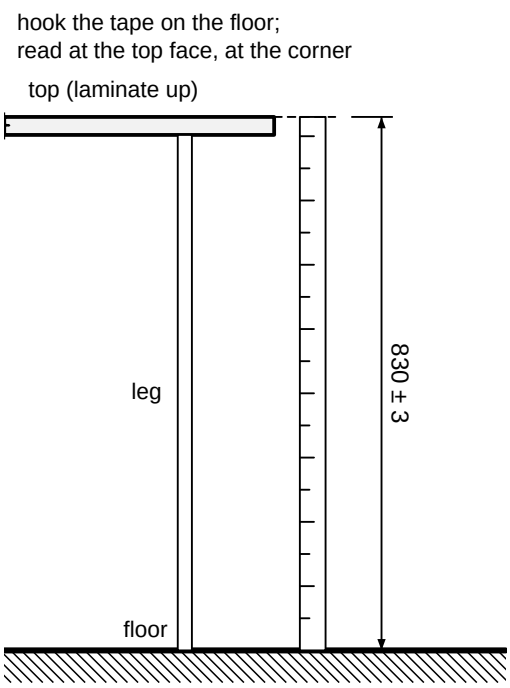


Fig. J-1b Tape from the floor to the top face, at the corner.

Corner	1	2	3	4	5	6
Reading (mm)						

Check 2 · Level and flush across the joint ACCEPTANCE CHECK

2

A level across the joint reads flat both ways: across it (1) and along it (2). A fingertip drawn across the joint (3) finds no step. Level the two tops with the OLOV feet and the round feet, never by packing anything under the joining plates.

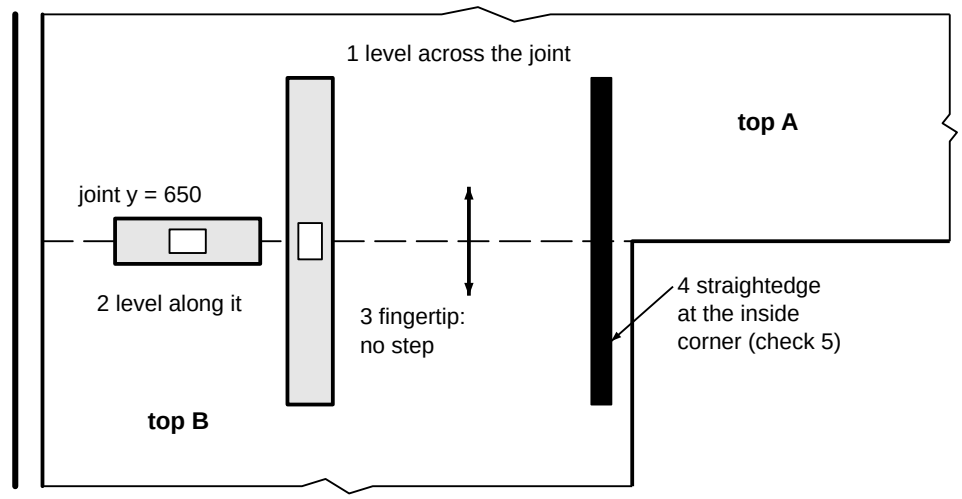


Fig. J-2 Plan of the joint at y = 650. 1, 2: level positions; 3: fingertip; 4: where the straightedge of check 5 goes, at the inside corner.

For the remaining checks you need



Tape
floor to top face, checks 1 and 8



Level, straightedge
checks 2, 3 and 5; a rule to read the gap



20 kg
a case of water plus a few books, check 3



Wrench
torque wrench at 25 N·m, or a 15 cm wrench, checks 5 and 10

Do the checks in this order, with the screens and everything else in place.

Check 3 · Corner load test ACCEPTANCE CHECK

NEVER STAND OR SIT ON THE DESK
The test load is a known weight: 20 kg of water and books. Do not stand or sit on the corner; that is not what the beam was sized for.

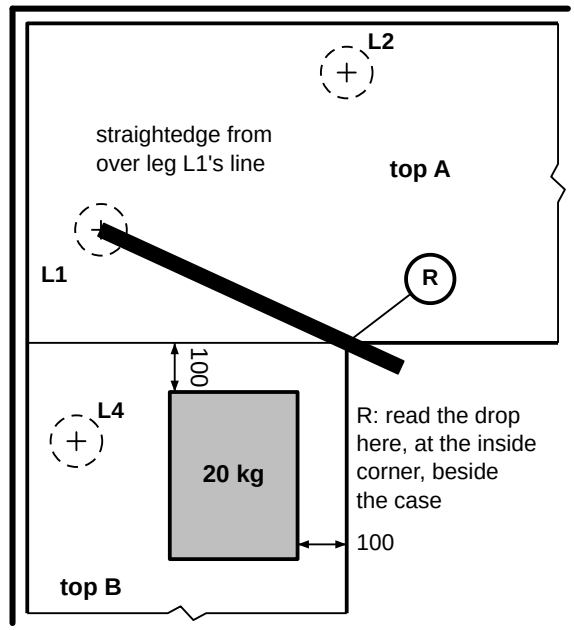


Fig. J-3a Plan. Case faces 100 in from top B's front edge and from the joint (case size schematic). R: reading point.

3

Set **20 kg** (a case of water plus a few books) on top B's inside corner, **100 (10 cm) in from both edges**.

Rest a straightedge from leg L1's line across the corner. Read the drop at the corner (R): **no more than 2**. The estimate is about 1. STRUCTURAL ESTIMATE

Lift the weight off. The corner returns to where it was, nothing creaks, no screw or bolt has moved.

Tests: sag at the inside corner (rail A carrying rail B's corner reaction) and corner-joint slip (the six fitting bolts holding by friction). STRUCTURAL ESTIMATE

Drop under 20 kg	Returned?	Creak / movement?

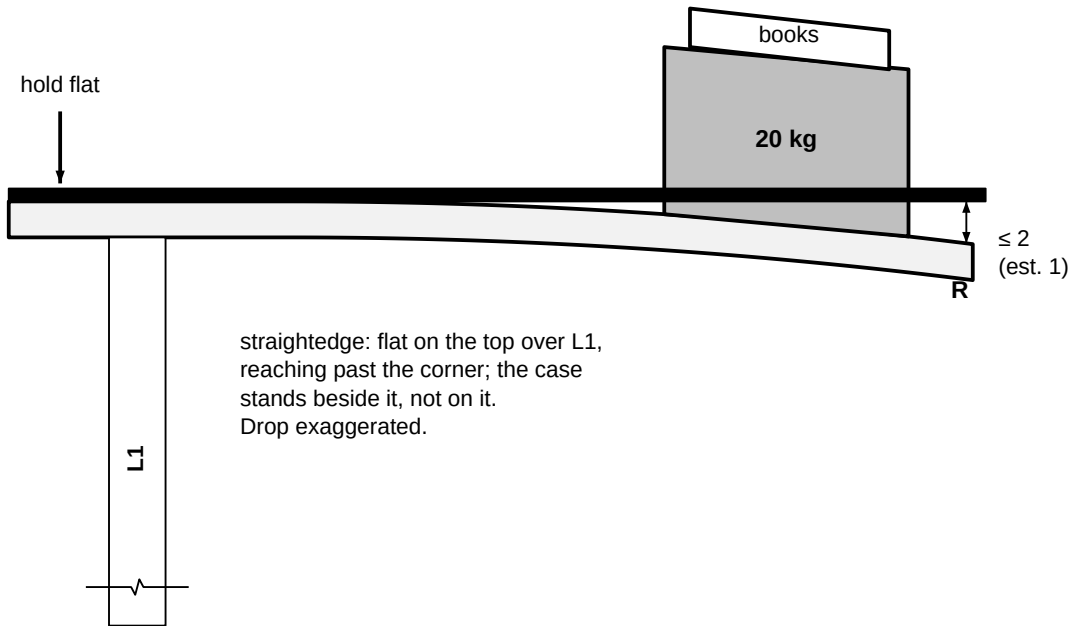


Fig. J-3b Elevation along the straightedge, from leg L1 to the corner. The case stands beside the straightedge, not under it.

Check 4 · All eight pads bear ACCEPTANCE CHECK

4

At each of the eight feet F1–F8, try to turn the pad in its ring by hand. None spins free or rocks. A pad that spins or rocks is not bearing: see chapter K, "Top moves on its ALEX".

Tests: the racking estimate (eight pads captive in their rings hold the frame against a sideways push) and the ALEX load share. STRUCTURAL ESTIMATE

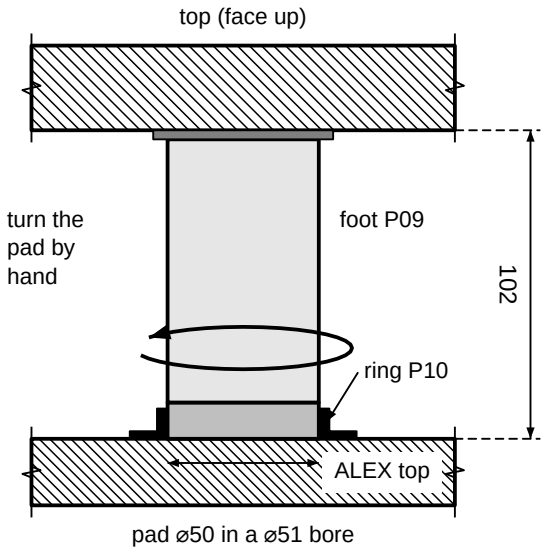


Fig. J-4 Foot P09 on its pad in ring P10 on the ALEX top.

Check 5 · Straightedge at the joint, now and later ACCEPTANCE CHECK

5

Lay a straightedge across the joint at the inside corner (Fig. J-2, position 4) and **write down any step**. Check it again after two weeks and after the first month. If the corner has settled, retighten the six fitting bolts (25 N·m, chapter K).

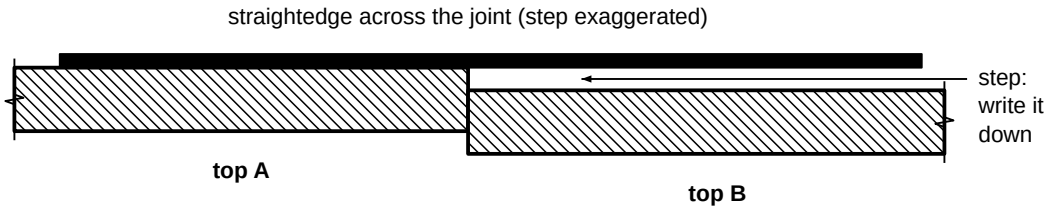


Fig. J-5 Section across the joint at the inside corner. Step drawn exaggerated.

Step at the joint	Day 0	Two weeks	One month	Bolts retightened?
mm, and which top is higher				

Tests: corner-joint slip, an estimate, not a measured capacity. STRUCTURAL ESTIMATE

Check 6 · Sideways push ACCEPTANCE CHECK

6

Push the desk sideways along each wall. No sway at the top beyond a few millimetres, and no top moves on its ALEX.

Tests: racking. The OLOV legs give almost no racking resistance; the two ALEX units, locked to the tops by the pads in their rings, do. STRUCTURAL ESTIMATE

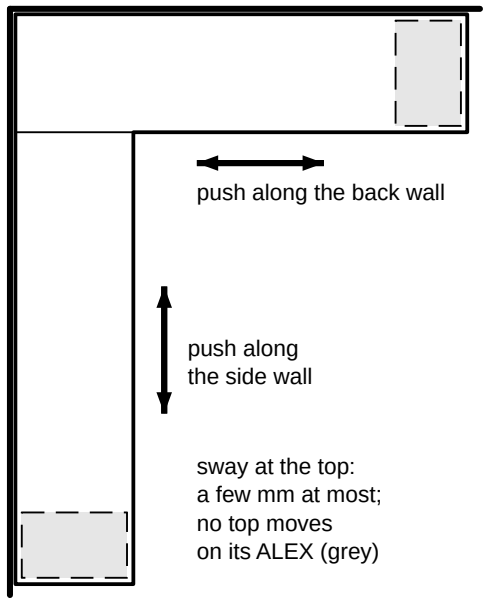


Fig. J-6 Push along the back wall, then along the side wall.

Check 7 · Screw tips ACCEPTANCE CHECK

7

No screw tip visible, or felt as a bump, on either laminate face. Run a hand over both tops, above every leg, foot plate, rail screw and joining plate.

Check 8 · Knees clear ACCEPTANCE CHECK

8

Knees clear. Sit at the corner and slide in: nothing within reach of a knee or thigh. The rails' lower edge is 751 (75.1 cm) from the floor, 120 (12 cm) back from the front edge.

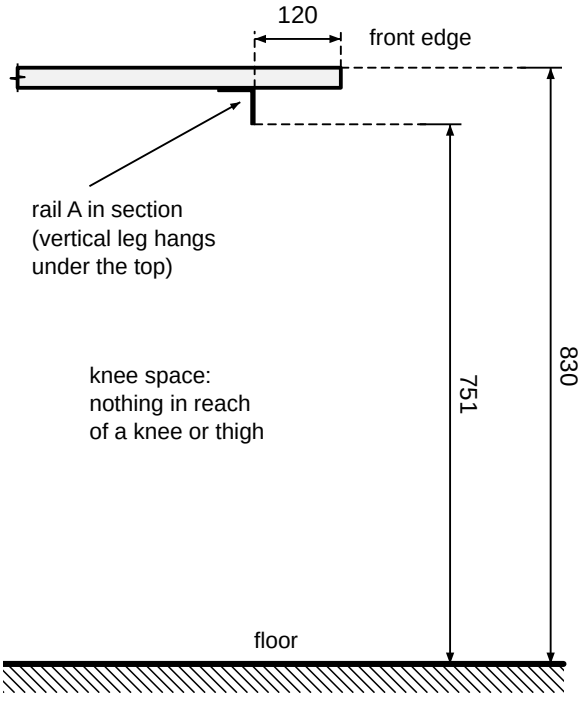


Fig. J-7 Section through top A's front at rail A.

Check 9 · ALEX fronts ACCEPTANCE CHECK

9

Both ALEX fronts open fully (ALEX 1's drawers, ALEX 2's door) without touching a foot or a leg.

Check 10 · Retighten ACCEPTANCE CHECK

10

Retighten every screw after two weeks, then repeat check 5.

Sign-off

#	Acceptance check	Pass when	Structural estimate it tests	Result
1	Surface height, six corners	830 ± 3 each (plus ALEX feet height if fitted)	—	
2	Level and fingertip across the joint	flat both ways; no step felt	—	
3	Corner load test, 20 kg	drop ≤ 2; returns; no creak; nothing moved	sag at the inside corner (estimate about 1); corner joint slip	
4	All eight pads bear	none spins free or rocks	racking; ALEX load share	
5	Straightedge at the joint	step written down; re-checked at two weeks and one month	corner joint slip (settling)	
6	Sideways push along each wall	sway a few mm at most; no top moves on its ALEX	racking	
7	Screw tips	none visible or felt on either laminate face	—	
8	Knees clear	nothing in reach; rail lower edge 751 from the floor, 120 back	—	
9	ALEX fronts	open fully, touch no foot or leg	—	
10	Retighten every screw	done after two weeks	—	

ACCEPTANCE CHECK pass or fail on this desk. STRUCTURAL ESTIMATE a number from the design review that the check confirms or refutes. If a check fails, see chapter K.

Checked by	Date	Two-week re-check	One-month re-check

K · Maintenance and troubleshooting

Schedule

When	Do	See
Day 0	Write down the step at the joint (straightedge at the inside corner).	Check 5, Fig. J-5
Two weeks	Retighten every screw. Straightedge at the joint again; if the corner has settled, retighten the six fitting bolts to 25 N·m. Look at each printed ring and its three screws.	Checks 5 and 10; troubleshooting
One month	Straightedge at the joint again; retighten the six fitting bolts if it settled.	Check 5
Any time	If the corner ever drops a millimetre, retighten the six fitting bolts.	Corner bolts, below

> **NEVER DEEPER THAN 21 MM INTO A TOP**
> The tops are 28 mm particleboard with a thin laminate skin; a screw reaching the skin lifts it. Replace a screw only with the same size. If a top measured 27, every screw into it is 1 mm shorter. SOURCE-DEFINED

Corner bolts

Bolts six 3/8-16 × 1-1/2 in hex bolts, grade 5 (H08), washer under head and nut (H09), nylon-insert nuts (H10): four through rail A's vertical leg, two through rail B's leg with a fitting on each side

Torque 25 N·m (18 ft·lb); without a torque wrench, pull as hard as one hand can on a 15 cm wrench, then stop

Why a friction joint: each fitting leg has two 21 × 12 mm slots, centred 29.15 and 78.45 from the heel VERIFIED; they leave each bolt 2.5 of play across and 11.5 along, and the rails' 7/16 in holes 1.6, so only the clamping force keeps rail B's corner from settling

Order tighten only with the two surfaces flush and rail B's end touching rail A

The mending plates across the joint align the tops; they do not carry the corner. SOURCE-DEFINED

Troubleshooting

Symptom	Cause	Fix
Corner has dropped; step at the joint grew	The six fitting bolts lost clamping force (friction joint); supports settling	Bring the surfaces flush with the OLOV feet and the round feet, never by packing anything under the joining plates. Check rail B's end still touches rail A. Retighten all six bolts to 25 N·m. Write the new step down (Fig. J-5).
Top moves on its ALEX, or a pad spins or rocks	A pad is out of its ring or not bearing	Lift that ALEX end a few millimetres and slide the ALEX unit, not the top, until the pads drop into their rings. Each foot set with the 102 gauge, no rock. Level along x and y; re-check 830 at the corners (Figs. J-1, J-4).
Printed ring cracked or its screws loose	Ring wall overloaded by a sideways push	Inspect at two weeks. A crack: reprint the ring (parts/structure/foot-ring.stl), PETG or ABS, flange-down: 51 bore, 3 wall, 75 × 2.5 flange. Refit with #6 × 5/8 in flat-head screws (H06), 5/64 in pilots 12 mm deep (12 ALEX puck). Never run the desk with a ring missing.
Leg moves or wobbles	OLOV plate screws loose	Retighten the five M4 × 16 mm screws (H01). A screw that spins: repair the hole (below), re-pilot 1/8 in, 15 mm deep (15 LEG puck). Re-set the OLOV to 802 (80.2 cm), or the ALEX-feet table's number.
Monitor clamp creeps	Clamp loosened over time	Retighten the clamp with the crossbar fitted. Keep the steel plate (P11) flat on its outline under the jaw. Never add a soft pad under the jaw.
Screw spins (stripped)	Hole worn in the particleboard	Back it out, fill the hole with wood glue and a couple of matchsticks, let it dry, re-pilot.
A mark is off by more than 2 mm	Marking error	Redraw it. A stray 1/8 in hole in the underside does no harm.
A fitting bolt will not go in	Holes not quite in line	Open the hole in the rail with a round file; never force it.

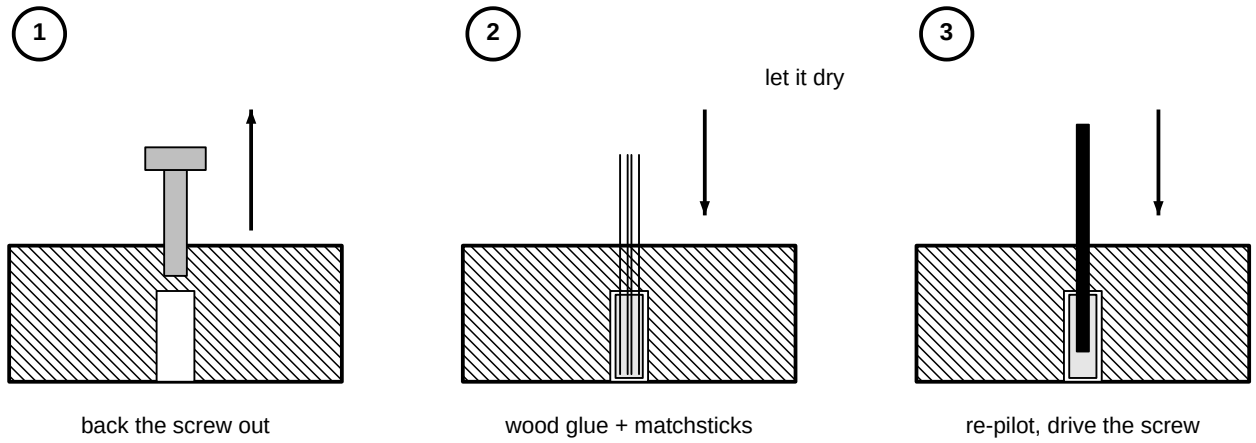


Fig. K-1 A screw that spins: back it out, fill the hole with wood glue and a couple of matchsticks, let it dry, re-pilot with the step's own bit and depth.

Every fix above is from the build manual and the build guide SOURCE-DEFINED.

Appendix · Outstanding measurements

These values were not measured on the purchased parts when this manual was built. Each is marked MEASURE ON HARDWARE at its point of use. Measure it before the step that uses it; if it differs, the step's own warning panel says what changes.

Measurement	Expected	Used in
ALEX top and bottom panel thickness	16 or more for the 5/8 in screws	Build manual step 5: ring screws, 12 mm pilots; optional ALEX feet screws
ALEX height, with and without its stock feet	700 (docs assume)	Underside 802 = 700 + foot 102; OLOV setting 802; surface 830 (check 1)
Feet pad diameter (calipers)	50; rings bored 51	Printed rings P10 before printing; check 4
EKBACKEN thickness (calipers)	28	Every screw into a top (21 max; 1 mm shorter if 27); clamp opening 28 + 3 = 31 (chapter I)
VIVO top arm reach and maximum opening	reach about 89; opening 31 or more	Chapter I: clamp zones, Fig. I-2
Delivered rail lengths and angle thickness	2130 layout (bought as 84 in = 2133.6); 4.8	Build manual step 1 hole positions; rail pilot depth 25 (4.8 steel + 20)
Room: wall to wall, and baseboard thickness	2520 and 3170	Build manual step 0 (252 × 317 cm of floor with the 30 wall gaps); step 8b wall-gap trim
Feet load rating	unknown; the design wants 50 kg or more each	Load path through the eight feet; check 3 corner load test

Confidence labels

VERIFIED	measured on the purchased part, or taken from its own geometry file.
SOURCE-DEFINED	defined by the project documents (docs/build-manual.md). A source-defined dimension is never called verified.
ASSUMPTION / FIELD CHECK	engineering estimate or field check.
MEASURE ON HARDWARE	not yet measured: measure it before the step that uses it.

Quality control

Every number in this manual comes from manual/data.json (extracted from guide.html's D object), manual/facts.json or docs/build-manual.md. Source: HEAD of https://github.com/scotthowson/desk-setup on 23 Oct 2026 (supersedes a2e99ea with measured facts). If you change a measured value, change it in the docs first, then rebuild the manual.